

Cattle Breeds of India

A Comprehensive Reference Covering 54 Indigenous & Exotic Breeds

Fully Expanded Edition — Complete Breed Profiles

India possesses one of the world's richest repositories of cattle genetic diversity, with over 50 officially recognised indigenous breeds developed over millennia to suit every agro-climatic zone — from the Thar desert to the Himalayas, from coastal wetlands to the Deccan plateau. This document provides complete breed-by-breed information including origin, physical characteristics, primary uses, milk yield, conservation status, history, adaptability, and population data. Indigenous breeds carry irreplaceable genetic traits for disease resistance, heat tolerance, and the ability to thrive on low-input systems. Several are critically endangered and urgently require conservation.

Dairy Breeds	12
Draft Breeds	18
Dual Purpose Breeds	24
Total Breeds Covered	54

Key: Each breed profile covers: Origin & Region · Physical Description (Coat, Horns, Body, Weight, Height) · Primary Uses · Milk Yield & Fat Content · Temperament · Disease Resistance · Special Traits · History · Feeding Habits · Adaptability · Conservation Status & Efforts

Amritmahal

Draft · Karnataka, India

Category	Draft
Origin	Karnataka, India
Region	Tumkur, Chitradurga, Hassan, and Mysuru districts of Karnataka
Alternative Names	Doddabasavana, Chital

Physical Characteristics

Coat Colour	Grey-white with darker grey neck and shoulders; black markings on the face and limbs
Horns	Long, lyre-shaped, pointing outward and upward
Hump	Prominent cervico-thoracic hump
Body Build	Compact, muscular, medium-to-large frame; short strong legs; tight, clean skin
Dewlap	Moderate
Weight	Males: 400–500 kg; Females: 280–350 kg
Height at Withers	130–140 cm at withers

Production & Performance

Primary Uses	Agricultural draught, military transport (historically)
Milk Yield	Low — 2–4 L/day; lactation length 200–270 days
Milk Fat Content	4.5–5.5%
Temperament	Spirited, energetic; requires experienced handlers
Disease Resistance	High resistance to foot-and-mouth disease and tick infestation

Special Traits

Exceptional stamina, speed, and agility on rugged terrain. Historically bred and maintained by the Mysore royal family as military cattle for fast troop movements. Renowned for covering long distances without fatigue.

History & Origin

The Amritmahal breed traces its lineage to the Hallikar and was selectively bred by the kings of the Mysore state during the 17th–19th centuries. The name 'Amritmahal' means 'Department of Milk' in Kannada, the administrative unit that managed the royal cattle herd. The breed was invaluable during Tipu Sultan's campaigns. After the abolition of the department post-independence, numbers fell sharply.

Feeding & Adaptability

Feeding Habits	Thrives on dry fodder, jowar straw, and coarse grasses; low input requirements
Climate Adaptability	Hot semi-arid Deccan conditions; tolerates temperatures up to 48°C
Conservation Status	Vulnerable; declining due to mechanisation
Population	Approximately 3–4 lakh (declining)

Conservation Efforts	NBAGR has registered the breed; Karnataka government runs conservation farms in Chitradurga
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Ayrshire

Dairy (Exotic) · Ayrshire, Scotland

Category	Dairy (Exotic)
Origin	Ayrshire, Scotland
Region	Originally Scotland; used in India primarily in Tamil Nadu, Karnataka, and Kerala crossbreeding programmes
Alternative Names	Ayr cattle

Physical Characteristics

Coat Colour	Red and white in irregular patches; can vary from mostly red to mostly white
Horns	Medium length, upward-curving with tips turned slightly forward
Hump	Absent (European taurine breed)
Body Build	Medium-sized, wedge-shaped dairy conformation; deep chest; angular, lean build
Dewlap	Minimal
Weight	Males: 550–700 kg; Females: 450–550 kg
Height at Withers	125–135 cm at withers

Production & Performance

Primary Uses	Dairy production; crossbreeding with local breeds for improved milk yields
Milk Yield	Moderate–High — 12–20 L/day in pure; 8–14 L/day in crossbreds in India
Milk Fat Content	3.8–4.2%; good protein content
Temperament	Active, sometimes nervous; can be difficult to handle in hot conditions
Disease Resistance	Moderate; susceptible to tropical diseases including theileriosis

Special Traits

Good udder conformation; adapts reasonably well to tropical conditions compared to Holstein-Friesian. Milk has good fat-to-protein balance.

History & Origin

Ayrshires were developed in Ayrshire County, Scotland during the 18th century by selective breeding of local cattle with Dutch and Channel Island bulls. They were introduced to India by British colonists and are now used in systematic crossbreeding programmes by state governments.

Feeding & Adaptability

Feeding Habits	Requires quality green fodder and balanced concentrate feed; not suited to low-input systems
Climate Adaptability	Moderate heat tolerance; performs best in cooler South Indian hill stations
Conservation Status	Not endangered; maintained at government dairy farms
Population	Pure populations limited in India; widespread in crossbred form
Conservation Efforts	Maintained at state government cattle breeding farms for pure line conservation

Bachaur

Dual Purpose · Sitamarhi and Sheohar districts, Bihar, India

Category	Dual Purpose
Origin	Sitamarhi and Sheohar districts, Bihar, India
Region	North Bihar terai belt, Sitamarhi, Sheohar, Muzaffarpur districts
Alternative Names	None widely recognised

Physical Characteristics

Coat Colour	White to light grey; some animals show pale yellowish tinge on neck
Horns	Short to medium, curving outward and upward
Hump	Well-developed
Body Build	Medium-sized, moderately muscular; compact frame suited to waterlogged terrain
Dewlap	Moderate
Weight	Males: 350–430 kg; Females: 260–320 kg

Height at Withers	120–130 cm at withers
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Production & Performance

Primary Uses	Draught in waterlogged paddy fields; moderate milk production
Milk Yield	Moderate — 4–7 L/day; lactation 240–280 days
Milk Fat Content	4.8–5.5%
Temperament	Docile, easy to manage
Disease Resistance	Good resistance to foot rot and water-borne infections common in flood-prone areas

Special Traits

Specially adapted to the flood-prone ecology of north Bihar. Bullocks work effectively in waterlogged paddy cultivation, a rare trait among Indian breeds.

History & Origin

Bachaur cattle evolved naturally in the sub-humid Gangetic plains near the Nepal border. Influenced by similar breeds from the terai belt of Nepal, the breed developed strong legs and hooves suited to working in wet, clayey soils.

Feeding & Adaptability

Feeding Habits	Thrives on paddy straw, mustard cake, and local grasses; low-input breed
Climate Adaptability	Hot, humid sub-tropical climate; tolerates seasonal flooding
Conservation Status	Recognised by NBAGR; population stable but small
Population	Approximately 1–2 lakh
Conservation Efforts	NBAGR has characterised the breed; Bihar State Livestock Development Agency maintains nucleus herds

Badri

Dual Purpose · Chamoli and Bageshwar districts, Uttarakhand, India

Category	Dual Purpose
Origin	Chamoli and Bageshwar districts, Uttarakhand, India
Region	Alpine meadows and mid-altitude valleys of Chamoli, Bageshwar, Pithoragarh, and Rudraprayag
Alternative Names	Pahadi cattle, Uttarakhand hill cattle

Physical Characteristics

Coat Colour	Variable — black, dark brown, or grey; thick, slightly long coat for insulation
Horns	Short to medium, variable curvature
Hump	Small to moderate
Body Build	Small-bodied, compact; strong legs for mountainous terrain; dense bone structure
Dewlap	Small
Weight	Males: 200–280 kg; Females: 150–220 kg
Height at Withers	95–115 cm at withers

Production & Performance

Primary Uses	Dairy, light agricultural draught, cultural significance
Milk Yield	Low in quantity but highly nutritious — 2–4 L/day; lactation 180–240 days
Milk Fat Content	5.5–7% fat; exceptionally high in omega-3 fatty acids and antioxidants
Temperament	Calm and hardy
Disease Resistance	Excellent resistance to respiratory and cold-climate diseases; highly adapted to high altitude

Special Traits

Graze at altitudes between 1,500 and 4,000 m on Himalayan herbs and medicinal plants. Their milk is among India's most nutritionally complete — rich in omega-3 fatty acids, CLA, antioxidants, and A2 beta-casein. Known as 'the cow of the gods' by local communities.

History & Origin

Badri cattle have grazed the Himalayan meadows for thousands of years, bred by the Garhwali and Kumaoni hill people. Named after the sacred Badrinath pilgrimage site, they are integral to Himalayan agro-pastoral culture. Traditional cattle fairs (pashu melas) in the region have historically been centred on Badri cattle.

Feeding & Adaptability

Feeding Habits	Grazes on diverse highland flora including medicinal herbs; seasonal transhumance from lower to alpine pastures
Climate Adaptability	Extreme cold (sub-zero winters); high altitude (up to 4,000 m); low-oxygen environment

Conservation Status	Vulnerable; undergoing decline due to crossbreeding with exotic breeds
Population	Approximately 1–2 lakh; declining
Conservation Efforts	Uttarakhand Livestock Development Board and ICAR-NBAGR have initiated conservation breeding programmes

Bargur

Draft · Bargur Hills, Erode district, Tamil Nadu, India

Category	Draft
Origin	Bargur Hills, Erode district, Tamil Nadu, India
Region	Bargur Hills, Bhavani forest area, Erode and Coimbatore districts, Tamil Nadu
Alternative Names	Bargur Madu (Tamil)

Physical Characteristics

Coat Colour	Reddish-brown with characteristic white stripes on the face and legs
Horns	Medium length, pointing upward and slightly backward
Hump	Moderate, well-defined
Body Build	Medium-sized, agile, well-muscled; slim legs suited to hilly terrain
Dewlap	Moderate
Weight	Males: 300–380 kg; Females: 220–270 kg
Height at Withers	115–125 cm at withers

Production & Performance

Primary Uses	Draught, pack transport on hilly terrain
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	4.5–5%
Temperament	Energetic and spirited; well-suited to mountain work
Disease Resistance	High resistance to tick-borne diseases; adapted to forest environments

Special Traits

Extraordinary sure-footedness on steep, rocky hill terrain. One of the fastest hill draught breeds in South India. The distinctive facial stripes make it visually unique among Indian cattle.

History & Origin

The Bargur breed developed in isolation in the forested hills of Erode district. Historically used by the tribal communities of the Bargur hills for pack transport and agricultural draught in terrain inaccessible to other breeds.

Feeding & Adaptability

Feeding Habits	Browses on forest shrubs, coarse hill grasses; requires minimal supplementary feeding
Climate Adaptability	Hot, dry hill climate; rugged, rocky terrain
Conservation Status	Endangered; confined to a small geographic area
Population	Less than 20,000; declining
Conservation Efforts	Tamil Nadu Livestock Development Agency has conservation efforts underway; breed registered with NBAGR

Bhelai

Draft · Chhattisgarh and Madhya Pradesh border region, India

Category	Draft
Origin	Chhattisgarh and Madhya Pradesh border region, India
Region	Rajnandgaon, Durg, and adjacent districts of Chhattisgarh and MP border
Alternative Names	Not widely documented

Physical Characteristics

Coat Colour	Grey to dark grey
Horns	Medium, curving upward
Hump	Moderate to prominent
Body Build	Sturdy, medium-sized; broad chest; adapted to forested working conditions
Dewlap	Moderate
Weight	Males: 320–400 kg; Females: 220–280 kg

Height at Withers	118–128 cm at withers
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Production & Performance

Primary Uses	Agricultural draught in forested terrain
Milk Yield	Low — 2–3 L/day
Milk Fat Content	4.5%
Temperament	Docile when trained; independent in nature
Disease Resistance	Good adaptation to tropical forest diseases

Special Traits

Suited to working in mixed forest-agriculture systems; maintained largely by Gond and other tribal communities of central India.

History & Origin

Bhelai cattle evolved in the forested margins of central India, developed through centuries of natural selection and managed breeding by tribal communities. Very little formal scientific documentation exists for this breed.

Feeding & Adaptability

Feeding Habits	Forest browse, coarse grasses, crop residues; low external input requirements
Climate Adaptability	Sub-humid tropical climate; forested terrain
Conservation Status	Critically endangered; poorly documented
Population	Very low; exact count unavailable
Conservation Efforts	Poorly supported; urgent need for government survey and conservation breeding

Brown Swiss

Dairy (Exotic) · Switzerland

Category	Dairy (Exotic)
Origin	Switzerland
Region	Originally Swiss Alps; used across India in crossbreeding, especially in Punjab, Haryana, and UP
Alternative Names	Braunvieh

Physical Characteristics

Coat Colour	Brown to grey-brown; lighter around the muzzle, eyes, and underbelly
Horns	Short, cream-coloured with dark tips
Hump	Absent
Body Build	Heavy-set, well-muscled for a dairy breed; deep chest; large, well-attached udder
Dewlap	Minimal
Weight	Males: 700–900 kg; Females: 550–650 kg
Height at Withers	135–145 cm at withers

Production & Performance

Primary Uses	Dairy production; heavy draught (historically)
Milk Yield	High — 15–25 L/day in pure animals; good production in crossbreds
Milk Fat Content	3.9–4.1% fat; 3.4% protein (excellent for cheese)
Temperament	Docile, calm; good temperament for dairy management
Disease Resistance	Moderate; better heat tolerance than Holstein-Friesian

Special Traits

One of the oldest dairy breeds in the world, with origins in the Swiss Alps dating back 4,000 years. Produces milk with superior protein content, making it ideal for cheese production. Adapts better to tropical conditions than most European breeds.

History & Origin

Brown Swiss cattle have been documented in Switzerland since the early medieval period. Introduced to India during the colonial era, they are now widely used in crossbreeding programmes to improve the productivity of native zebu cattle without sacrificing adaptability.

Feeding & Adaptability

Feeding Habits	Requires good-quality green fodder and concentrate supplementation
Climate Adaptability	Tolerates heat better than most European breeds; suited to Indo-Gangetic plains and Deccan plateau crossbreeding
Conservation Status	Not endangered; widely used globally
Population	Maintained at government dairy farms and private herds across India

Conservation Efforts	Maintained by National Dairy Development Board (NDDB) for crossbreeding
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Dagri

Draft · Kangra and Hamirpur districts, Himachal Pradesh, India

Category	Draft
Origin	Kangra and Hamirpur districts, Himachal Pradesh, India
Region	Lower foothills of Himachal Pradesh — Kangra, Hamirpur, Una, Bilaspur districts
Alternative Names	Lower Himalayan hill cattle

Physical Characteristics

Coat Colour	Variable — grey or brown; short, close coat
Horns	Short to medium, simple curve
Hump	Small but present
Body Build	Small, compact, sure-footed; strong hindquarters for mountain work
Dewlap	Small
Weight	Males: 200–270 kg; Females: 150–200 kg
Height at Withers	95–115 cm at withers

Production & Performance

Primary Uses	Pack transport on mountain paths, light agricultural draught
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5–6%
Temperament	Calm and obedient
Disease Resistance	Good resistance to respiratory diseases; adapted to cool-climate pathogens

Special Traits

Exceptional balance and agility on narrow mountain paths. Historically served as the primary means of transporting goods between lower Himachal villages before roads were built.

History & Origin

Dagri cattle developed in the lower Himalayan foothills through centuries of natural and managed selection for pack-carrying ability. They were central to the subsistence economy of Himachal hill communities before modern road infrastructure.

Feeding & Adaptability

Feeding Habits	Hill grasses, tree leaves, agricultural residues
Climate Adaptability	Cool to temperate Himalayan foothills (600–1,500 m altitude)
Conservation Status	Endangered; declining due to road development reducing demand for pack animals
Population	Very small; under 10,000
Conservation Efforts	NBAGR has surveyed the breed; conservation largely dependent on traditional farming households

Dangi

Draft · Ahmednagar and Nashik districts, Maharashtra, India

Category	Draft
Origin	Ahmednagar and Nashik districts, Maharashtra, India
Region	Sahyadri foothills — Ahmednagar, Nashik, Thane, and Raigad districts
Alternative Names	Dhangar cattle

Physical Characteristics

Coat Colour	Black or black-and-white; sometimes all-white; short, dense coat
Horns	Short, thick, pointing outward
Hump	Moderate to prominent
Body Build	Medium-sized, deep chest, very sturdy; heavily built for working in rain and mud
Dewlap	Moderate
Weight	Males: 380–450 kg; Females: 260–330 kg
Height at Withers	120–130 cm at withers

Production & Performance

Primary Uses	Heavy agricultural draught, especially in wet conditions
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Milk Yield	Low — 2–4 L/day
Milk Fat Content	4.5–5%
Temperament	Docile, steady workers
Disease Resistance	Excellent resistance to waterlogged diseases, foot rot, and tropical infections

Special Traits

Specifically bred for working in the heavy monsoon conditions of the Sahyadri. Can work in continuous rain and thick mud where other breeds fail. Disease-resistant and thrives on coarse fodder.

History & Origin

Dangi cattle were developed by the Dhargar pastoral community of western Maharashtra, who have maintained the breed for over a thousand years. The cattle were central to farming in the Western Ghats region, where monsoon rains render normal draught animals ineffective.

Feeding & Adaptability

Feeding Habits	Coarse grass, jowar straw, forest browse; very low input requirements
Climate Adaptability	Extremely high rainfall zones (2,000–4,000 mm/year); hilly terrain
Conservation Status	Vulnerable; declining due to mechanisation
Population	Approximately 2–3 lakh
Conservation Efforts	Maharashtra government maintains conservation herds; breed included in national gene bank programme

Deoni

Dual Purpose · Bidar and Osmanabad districts, Karnataka/Maharashtra, India

Category	Dual Purpose
Origin	Bidar and Osmanabad districts, Karnataka/Maharashtra, India
Region	Bidar (Karnataka), Osmanabad, Latur (Maharashtra), and parts of Andhra Pradesh
Alternative Names	Dongarpati, Wannera, Shevera, Balankya

Physical Characteristics

Coat Colour	White with bold black patches, especially on the forequarters; pink or black muzzle
Horns	Medium length, pointing outward and upward
Hump	Moderate
Body Build	Medium to large, well-proportioned; good milk lines and muscular build
Dewlap	Moderate
Weight	Males: 380–480 kg; Females: 280–360 kg
Height at Withers	125–138 cm at withers

Production & Performance

Primary Uses	Dairy production, light draught
Milk Yield	Moderate — 4–8 L/day; lactation 280–320 days
Milk Fat Content	3.8–4.5%
Temperament	Docile, easy to manage
Disease Resistance	Good natural resistance to tropical diseases

Special Traits

One of the better milk-producing native breeds of the Deccan plateau. The striking black-and-white coat pattern (similar to Holstein but of Indian origin) makes it visually distinctive.

History & Origin

The Deoni breed was developed in the Deccan plateau region and is named after the Deoni taluka in Latur district. It was historically maintained by settled farming communities of the region and has been known as a reliable dual-purpose animal for centuries.

Feeding & Adaptability

Feeding Habits	Jowar straw, groundnut cake, green fodder; moderate supplementation improves milk yield
Climate Adaptability	Semi-arid Deccan plateau; moderate heat tolerance
Conservation Status	Vulnerable; crossbreeding pressure reducing pure numbers
Population	Approximately 5–7 lakh
Conservation Efforts	Maharashtra and Karnataka state animal husbandry departments maintain conservation herds

Gangatari

Dual Purpose · Varanasi and Ghazipur districts, Uttar Pradesh, India

Category	Dual Purpose
Origin	Varanasi and Ghazipur districts, Uttar Pradesh, India
Region	Varanasi, Ghazipur, Chandauli, and Mirzapur districts of eastern UP along the Ganga
Alternative Names	Gangatlri, Middle Ganga plains cattle

Physical Characteristics

Coat Colour	White; smooth, short coat
Horns	Short to medium, slightly upward
Hump	Well-developed
Body Build	Medium to large; moderately muscular; adapted to alluvial plains
Dewlap	Moderate
Weight	Males: 380–460 kg; Females: 270–340 kg
Height at Withers	122–132 cm at withers

Production & Performance

Primary Uses	Dairy, agricultural draught in Gangetic plains
Milk Yield	Moderate — 4–7 L/day; lactation 270–300 days
Milk Fat Content	4.5–5%
Temperament	Docile
Disease Resistance	Adapted to humid river-valley conditions

Special Traits

Well-adapted to the alluvial, river-valley environment of the Ganga plains. Bullocks perform well in the black, heavy soils of the region.

History & Origin

Gangatari cattle developed in the middle Ganga basin, one of India's most densely populated agricultural landscapes. The breed evolved in river-valley conditions and has affinities with Haryana but is distinct in its origin and adaptation.

Feeding & Adaptability

Feeding Habits	Paddy straw, mustard cake, green fodder; moderate feed efficiency
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Climate Adaptability	Hot, humid sub-tropical Gangetic plains; seasonal flood-adaptation
Conservation Status	Recognised by NBAGR; population declining
Population	Approximately 2–3 lakh
Conservation Efforts	UP State Livestock Development Agency maintains records; NBAGR characterised

Gaolao

Dual Purpose · Wardha and Nagpur districts, Maharashtra/MP border, India

Category	Dual Purpose
Origin	Wardha and Nagpur districts, Maharashtra/MP border, India
Region	Wardha, Nagpur, Yavatmal (Maharashtra) and Chhindwara, Betul (MP)
Alternative Names	Gaolavi

Physical Characteristics

Coat Colour	White with a distinctive pinkish tinge on muzzle; pale yellowish-white on legs and neck
Horns	Long, white, pointing upward with inward tips
Hump	Moderate to prominent
Body Build	Medium to large; broad forehead; deep body; elegant appearance
Dewlap	Moderate to large
Weight	Males: 400–500 kg; Females: 280–360 kg
Height at Withers	128–140 cm at withers

Production & Performance

Primary Uses	Draught, dairy
Milk Yield	Moderate — 4–7 L/day; lactation 270–300 days
Milk Fat Content	4.8–5.2%
Temperament	Docile, good-natured
Disease Resistance	High resistance to foot-and-mouth disease

Special Traits

Distinctive pinkish muzzle and white horn colouration make Gaolao instantly recognisable. Well-regarded for stamina in the black cotton soils of Vidarbha.

History & Origin

Gaolao is one of the recognised breeds of central India, developed in the Vidarbha region of Maharashtra. The breed has Gir and local cattle influence and has been maintained by farming communities of the cotton-growing belt for centuries.

Feeding & Adaptability

Feeding Habits	Cotton seed cake, sorghum straw; adapted to Vidarbha agricultural by-products
Climate Adaptability	Hot, semi-arid Vidarbha and MP regions; works well in black cotton soils
Conservation Status	Vulnerable; declining
Population	Approximately 3–5 lakh
Conservation Efforts	ICAR and Maharashtra animal husbandry department running conservation programme

Ghumsari

Draft · Ganjam district, Odisha, India

Category	Draft
Origin	Ganjam district, Odisha, India
Region	Ghumsur sub-division of Ganjam district, Eastern Ghats foothills
Alternative Names	Ghumsur cattle

Physical Characteristics

Coat Colour	Variable — grey, brown, or reddish
Horns	Short, outward-pointing
Hump	Moderate
Body Build	Small to medium; compact, well-built for hilly terrain
Dewlap	Small
Weight	Males: 260–350 kg; Females: 180–250 kg
Height at Withers	110–122 cm at withers

Production & Performance

Primary Uses	Agricultural draught, pack transport in hilly terrain
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Hardy and independent
Disease Resistance	Good adaptation to forest-related diseases

Special Traits

Adapted to the Eastern Ghats hill ecosystem; serves tribal farming communities of Ganjam who rely on indigenous breeds for all agricultural operations.

History & Origin

Ghumsari cattle evolved in the isolated hilly tracts of southern Odisha, developed by the tribal communities of the region. Named after the Ghumsur area, the breed is critically under-documented.

Feeding & Adaptability

Feeding Habits	Forest browse, hill grasses, crop residues; very low input
Climate Adaptability	Sub-humid Eastern Ghats; rugged hilly terrain
Conservation Status	Endangered; limited to small geographic area
Population	Very small; under 10,000
Conservation Efforts	Odisha state animal husbandry department is conducting breed surveys; NBAGR registered

Gir

Dairy · Gir forests, Saurashtra, Gujarat, India

Category	Dairy
Origin	Gir forests, Saurashtra, Gujarat, India
Region	Gir forest area, Junagarh, Amreli, Bhavnagar, and Rajkot districts, Gujarat
Alternative Names	Bhadawari, Desan, Gujarati, Kathiawari, Sorthi, Surti

Physical Characteristics

Coat Colour	Reddish-yellow, or spotted red/white; sometimes dark red patches on white
Horns	Thick at base, curved downward then upward with sharp tips; very characteristic
Hump	Prominent, well-developed
Body Build	Medium to large; broad, domed forehead (most distinctive feature); pendulous ears; deep body
Dewlap	Large, pendulous
Weight	Males: 550–650 kg; Females: 380–450 kg
Height at Withers	130–145 cm at withers

Production & Performance

Primary Uses	Dairy (A2 milk), breeding improvement
Milk Yield	High — 10–16 L/day; up to 20 L in elite cows; lactation 300+ days
Milk Fat Content	4.5–5.5% fat; exclusively A2 beta-casein
Temperament	Gentle, calm; very docile
Disease Resistance	Exceptional — highly tick-resistant; heat-tolerant; resistant to foot-and-mouth disease

Special Traits

One of India's most globally recognised cattle breeds. The Gir is the foundation breed for the Brazilians' Gir Leiteiro (dairy type), which now numbers over 5 million in Brazil — ironically outnumbering India's own population. The breed's domed forehead, pendulous ears, and deep dewlap are unmistakable. Produces exclusively A2 beta-casein milk, highly sought after for health benefits.

History & Origin

The Gir breed has been selectively bred for over 2,000 years in the Saurashtra region. Named after the Gir forest (home of the Asiatic lion), the breed was historically maintained by the Maldharis (pastoral nomads) of Gujarat. The breed was exported to Brazil and Israel in the mid-20th century, where it transformed local dairy industries. Brazilian Gir cattle now rival the best Holstein herds in yield.

Feeding & Adaptability

Feeding Habits	Thrives on coarse fodder; highly efficient feed converter; suited to low-input systems
Climate Adaptability	Hot, semi-arid tropical conditions; extremely heat-tolerant; tick-resistant
Conservation Status	Stable in Gujarat; thriving globally due to export demand

Population	Approximately 5–7 lakh in India; several million in Brazil
Conservation Efforts	Gir Cattle Breeders Association of Gujarat; National Bureau of Animal Genetic Resources; breed exported under government supervision

Hallikar

Draft · Mysuru and Tumkur districts, Karnataka, India

Category	Draft
Origin	Mysuru and Tumkur districts, Karnataka, India
Region	Mysuru, Tumkur, Hassan, and Mandya districts of Karnataka
Alternative Names	Halli (village in Kannada)

Physical Characteristics

Coat Colour	Grey to dark grey; darkening on face, neck, and limbs
Horns	Long, swept-back, parallel to the neck; distinctive and elegant
Hump	Moderate
Body Build	Compact, athletic build; narrow face; deep chest; tight, well-knit body
Dewlap	Moderate
Weight	Males: 380–480 kg; Females: 260–340 kg
Height at Withers	125–135 cm at withers

Production & Performance

Primary Uses	Agricultural draught — widely regarded as India's finest light draught breed
Milk Yield	Low — 2–3 L/day
Milk Fat Content	4.5–5%
Temperament	Spirited; requires skilled handlers; very responsive once trained
Disease Resistance	High resistance to tick-borne diseases and foot-and-mouth disease

Special Traits

The progenitor of several important South Indian draught breeds including Amritmahal and Khillari. Legendary for speed, endurance, and precision in field operations. Hallikar bullocks won numerous cart-racing championships at Indian cattle shows historically.

History & Origin

The Hallikar is considered one of India's oldest and most pure draught breeds, developed by the farming communities of the Old Mysore State over millennia. The breed has been meticulously maintained through selective breeding for speed and endurance. It is believed to be one of the parent breeds for the Khillari and Amritmahal.

Feeding & Adaptability

Feeding Habits	Coarse fodder, ragi straw, jowar; low-input; thrives on poor-quality roughage
Climate Adaptability	Hot semi-arid Deccan plateau; works well in red laterite and black cotton soils
Conservation Status	Vulnerable; mechanisation has reduced economic importance
Population	Approximately 4–6 lakh; declining
Conservation Efforts	Karnataka government maintains conservation herds; Hallikar Cattle Breeders Association active in Mysuru

Hariana

Dual Purpose · Haryana and western Uttar Pradesh, India

Category	Dual Purpose
Origin	Haryana and western Uttar Pradesh, India
Region	Rohtak, Hissar, Gurgaon (Haryana), and Meerut, Bulandshahr (UP)
Alternative Names	Hissar, Delhi cattle

Physical Characteristics

Coat Colour	White to light grey; fine, smooth, short hair
Horns	Short, straight, pointing upward or slightly outward
Hump	Moderate
Body Build	Medium to large; comparatively narrow body; long neck; fine-boned
Dewlap	Moderate
Weight	Males: 420–520 kg; Females: 300–380 kg

Height at Withers	130–142 cm at withers
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Production & Performance

Primary Uses	Dairy production, fast agricultural draught
Milk Yield	Moderate — 4–8 L/day; up to 10 L in elite cows; lactation 280–310 days
Milk Fat Content	4.0–4.5%
Temperament	Docile, easy to manage; good disposition
Disease Resistance	Good resistance to tick-borne diseases; adapted to semi-arid North India

Special Traits

One of North India's most important dual-purpose breeds. Haryana bullocks are celebrated for their speed and responsiveness — historically used in bullock cart races. The breed is widely distributed across northern India and forms the base for several commercial dairy herds.

History & Origin

Haryana is a very old breed that evolved in the Indo-Gangetic plains of present-day Haryana. It was historically the mainstay of farming in the rich agricultural belt around Delhi. The breed has influenced several neighbouring cattle types and remains a foundation breed for north Indian dairy improvement.

Feeding & Adaptability

Feeding Habits	Wheat straw, bajra stalks, green fodder; moderate concentrate needs
Climate Adaptability	Hot, semi-arid North Indian plains; tolerates extreme heat in summer
Conservation Status	Stable but declining in purity due to Holstein crossbreeding
Population	Approximately 5–8 lakh pure animals
Conservation Efforts	NBAGR maintains breed records; Haryana Livestock Development Board runs conservation farms

Himachali Pahari

Dual Purpose · Himachal Pradesh, India

Category	Dual Purpose
Origin	Himachal Pradesh, India

Region	Widespread across Himachal Pradesh — Chamba, Mandi, Kangra, Kullu, Shimla districts
Alternative Names	Pahari cattle, Hill cattle of HP

Physical Characteristics

Coat Colour	Variable — brown, black, grey, or white; medium-length coat for insulation
Horns	Short to medium, variable
Hump	Small
Body Build	Small, compact; strong for its size; well-adapted to slopes
Dewlap	Small
Weight	Males: 180–250 kg; Females: 130–190 kg
Height at Withers	90–110 cm at withers

Production & Performance

Primary Uses	Dairy for household use, light agricultural draught
Milk Yield	Low — 2–4 L/day; lactation 180–220 days
Milk Fat Content	5.5–6.5% fat; high-quality mountain milk
Temperament	Hardy, docile
Disease Resistance	Well-adapted to respiratory diseases of cold climates

Special Traits

Small body size is an ecological adaptation to scarce high-altitude resources. Their milk is rich in fat and nutrients due to varied alpine diet. Central to subsistence farming in Himachal's mountain villages.

History & Origin

Hill cattle of Himachal Pradesh have not been selectively developed as a formal breed but represent an ecotype adapted over centuries to the specific agro-ecological conditions of the Himalayan foothills and higher mountains.

Feeding & Adaptability

Feeding Habits	Hill grasses, forest browse, tree fodder; minimal supplementation
Climate Adaptability	Cool to cold mountain climate (600–3,500 m altitude); strong seasonal variation
Conservation Status	Not officially recognised as a separate breed; classified as nondescript

Population	Several lakh across HP
Conservation Efforts	Being considered for formal characterisation by HP Animal Husbandry Department

Holstein Friesian

Dairy (Exotic) · Friesland, Netherlands

Category	Dairy (Exotic)
Origin	Friesland, Netherlands
Region	Originally Netherlands; now the world's dominant dairy breed; widely used in Punjab, Haryana, Gujarat, Tamil Nadu
Alternative Names	Holstein (US), Friesian (UK/Europe), Black-and-White

Physical Characteristics

Coat Colour	Black and white in irregular patches; very distinctive
Horns	Short, often dehorned
Hump	Absent
Body Build	Large, angular, dairy-type conformation; long neck; large prominent udder
Dewlap	Minimal
Weight	Males: 900–1,100 kg; Females: 550–680 kg
Height at Withers	140–155 cm at withers

Production & Performance

Primary Uses	Commercial dairy production
Milk Yield	Very high — 20–35 L/day in crossbreds in India; pure animals in Europe: 30–45 L/day; lactation 305 days
Milk Fat Content	3.4–3.8% fat
Temperament	Docile; requires regular human contact and intensive management
Disease Resistance	Poor — highly susceptible to tropical diseases, ticks, heat stress; requires intensive veterinary care in India

Special Traits

The world's highest-yielding dairy breed. Produces more milk than any other breed in history. However, pure Holsteins in India suffer severely from heat stress and require expensive management (fans, misters, quality feed, frequent veterinary attention). Crossbreds (50% HF + 50% native) represent the most practical application in Indian dairy farming.

History & Origin

Holstein Friesian cattle were developed over centuries in the lowlands of the Netherlands and Germany. The Holstein branch was developed separately in North America for maximum milk yield. The breed was introduced to India by colonial authorities and has been systematically used in crossbreeding since the 1970s through the National Dairy Development Board's Operation Flood programme.

Feeding & Adaptability

Feeding Habits	Requires high-energy, high-protein diet; 15–20 kg green fodder + 3–5 kg concentrate daily
Climate Adaptability	Poor in tropical heat; requires shading, fans, cool water; functions best in cooler Indian regions (Punjab, HP hills)
Conservation Status	Not endangered; the world's most numerous dairy breed
Population	Hundreds of millions globally; tens of millions of crossbreds in India
Conservation Efforts	No conservation concerns; maintained globally for commercial purposes

Jersey

Dairy (Exotic) · Jersey, Channel Islands

Category	Dairy (Exotic)
Origin	Jersey, Channel Islands
Region	Originally Channel Islands; used across India in crossbreeding, especially in South India and hilly areas
Alternative Names	Jersey cattle

Physical Characteristics

Coat Colour	Fawn, light brown, or cream; sometimes with darker shading; large, gentle eyes
Horns	Small, fine; often dehorned
Hump	Absent

Body Build	Small-to-medium, refined dairy conformation; angular; large, well-formed udder
Dewlap	Small
Weight	Males: 550–700 kg; Females: 380–450 kg
Height at Withers	118–128 cm at withers

Production & Performance

Primary Uses	Dairy — especially high-fat milk
Milk Yield	High — 10–20 L/day; lactation 305 days
Milk Fat Content	5.0–6.0% fat; highest fat % among major dairy breeds
Temperament	Docile, affectionate; popular as family cows
Disease Resistance	Better tropical adaptability than Holstein; moderate resistance to heat stress

Special Traits

Produces the richest milk of any major dairy breed in terms of fat content (5–6%), making it ideal for ghee, butter, and paneer production. Jersey crossbreds are popular among small farmers in south India because of their manageable size and high-quality milk. They tolerate heat better than Holsteins.

History & Origin

Jersey cattle were developed on the Island of Jersey for over 600 years, prized for extraordinarily rich milk. Introduced to India in the 19th century by the British, Jerseys became the most popular exotic breed for small-scale dairy farmers due to their small size and adaptability.

Feeding & Adaptability

Feeding Habits	Moderate concentrate needs; less demanding than Holstein; adapts to varied fodder
Climate Adaptability	Good heat tolerance relative to European breeds; widely suited across Indian conditions
Conservation Status	Not endangered
Population	Widely distributed across India as crossbreds
Conservation Efforts	Maintained at National Dairy Research Institute and state farms

Kangayam

Dual Purpose · Kangayam, Erode district, Tamil Nadu, India

Category	Dual Purpose
Origin	Kangayam, Erode district, Tamil Nadu, India
Region	Kangayam, Dharapuram, Tirupur, and Coimbatore districts of Tamil Nadu
Alternative Names	Kanganad, Kongu cattle

Physical Characteristics

Coat Colour	Grey with black marking on face (nose, eyelids, lips); white patches sometimes on legs
Horns	Medium length, pointing outward and slightly upward
Hump	Moderate, well-defined
Body Build	Medium-sized, well-built; strong legs; deep, compact body
Dewlap	Moderate
Weight	Males: 380–460 kg; Females: 260–340 kg
Height at Withers	122–133 cm at withers

Production & Performance

Primary Uses	Agricultural draught, moderate dairy production
Milk Yield	Low–Moderate — 3–6 L/day; lactation 240–280 days
Milk Fat Content	4.5–5%
Temperament	Spirited but manageable; hardworking
Disease Resistance	High — excellent tick resistance; adapted to hot dry climate

Special Traits

One of Tamil Nadu's most important working breeds. Known for exceptional endurance in the hot, semi-arid Kongu Nadu region. The black facial markings are a breed hallmark. Kangayam cattle play an important role in regional agricultural identity.

History & Origin

The Kangayam breed was developed and maintained over centuries by the Vellala Gounder community of the Kongu Nadu region of Tamil Nadu. Selective breeding focused on draught performance in the dry, heat-intense conditions of the region. The breed was widely used on cotton and groundnut farms.

Feeding & Adaptability

Feeding Habits	Cotton seed cake, groundnut straw, coarse grasses; thrives on semi-arid forage
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Climate Adaptability	Hot semi-arid South India; excellent heat and drought tolerance
Conservation Status	Vulnerable; pressure from crossbreeding
Population	Approximately 5–7 lakh
Conservation Efforts	Kangayam Cattle Breeders Association; Tamil Nadu Livestock Development Agency

Kankrej

Dual Purpose · Rann of Kutch, Gujarat and Rajasthan, India

Category	Dual Purpose
Origin	Rann of Kutch, Gujarat and Rajasthan, India
Region	Kankrej, Radhanpur, Tharad talukas of Gujarat and Barmer, Jalore districts of Rajasthan
Alternative Names	Wadad, Waged, Nagar, Talabda, Vaghania, Bannai

Physical Characteristics

Coat Colour	Grey to silver-grey; lighter on body; darker grey to nearly black on forequarters, neck, and legs
Horns	Large, lyre-shaped (U-shaped when viewed from front); one of the most distinctive horn shapes in India
Hump	Large, firm, well-developed
Body Build	Large-framed; powerful muscular build; broad forehead; deep chest; long neck
Dewlap	Large
Weight	Males: 550–700 kg; Females: 380–480 kg
Height at Withers	140–152 cm at withers (one of India's tallest breeds)

Production & Performance

Primary Uses	Heavy agricultural draught, dairy
Milk Yield	Moderate — 4–8 L/day; lactation 280–310 days
Milk Fat Content	4.5–5.5%
Temperament	Docile but energetic; very intelligent

Disease Resistance	Outstanding — highly tick-resistant; adapted to extreme heat and aridity
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Special Traits

Kankrej bullocks are among the fastest working bullocks in all of India and are celebrated for their ability to cover long distances at speed. The breed was one of the foundational breeds for the American Brahman (along with Gir, Ongole, and Sahiwal). It remains one of India's most important genetic resources for tropical cattle improvement globally.

History & Origin

The Kankrej breed has been developed over centuries in the extreme arid conditions of the Rann of Kutch and surrounding areas. Historical records mention the breed being traded to the Mughal court. Kankrej bulls were exported to USA in the early 20th century to help develop the American Brahman breed, which now numbers tens of millions.

Feeding & Adaptability

Feeding Habits	Adapted to sparse, coarse forage; highly efficient at extracting nutrients from low-quality feed
Climate Adaptability	Extreme heat (45°C+); high salinity soils; drought conditions; one of India's most climate-resilient breeds
Conservation Status	Stable; internationally valued
Population	Approximately 8–10 lakh in India; used extensively in Brazil
Conservation Efforts	Kankrej Cattle Breeders Association of Gujarat; widely maintained by private breeders

Kenkatha

Draft · Ken river valley, Bundelkhand, Uttar Pradesh/MP, India

Category	Draft
Origin	Ken river valley, Bundelkhand, Uttar Pradesh/MP, India
Region	Banda, Hamirpur, Mahoba districts (UP) and Chhatarpur, Panna (MP)
Alternative Names	Kenwariya

Physical Characteristics

Coat Colour	White to pale grey; short coat
Horns	Short, thick, pointing outward
Hump	Moderate

Body Build	Small, compact, deep-chested; very strong for its size
Dewlap	Small
Weight	Males: 280–360 kg; Females: 200–260 kg
Height at Withers	110–120 cm at withers

Production & Performance

Primary Uses	Agricultural draught on rocky, dry soils
Milk Yield	Low — 1–3 L/day
Milk Fat Content	5%
Temperament	Docile and hardy
Disease Resistance	Good resistance to drought-related diseases

Special Traits

Exceptionally economical to maintain — thrives on very poor-quality fodder. Perfectly suited to the rocky, drought-prone Bundelkhand terrain. Small size means lower feed requirements, making it ideal for marginal farmers.

History & Origin

Kenkatha cattle evolved in the Ken-Betwa river valley of Bundelkhand, one of India's most drought-affected regions. The breed developed through centuries of natural selection under conditions of severe fodder scarcity, producing an extremely hardy, efficient animal.

Feeding & Adaptability

Feeding Habits	Coarse grasses, shrub browse, crop residues; survives on minimum feed
Climate Adaptability	Hot, dry, rocky terrain of Bundelkhand; extreme drought tolerance
Conservation Status	Endangered; declining
Population	Approximately 50,000–1 lakh
Conservation Efforts	MP and UP animal husbandry departments have surveyed the breed; NBAGR registered

Khariar

Draft · Khariar region, Nuapada district, Odisha, India

Category	Draft
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Origin	Khariar region, Nuapada district, Odisha, India
Region	Nuapada, Bolangir, Kalahandi districts of western Odisha
Alternative Names	Khariyar cattle

Physical Characteristics

Coat Colour	Variable — grey, brown, or black
Horns	Short, outward-pointing
Hump	Moderate
Body Build	Small to medium; compact; strong for working in rocky plateau conditions
Dewlap	Small
Weight	Males: 260–340 kg; Females: 180–240 kg
Height at Withers	108–120 cm at withers

Production & Performance

Primary Uses	Agricultural draught, serving tribal communities
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	4.5–5%
Temperament	Hardy and docile
Disease Resistance	Adapted to drought-prone conditions

Special Traits

Critical resource for tribal farming communities of western Odisha who have no access to mechanised farming. The breed's ability to survive on minimal feed in drought-prone areas makes it invaluable.

History & Origin

Khariar cattle evolved in the Khariar plateau of western Odisha, maintained by the tribal farming communities of the Nuapada region. The breed has had minimal outside influence and retains high genetic distinctiveness.

Feeding & Adaptability

Feeding Habits	Coarse grasses, shrub browse; very low input requirements
Climate Adaptability	Hot, drought-prone plateau climate
Conservation Status	Endangered; very limited population
Population	Under 20,000

Conservation Efforts	Odisha Animal Husbandry Department and NBAGR are conducting surveys
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Kherigarh

Dual Purpose · Lakhimpur Kheri district, Uttar Pradesh, India

Category	Dual Purpose
Origin	Lakhimpur Kheri district, Uttar Pradesh, India
Region	Lakhimpur Kheri, Pilibhit, Shahjahanpur districts of northern UP, along the Nepal border
Alternative Names	Terai cattle of Kheri

Physical Characteristics

Coat Colour	White to grey
Horns	Short to medium
Hump	Moderate
Body Build	Medium-sized; adapted to forested terai conditions
Dewlap	Moderate
Weight	Males: 360–440 kg; Females: 260–320 kg
Height at Withers	120–130 cm at withers

Production & Performance

Primary Uses	Dairy, draught
Milk Yield	Moderate — 4–7 L/day
Milk Fat Content	4.5–5%
Temperament	Docile
Disease Resistance	Adapted to humid terai environment diseases

Special Traits

Well-adapted to the Terai's unique combination of agricultural land, dense forest, and marshy terrain. Close in type to the Haryana but distinct in its forest-edge adaptation.

History & Origin

Kherigarh cattle were developed in the Terai belt bordering Nepal, in the ecologically rich Lakhimpur Kheri district. The area, known for its forests and farmland, naturally selected for cattle capable of working in varied terrain.

Feeding & Adaptability

Feeding Habits	Green fodder from terai, paddy straw, tree leaves
Climate Adaptability	Humid sub-tropical terai; seasonal flooding tolerance
Conservation Status	Recognised by NBAGR; declining
Population	Approximately 1–2 lakh
Conservation Efforts	NBAGR has characterised the breed; local farmers maintain herds

Khillari

Draft · Sholapur and Satara districts, Maharashtra, India

Category	Draft
Origin	Sholapur and Satara districts, Maharashtra, India
Region	Sholapur, Satara, Sangli, Kolhapur districts of Maharashtra and parts of northern Karnataka
Alternative Names	Khillara, Nakali Khillari

Physical Characteristics

Coat Colour	Grey-white; neck and shoulder may be darker grey
Horns	Long, swept-back, nearly parallel to the neck and spine; the most distinctive feature
Hump	Moderate
Body Build	Compact, lean, muscular; narrow forehead; clean-cut features; very athletic build
Dewlap	Small
Weight	Males: 380–460 kg; Females: 260–330 kg
Height at Withers	125–136 cm at withers

Production & Performance

Primary Uses	High-speed draught — cart-pulling, racing, agricultural operations
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Milk Yield	Low — 1–3 L/day
Milk Fat Content	4.5–5%
Temperament	Spirited, energetic; one of the most athletic Indian breeds
Disease Resistance	Good natural resistance; suited to semi-arid Deccan conditions

Special Traits

One of the fastest draught cattle breeds in India. Khillari bullocks were the champions of bullock cart races across Maharashtra and Karnataka. The distinctive long, swept-back horns are their hallmark. The breed is a descendant of the Hallikar and has influenced other South-Western breeds.

History & Origin

The Khillari breed was developed by farming communities of the Sholapur plateau, selectively bred for generations for maximum speed and agility. The bullock cart racing tradition of Maharashtra centred on Khillari bulls, and breeders took immense pride in their animals' racing performance.

Feeding & Adaptability

Feeding Habits	Jowar straw, groundnut cake, coarse grasses; efficient converter of dry fodder
Climate Adaptability	Hot, semi-arid Deccan plateau; works in black cotton and red laterite soils
Conservation Status	Endangered; numbers have crashed due to mechanisation
Population	Under 1 lakh; sharply declining
Conservation Efforts	Maharashtra Animal Husbandry Department; Khillari Cattle Breeders Association in Sholapur

Konkan Kapila

Dual Purpose · Konkan coast, Maharashtra, India

Category	Dual Purpose
Origin	Konkan coast, Maharashtra, India
Region	Ratnagiri, Sindhudurg, Raigad, and coastal Thane districts
Alternative Names	Kapila

Physical Characteristics

Coat Colour	Reddish or dark brown; often with a darker dorsal stripe
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Horns	Short to medium, curving upward
Hump	Small to moderate
Body Build	Small, compact; well-adapted to hilly coastal terrain
Dewlap	Small
Weight	Males: 250–320 kg; Females: 180–240 kg
Height at Withers	105–118 cm at withers

Production & Performance

Primary Uses	Dairy (household), light draught in coconut plantations
Milk Yield	Low — 2–4 L/day; lactation 200–240 days
Milk Fat Content	5–6%
Temperament	Docile, easy to manage
Disease Resistance	Good adaptation to high-humidity coastal diseases

Special Traits

Specifically adapted to the hot, humid, hilly terrain of the Konkan coast. Their small size makes them ideal for working between coconut palms and navigating the hilly, laterite coastal terrain.

History & Origin

Konkan Kapila cattle evolved in the coastal Konkan belt and have been maintained by subsistence farmers of the region for centuries. The breed remains relatively pure due to geographic isolation of the coastal strip.

Feeding & Adaptability

Feeding Habits	Coconut by-products, paddy straw, coastal grasses; low input
Climate Adaptability	High humidity; hilly coastal terrain; monsoon conditions
Conservation Status	Vulnerable; under threat from crossbreeding
Population	Approximately 50,000–1 lakh
Conservation Efforts	NBAGR registered; Maharashtra Animal Husbandry Department conducting surveys

Kosali

Draft · Chhattisgarh (Kosala region), India

Category	Draft
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Origin	Chhattisgarh (Kosala region), India
Region	Raipur, Durg, Rajnandgaon, Bilaspur districts of Chhattisgarh
Alternative Names	Chhattisgarhi cattle

Physical Characteristics

Coat Colour	Grey-white; uniform colour
Horns	Short to medium, curving upward
Hump	Moderate
Body Build	Medium-sized; moderately muscular; well-suited to paddy cultivation
Dewlap	Moderate
Weight	Males: 320–400 kg; Females: 220–280 kg
Height at Withers	115–125 cm at withers

Production & Performance

Primary Uses	Agricultural draught, particularly in paddy cultivation
Milk Yield	Low — 1–3 L/day
Milk Fat Content	4.5%
Temperament	Docile
Disease Resistance	Adapted to sub-humid tropical conditions

Special Traits

Central to the agricultural system of Chhattisgarh, particularly paddy farming. The breed is maintained by tribal and farming communities who rely on it for all agricultural operations.

History & Origin

Kosali cattle evolved in the Kosala region (ancient name for Chhattisgarh), one of India's major rice-growing regions. The breed has been poorly documented despite being the primary working animal for millions of Chhattisgarh farmers.

Feeding & Adaptability

Feeding Habits	Paddy straw, coarse grasses; low supplementation requirement
Climate Adaptability	Hot, sub-humid tropical climate of central India
Conservation Status	Poorly characterised; at risk of genetic dilution
Population	Estimated 3–5 lakh but mixed with nondescript

Conservation Efforts	Chhattisgarh government has initiated breed survey; NBAGR registration in progress
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Krishna Valley

Dual Purpose · Krishna river valley, Karnataka/Maharashtra, India

Category	Dual Purpose
Origin	Krishna river valley, Karnataka/Maharashtra, India
Region	Sangli, Kolhapur, Solapur (Maharashtra) and Bijapur, Dharwad (Karnataka)
Alternative Names	None

Physical Characteristics

Coat Colour	White to pale grey; large frame, impressive appearance
Horns	Short to medium, thick
Hump	Large, well-developed
Body Build	Very large, massive; one of the heaviest Indian breeds; broad forehead; deep body
Dewlap	Large
Weight	Males: 600–750 kg; Females: 380–480 kg
Height at Withers	140–152 cm at withers

Production & Performance

Primary Uses	Heavy agricultural draught, dairy
Milk Yield	Moderate — 4–7 L/day; lactation 260–290 days
Milk Fat Content	4.5–5%
Temperament	Docile given its size
Disease Resistance	Good; adapted to the Deccan plateau conditions

Special Traits

One of India's largest and heaviest cattle breeds, developed specifically for heavy-duty agricultural work in the deep black cotton soils of the Krishna basin. The massive body size — result of Gir, Ongole, and local breed crossing — makes it unique.

History & Origin

The Krishna Valley breed was systematically developed in the 19th century by the progressive rulers of the Sangli princely state by crossing Gir bulls from Gujarat with Ongole bulls from Andhra and local Deccan cows. The resulting breed was purpose-built for heavy agricultural work in the fertile Krishna river belt.

Feeding & Adaptability

Feeding Habits	Large body requires substantial fodder: jowar, sugarcane tops, groundnut cake
Climate Adaptability	Hot semi-arid Deccan plateau; black cotton soils
Conservation Status	Vulnerable; declining
Population	Approximately 1–2 lakh
Conservation Efforts	Maharashtra and Karnataka state animal husbandry departments working on conservation

Ladakhi

Dual Purpose · Ladakh, Jammu and Kashmir, India

Category	Dual Purpose
Origin	Ladakh, Jammu and Kashmir, India
Region	Leh, Kargil, Zaskar valleys of Ladakh
Alternative Names	Ladakhi cattle, Zomo (cross with yak)

Physical Characteristics

Coat Colour	Variable — black, dark brown, or grey; long, shaggy coat for insulation
Horns	Short to medium, variable
Hump	Small
Body Build	Small-bodied with compact, heavy-boned structure; shaggy coat; adapted for cold
Dewlap	Small
Weight	Males: 200–280 kg; Females: 140–200 kg
Height at Withers	95–115 cm at withers

Production & Performance

Primary Uses	Dairy, draught, and occasionally meat
Milk Yield	Low — 1–3 L/day; lactation 150–200 days (restricted by cold season)
Milk Fat Content	6–8% fat; very rich milk to sustain calves in cold
Temperament	Hardy, somewhat independent
Disease Resistance	Exceptional cold-weather adaptation; immune to high-altitude respiratory conditions

Special Traits

Survives at altitudes exceeding 3,500 m with sub-zero winters. Produces highly rich milk to sustain calves in extreme cold. Often crossed with yaks to produce the 'Dzo' — a powerful hybrid used for both dairy and draught. Essential to remote Ladakhi communities cut off from modern systems.

History & Origin

Ladakhi cattle have co-evolved with the unique high-altitude desert ecosystem of Ladakh over thousands of years, influenced by Central Asian cattle genetics. They are adapted to the cold desert environment of the Trans-Himalayan plateau and remain integral to the traditional agro-pastoral system of Ladakh.

Feeding & Adaptability

Feeding Habits	Scarce highland vegetation — coarse mountain grasses, willow leaves, crop residues
Climate Adaptability	Extreme cold (down to -30°C); high altitude (3,500–5,000 m); sub-zero winters
Conservation Status	Vulnerable; threatened by improved exotic breeds and yak hybrids
Population	Under 1 lakh; declining
Conservation Efforts	ICAR-NBAGR and J&K Animal Husbandry Department have surveyed the breed

Lakhimi

Dairy · Assam, India

Category	Dairy
Origin	Assam, India
Region	Lakhimpur, Dhemaji, Dibrugarh districts and the Brahmaputra valley of Assam

Alternative Names	Assam hill cattle, Lakhimpur cattle
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Physical Characteristics

Coat Colour	Reddish or brown; sometimes light grey; short coat
Horns	Short to medium, curved
Hump	Small to moderate
Body Build	Small to medium; compact; adapted to wet, humid conditions
Dewlap	Small
Weight	Males: 240–320 kg; Females: 160–220 kg
Height at Withers	100–115 cm at withers

Production & Performance

Primary Uses	Dairy for household consumption
Milk Yield	Low–Moderate — 3–6 L/day; lactation 230–260 days
Milk Fat Content	4.8–5.5%
Temperament	Docile, easy to manage
Disease Resistance	Adapted to humid tropical diseases; relatively resistant to foot-and-mouth disease

Special Traits

The best indigenous dairy breed of Assam, relatively higher milk-producing than most other north-eastern cattle. Adapted to the hot, humid Brahmaputra valley ecosystem.

History & Origin

Lakhimi cattle developed in the upper Brahmaputra valley region of Assam through centuries of natural selection and selective breeding by Assamese farming communities. The breed was developed in relative isolation from peninsular Indian breeds.

Feeding & Adaptability

Feeding Habits	Green grasses, paddy straw, mustard cake; suited to Brahmaputra valley forage
Climate Adaptability	Hot, humid tropical climate; high annual rainfall (2,000+ mm); flooded paddy environments
Conservation Status	Recognised by NBAGR; declining due to crossbreeding
Population	Approximately 50,000–1 lakh pure animals
Conservation Efforts	NBAGR has characterised the breed; Assam Animal Husbandry Department conducting conservation

Malnad Gidda

Dual Purpose · Malnad (Western Ghats forest belt), Karnataka, India

Category	Dual Purpose
Origin	Malnad (Western Ghats forest belt), Karnataka, India
Region	Chikkamagaluru, Shivamogga, Hassan, Kodagu districts — Western Ghats belt
Alternative Names	Malnad dwarf cattle, Gidda (Kannada for dwarf)

Physical Characteristics

Coat Colour	Light grey to dark grey; short, smooth coat
Horns	Very short, pointed
Hump	Small
Body Build	Very small, compact; the second-smallest cattle breed in India; lightweight, agile
Dewlap	Very small
Weight	Males: 130–180 kg; Females: 90–140 kg
Height at Withers	75–95 cm at withers

Production & Performance

Primary Uses	Dairy (high-quality milk), light draught, manure production
Milk Yield	Very low in quantity — 2–3 L/day; but extremely high quality; lactation 180–220 days
Milk Fat Content	5.5–7% fat; high A2 beta-casein; rich in medicinal phytonutrients from forest diet
Temperament	Extremely docile, friendly
Disease Resistance	Outstanding — highly resistant to tick-borne diseases, mastitis, and tropical infections due to forest environment co-evolution

Special Traits

One of India's smallest breeds and one of its most extraordinary. The milk of Malnad Gidda contains phytonutrients, antioxidants, and secondary plant compounds absorbed from their diverse forest diet of over 100 plant species. Their disease resistance is exceptional. The breed is critically endangered and urgent conservation is underway.

History & Origin

Malnad Gidda cattle evolved in the dense rainforests of the Western Ghats, developing small body size as an adaptation to forested terrain where large animals cannot forage efficiently. They have been maintained for centuries by the Malnad farming communities as household animals providing milk and manure in a low-input, forest-integrated farming system.

Feeding & Adaptability

Feeding Habits	Grazes freely in forest, browsing 100+ plant species; minimal supplementary feeding required
Climate Adaptability	Hot and humid Western Ghats (high rainfall 2,500–5,000 mm/year); forested terrain
Conservation Status	Endangered; one of India's most threatened cattle breeds
Population	Under 10,000 pure animals; critically low
Conservation Efforts	KVAFSU Bidar, NBAGR, and Karnataka government have emergency conservation programmes; breed registered as endangered

Malvi

Dual Purpose · Malwa plateau, Madhya Pradesh, India

Category	Dual Purpose
Origin	Malwa plateau, Madhya Pradesh, India
Region	Ujjain, Indore, Mandsaur, Ratlam districts of western MP (Malwa plateau)
Alternative Names	Mahadeopur

Physical Characteristics

Coat Colour	Grey to white; neck and shoulder slightly darker
Horns	Medium length, outward-pointing with slight upward curve
Hump	Moderate
Body Build	Medium to large; well-built for working in the deep black cotton soils of Malwa
Dewlap	Moderate
Weight	Males: 400–500 kg; Females: 280–360 kg
Height at Withers	126–137 cm at withers

Production & Performance

Primary Uses	Draught, dairy
Milk Yield	Moderate — 4–7 L/day; lactation 260–290 days
Milk Fat Content	4.5–5%
Temperament	Docile, good for farm work
Disease Resistance	Adapted to the semi-arid Malwa climate

Special Traits

The Malvi is considered the ancestral breed of the Nimari. Exceptionally well-adapted to working in the heavy black cotton soils of the Malwa plateau, where its powerful build is essential for deep ploughing.

History & Origin

Malvi cattle are one of the old breeds of central India, developed on the fertile Malwa plateau. The breed has affinities with the Gir type, suggesting historical crossbreeding with Gujarati cattle traded through the region.

Feeding & Adaptability

Feeding Habits	Soybean straw, wheat straw, coarse grasses; adapted to Malwa agricultural by-products
Climate Adaptability	Hot, semi-arid Malwa plateau; black cotton soils
Conservation Status	Declining; pressure from crossbreeding
Population	Approximately 3–5 lakh
Conservation Efforts	MP Animal Husbandry Department maintains conservation herds at Sagar

Mewati

Dual Purpose · Mewat region, Haryana and Rajasthan, India

Category	Dual Purpose
Origin	Mewat region, Haryana and Rajasthan, India
Region	Nuh (Mewat), Faridabad, Palwal (Haryana) and Alwar (Rajasthan)
Alternative Names	Mewat cattle

Physical Characteristics

Coat Colour	White to light grey; uniform colour
Horns	Short, straight, upward
Hump	Moderate
Body Build	Medium to large; compact, well-muscled; resembles Haryana type
Dewlap	Moderate
Weight	Males: 420–520 kg; Females: 300–380 kg
Height at Withers	130–140 cm at withers

Production & Performance

Primary Uses	Dairy, draught — classic dual purpose
Milk Yield	Moderate — 4–8 L/day; lactation 270–300 days
Milk Fat Content	4.2–5%
Temperament	Docile, responsive
Disease Resistance	Adapted to the semi-arid Aravalli belt

Special Traits

A highly versatile dual-purpose breed — bullocks perform sustained field work over long hours and cows give moderate but consistent milk yields. Well-suited to the challenging semi-arid ecology of the Mewat region.

History & Origin

Mewati cattle developed in the Mewat region south of Delhi, one of India's historically important cattle-keeping areas. The breed has affinities with Haryana but is geographically and genetically distinct, influenced by the Thar desert cattle of Rajasthan.

Feeding & Adaptability

Feeding Habits	Wheat straw, bajra stover, green fodder; moderate inputs
Climate Adaptability	Hot semi-arid conditions; Aravalli terrain
Conservation Status	Declining; under crossbreeding pressure
Population	Approximately 2–3 lakh
Conservation Efforts	Haryana State Animal Husbandry Department conducting surveys

Draft · Malkangiri and Koraput districts, Odisha, India

Category	Draft
Origin	Malkangiri and Koraput districts, Odisha, India
Region	Malkangiri, Koraput, Nabarangpur districts — southernmost Odisha
Alternative Names	Motu cattle, southern Odisha tribal cattle

Physical Characteristics

Coat Colour	Variable — grey, black, or brown
Horns	Short, outward-pointing
Hump	Small to moderate
Body Build	Small, very compact; hardy; adapted to dense forest terrain
Dewlap	Small
Weight	Males: 220–300 kg; Females: 160–220 kg
Height at Withers	100–115 cm at withers

Production & Performance

Primary Uses	Agricultural draught, religious/ceremonial use in tribal culture
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Semi-wild; requires gentle handling
Disease Resistance	High natural immunity; forest-adapted

Special Traits

Maintained almost exclusively by Kondh, Koya, and Gadaba tribal communities for whom the cattle hold religious significance. Used as sacrificial animals in important tribal festivals. One of Odisha's most culturally significant cattle breeds.

History & Origin

Motu cattle evolved in the forested mountains of southernmost Odisha, among India's most tribally diverse and remote regions. The breed has developed in near-complete isolation and carries unique genetic material.

Feeding & Adaptability

Feeding Habits	Forest browse; minimal supplementation; seasonal grazing
Climate Adaptability	Humid tropical forests; hilly terrain

Conservation Status	Endangered; threatened by crossbreeding and tribal economic change
Population	Under 15,000
Conservation Efforts	Odisha government has initiated surveys; urgent conservation needed

Nagori

Draft · Nagaur district, Rajasthan, India

Category	Draft
Origin	Nagaur district, Rajasthan, India
Region	Nagaur, Jodhpur, Pali, and Ajmer districts of Rajasthan
Alternative Names	Nagauri

Physical Characteristics

Coat Colour	White; sometimes with grey patches on neck or shoulders
Horns	Long, swept forward, slightly upward; elegant appearance
Hump	Moderate
Body Build	Large, tall, powerful; long neck; narrow but deep chest; clean, elegant lines
Dewlap	Moderate to large
Weight	Males: 500–650 kg; Females: 340–440 kg
Height at Withers	140–155 cm at withers (one of India's tallest breeds)

Production & Performance

Primary Uses	Heavy and fast draught — cart-pulling, agricultural operations
Milk Yield	Low — 2–4 L/day
Milk Fat Content	4.5–5%
Temperament	Energetic, proud; requires skilled handling
Disease Resistance	Excellent — highly heat and drought tolerant; adapted to Thar Desert conditions

Special Traits

The Nagaur bull is the star of the Nagaur cattle fair (Ramdeoji Pashu Mela), one of India's largest cattle fairs attracting over 70,000 animals and 100,000+ visitors annually. Nagori bullocks are among the finest-looking working cattle in India — tall, powerful, and elegant. They have historically been gifted between kings and nobles.

History & Origin

The Nagori breed was developed in the Nagaur district, historically a major livestock trading centre. The breed was maintained by the Nagori community (whose name is associated with cattle trading) and by Rajput and Jat farmers who valued impressive, high-performance bulls. The Nagaur fair has been held for centuries.

Feeding & Adaptability

Feeding Habits	Coarse desert grasses, khejri (Prosopis cineraria) pods, bajra stover; drought-resistant diet
Climate Adaptability	Extreme heat (50°C+); very low rainfall (< 300 mm/year); desert and semi-arid conditions
Conservation Status	Stable; supported by Nagaur cattle fair tradition
Population	Approximately 3–5 lakh
Conservation Efforts	Nagaur Cattle Fair; Rajasthan government maintains breed records; NBAGR registered

Nari

Draft · Sindh (now Pakistan); found in Rajasthan, India

Category	Draft
Origin	Sindh (now Pakistan); found in Rajasthan, India
Region	Barmer, Jaisalmer, Jodhpur districts of Rajasthan
Alternative Names	Nari cattle

Physical Characteristics

Coat Colour	White to grey; compact coat
Horns	Medium, curving outward and upward
Hump	Moderate
Body Build	Medium-sized; lean but strong; excellent for covering long distances
Dewlap	Moderate
Weight	Males: 320–420 kg; Females: 220–300 kg

Height at Withers	118–128 cm at withers
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Production & Performance

Primary Uses	Draught, pack transport across desert distances
Milk Yield	Low — 1–3 L/day
Milk Fat Content	5%
Temperament	Hardy, independent
Disease Resistance	Excellent desert adaptations; heat and drought tolerant

Special Traits

Historically associated with nomadic pastoral communities of Sindh and Rajasthan, capable of covering 20–30 km per day with pack loads in extreme desert heat. The breed's migration capacity across the Thar desert is remarkable.

History & Origin

Nari cattle were historically the cattle of the nomadic Sindhi pastoral communities. After Partition (1947), the breed's main breeding grounds were in present-day Pakistan, and populations in India became isolated in the western Thar desert. The breed has been maintained by Rajasthani pastoral communities since then.

Feeding & Adaptability

Feeding Habits	Desert shrubs, Prosopis, coarse grasses; exceptionally water-efficient
Climate Adaptability	Extreme desert conditions; long-distance travel capacity; minimal water needs
Conservation Status	Endangered; partition-era disruption reduced population
Population	Under 20,000
Conservation Efforts	Rajasthan Animal Husbandry Department surveying breed; NBAGR has characterised

Nimari

Dual Purpose · Nimar region, Madhya Pradesh, India

Category	Dual Purpose
Origin	Nimar region, Madhya Pradesh, India
Region	Khandwa, Khargone, Barwani districts of MP along the Narmada valley

Alternative Names	None widely used
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Physical Characteristics

Coat Colour	Red to reddish-brown; uniform colour
Horns	Medium, outward-pointing
Hump	Moderate
Body Build	Medium to large; well-muscled; deep body
Dewlap	Moderate
Weight	Males: 400–500 kg; Females: 280–360 kg
Height at Withers	126–138 cm at withers

Production & Performance

Primary Uses	Draught, dairy
Milk Yield	Moderate — 4–7 L/day; lactation 260–290 days
Milk Fat Content	4.8–5.2%
Temperament	Docile
Disease Resistance	Good; adapted to Narmada valley conditions

Special Traits

A dual-purpose breed developed from Gir and Khillari crosses, combining good milk production with solid draught capacity. The distinctive reddish coat sets it apart from other central Indian breeds.

History & Origin

The Nimari breed developed along the Narmada river basin, an area historically known for cattle trading between Gujarat and central India. The breed incorporates Gir genetics from Gujarat and Khillari influence from Maharashtra.

Feeding & Adaptability

Feeding Habits	Soybean straw, wheat straw, sorghum; adapted to Narmada valley agriculture
Climate Adaptability	Hot, semi-arid Narmada valley; adapts to both black cotton and red soils
Conservation Status	Officially recognised breed; declining
Population	Approximately 2–3 lakh
Conservation Efforts	MP Animal Husbandry Department; NBAGR registered

Ongole

Dual Purpose · Ongole region, Prakasam district, Andhra Pradesh, India

Category	Dual Purpose
Origin	Ongole region, Prakasam district, Andhra Pradesh, India
Region	Prakasam, Guntur, Nellore districts of coastal Andhra Pradesh
Alternative Names	Nellore (in USA/Brazil), American Brahman base

Physical Characteristics

Coat Colour	White to pale grey; skin pigmentation protects against solar radiation
Horns	Medium to long, upward-curving
Hump	Very large, prominent (one of the largest humps among Indian breeds)
Body Build	Very large, massive; broad forehead; large dewlap; deep body; powerful build
Dewlap	Very large, pendulous
Weight	Males: 600–800 kg; Females: 380–500 kg
Height at Withers	140–155 cm at withers

Production & Performance

Primary Uses	Heavy agricultural draught, dairy, beef (globally as Nelore)
Milk Yield	Moderate — 5–9 L/day; lactation 260–290 days
Milk Fat Content	4.5–5%
Temperament	Docile despite massive size
Disease Resistance	Exceptional — highly resistant to heat stress, ticks, and tropical diseases

Special Traits

The Ongole is the foundation of the Nelore breed in Brazil, which has become the world's most numerous beef breed (over 200 million animals). The Ongole's massive hump, large dewlap, and tick-resistant hide made it ideal for tropical beef production. In India, the breed is valued both as a draught animal and for A2 milk.

History & Origin

The Ongole breed has been developed for centuries in the fertile plains of the Krishna-Godavari delta. It was systematically selected for large size and dual-purpose performance by farmers of the coastal Andhra region. Ongole bulls were exported to Brazil and the USA in the late 19th and early 20th centuries, where they became foundational to New World tropical cattle.

Feeding & Adaptability

Feeding Habits	Thrives on rice straw, groundnut cake, coarse grasses; efficient in low-input conditions
Climate Adaptability	Tropical heat; highly tick-resistant; salt-tolerant in coastal areas
Conservation Status	Stable in India; thriving globally
Population	Approximately 5–7 lakh in India; tens of millions as Nelore in Brazil
Conservation Efforts	Andhra Pradesh Livestock Development Agency maintains conservation herds; internationally valued

Poda Thirupu

Draft · Coastal Andhra Pradesh, India

Category	Draft
Origin	Coastal Andhra Pradesh, India
Region	East Godavari, West Godavari, Krishna, and Guntur districts of coastal AP
Alternative Names	Red Eastern cattle (literal Telugu translation)

Physical Characteristics

Coat Colour	Reddish coat — the name literally means 'red eastern' in Telugu
Horns	Short, outward-curving
Hump	Moderate
Body Build	Small to medium; compact for paddy field work
Dewlap	Small
Weight	Males: 260–340 kg; Females: 180–250 kg
Height at Withers	108–120 cm at withers

Production & Performance

Primary Uses	Agricultural draught — specifically paddy cultivation in deltaic plains
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Docile
Disease Resistance	Adapted to humid delta conditions

Special Traits

Small size ideal for paddy field operations in the deltaic plains of coastal Andhra Pradesh. One of the lesser-known breeds facing extinction pressure from mechanisation.

History & Origin

Poda Thirupu cattle evolved in the Godavari and Krishna delta regions, one of India's most productive rice-growing areas. The breed has been maintained by delta farming communities for paddy cultivation.

Feeding & Adaptability

Feeding Habits	Paddy straw, coarse grasses, aquatic vegetation; deltaic diet
Climate Adaptability	Humid, deltaic tropical climate; waterlogged paddy conditions
Conservation Status	Endangered; very poorly documented
Population	Under 15,000
Conservation Efforts	AP Animal Husbandry Department conducting surveys; NBAGR registered

Pulikulam

Draft · Sivaganga and Madurai districts, Tamil Nadu, India

Category	Draft
Origin	Sivaganga and Madurai districts, Tamil Nadu, India
Region	Sivaganga, Madurai, Virudhunagar, Ramanathapuram districts of southern Tamil Nadu
Alternative Names	Jalli Kattu madu (Jallikattu cattle), Mattu

Physical Characteristics

Coat Colour	Grey to dark grey; muscular, compact body
Horns	Medium, sharply pointed; ideal for Jallikattu

Hump	Moderate to prominent
Body Build	Small to medium but extremely muscular; powerful hindquarters; fast-twitch muscles dominant
Dewlap	Moderate
Weight	Males: 280–380 kg; Females: 200–270 kg
Height at Withers	112–125 cm at withers

Production & Performance

Primary Uses	Jallikattu (bull-taming sport), agricultural draught
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Bulls extremely ferocious and fast; cows docile
Disease Resistance	High — adapted to the dry heat of southern Tamil Nadu

Special Traits

The Pulikulam is Tamil Nadu's most iconic breed, central to the ancient festival of Jallikattu (Pongal bull-taming). The breed is selected for explosive speed, aggression, and resistance to being caught — the opposite of desirable traits in a dairy breed. Its cultural role as a symbol of Tamil identity drove massive public protests when Jallikattu was briefly banned (2017).

History & Origin

Pulikulam cattle have been selectively bred for Jallikattu performance for at least 2,000 years in southern Tamil Nadu. The bull's ability to resist being held and to run at high speed has been the sole selection criterion. The 2017 public protests that led to the lifting of the Jallikattu ban brought global attention to this breed and its cultural significance.

Feeding & Adaptability

Feeding Habits	Coarse grasses, cotton seed cake, paddy straw; suited to semi-arid southern conditions
Climate Adaptability	Hot semi-arid South Tamil Nadu; drought-tolerant
Conservation Status	Culturally protected; numbers improving after Jallikattu reinstatement
Population	Approximately 1–2 lakh; recovering
Conservation Efforts	Tamil Nadu Jallikattu Conservation Organisations; Tamil Nadu government breed register

Punganur

Dairy · Punganur, Chittoor district, Andhra Pradesh, India

Category	Dairy
Origin	Punganur, Chittoor district, Andhra Pradesh, India
Region	Punganur, Palamaner talukas of Chittoor district
Alternative Names	Punganur dwarf, Nati

Physical Characteristics

Coat Colour	White or light grey; soft skin
Horns	Short, slightly outward-curving
Hump	Small
Body Build	Extremely small — stands only 70–90 cm at withers; the world's smallest cattle breed
Dewlap	Small
Weight	Males: 100–150 kg; Females: 70–120 kg
Height at Withers	70–90 cm at withers — world's smallest

Production & Performance

Primary Uses	Dairy (high-fat sacred/medicinal milk)
Milk Yield	Very low in volume — 1–3 L/day; but extraordinary fat content
Milk Fat Content	8% fat; exclusively A2 beta-casein; considered sacred in Hindu traditions
Temperament	Very gentle and docile; popular as pets and temple animals
Disease Resistance	Moderate; requires careful management due to small size

Special Traits

The world's smallest cattle breed, standing under 90 cm. Despite tiny size, produces extraordinarily rich milk (8% fat — nearly double that of a Jersey). The milk is considered sacred in Hinduism and was traditionally used by the Nawabs of Punganur as a luxury product. Maintained as temple cattle due to religious significance.

History & Origin

The Punganur breed evolved in relative isolation in the dry, rocky terrain of Chittoor district over centuries. Historically maintained by the Nawabs of Punganur and by local Brahmin communities for religious use. The breed nearly went extinct in the 1970s–1980s but was saved by local breeders and the AP government. ICAR-NBAGR has since led systematic conservation.

Feeding & Adaptability

Feeding Habits	Low-quality dry fodder — their tiny bodies need very little feed; efficient A2 milk producers
Climate Adaptability	Hot, rocky, semi-arid conditions of Chittoor
Conservation Status	Critically endangered; under active conservation
Population	Under 1,500 pure animals; critically endangered
Conservation Efforts	Andhra Pradesh government maintains conservation farm; Punganur Cattle Conservation Society; NBAGR gene bank

Purnea

Dual Purpose · Purnea district, Bihar, India

Category	Dual Purpose
Origin	Purnea district, Bihar, India
Region	Purnea, Kishanganj, Katihar, Araria districts of north-eastern Bihar
Alternative Names	North Bihar terai cattle

Physical Characteristics

Coat Colour	Variable — white, grey, or brown
Horns	Short to medium
Hump	Moderate
Body Build	Medium-sized; adapted to terai and flood-prone areas
Dewlap	Moderate
Weight	Males: 320–420 kg; Females: 220–300 kg
Height at Withers	115–125 cm at withers

Production & Performance

Primary Uses	Dairy, draught in terai conditions
Milk Yield	Moderate — 4–8 L/day
Milk Fat Content	4.5–5%
Temperament	Docile
Disease Resistance	Adapted to humid, flood-prone Bihar conditions

Special Traits

Influenced by cattle genetics from Nepal and West Bengal due to geographic proximity. Adapted to the challenging ecology of the Kosi-Mahananda river belt, one of India's most flood-prone regions.

History & Origin

Purnea cattle developed in the north-eastern corner of Bihar where the rivers Kosi, Mahananda, and their tributaries regularly flood the plains. The cattle's adaptation to waterlogged conditions and their dual-purpose nature made them invaluable to the communities of the region.

Feeding & Adaptability

Feeding Habits	Paddy straw, mustard cake, green fodder; moderate inputs
Climate Adaptability	Hot humid climate; seasonal flooding
Conservation Status	Poorly characterised; under crossbreeding pressure
Population	Approximately 2–3 lakh
Conservation Efforts	Bihar Animal Husbandry Department has initiated characterisation

Rathi

Dairy · Bikaner and Ganganagar districts, Rajasthan, India

Category	Dairy
Origin	Bikaner and Ganganagar districts, Rajasthan, India
Region	Bikaner, Ganganagar, Hanumangarh, Churu districts — north-west Rajasthan
Alternative Names	Rath cattle

Physical Characteristics

Coat Colour	Brown with white patches; irregular markings
Horns	Short to medium, curved outward
Hump	Moderate
Body Build	Medium-sized, well-proportioned; deep body; good udder development
Dewlap	Moderate
Weight	Males: 420–520 kg; Females: 300–380 kg

Height at Withers	126–136 cm at withers
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Production & Performance

Primary Uses	Dairy in desert conditions
Milk Yield	High for a desert breed — 6–12 L/day; lactation 280–310 days
Milk Fat Content	4.8–5.5% fat
Temperament	Docile, manageable
Disease Resistance	Excellent — heat tolerant, drought tolerant, tick resistant

Special Traits

Known as the 'Friesian of the East' for its relatively high milk yield in extreme desert conditions. The Rathi's combination of reasonable milk production and desert hardiness makes it uniquely valuable. It has been used in crossbreeding with Sahiwal and exotic breeds to create productive desert dairy cattle.

History & Origin

Rathi cattle developed in the Bikaner region of north-west Rajasthan, historically the home of influential cattle-breeding communities. The breed was developed by the Rath community of Rajasthan, who gave the breed its name. It has Sahiwal influence in its ancestry.

Feeding & Adaptability

Feeding Habits	Coarse desert grasses, Prosopis pods, khejri leaves; highly water-efficient
Climate Adaptability	Extreme heat (48°C+); very low rainfall (< 400 mm/year); desert conditions
Conservation Status	Stable; valued for desert dairy
Population	Approximately 5–7 lakh
Conservation Efforts	Rajasthan Animal Husbandry Department; NBAGR registered; Central Sheep and Wool Research Institute conducts surveys

Red Dane

Dairy (Exotic) · Denmark

Category	Dairy (Exotic)
Origin	Denmark
Region	Originally Denmark; used in India particularly in Tamil Nadu, Kerala, and Karnataka crossbreeding

Alternative Names	Rød Dansk Malkerace (Danish Red), RDM
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Physical Characteristics

Coat Colour	Solid deep red; no white markings in purebred animals
Horns	Short to medium, often dehorned
Hump	Absent
Body Build	Medium to large; typical dairy conformation; good udder development
Dewlap	Minimal
Weight	Males: 700–850 kg; Females: 500–600 kg
Height at Withers	130–143 cm at withers

Production & Performance

Primary Uses	Dairy production; crossbreeding for improved tropical dairy output
Milk Yield	High — 15–22 L/day; lactation 305 days
Milk Fat Content	4.2–4.5% fat — higher than Holstein
Temperament	Docile, easy to manage
Disease Resistance	Moderate; better than Holstein in tropical conditions

Special Traits

Produces milk with higher fat content than Holstein-Friesian, making it preferred for milk-fat-based products. Red Danes have better heat tolerance and adaptability to varied climates than Holsteins, making them useful for crossbreeding in India's diverse agro-climatic zones.

History & Origin

Danish Red cattle were developed over centuries in Denmark, with selection for dual-purpose utility (milk and beef). They were introduced to India as part of systematic crossbreeding programmes to improve indigenous cattle productivity while maintaining better adaptability than pure Holsteins.

Feeding & Adaptability

Feeding Habits	Requires good-quality concentrate and green fodder; less demanding than Holstein
Climate Adaptability	Moderate heat tolerance; adapts to South Indian hill stations and moderate climates
Conservation Status	Not endangered globally; limited pure numbers in India
Population	Mainly used as crossbreeding sires; limited pure herds in India

Conservation Efforts	Maintained at NDDDB farms and state animal husbandry departments
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Red Kandhari

Dual Purpose · Nanded and Latur districts, Maharashtra, India

Category	Dual Purpose
Origin	Nanded and Latur districts, Maharashtra, India
Region	Nanded, Latur, Osmanabad districts of Marathwada, Maharashtra
Alternative Names	Kandhari cattle

Physical Characteristics

Coat Colour	Deep red — uniform, rich colour; one of the most distinctively coloured Indian breeds
Horns	Medium length, outward-pointing
Hump	Moderate
Body Build	Medium to large; sturdy, well-built for Deccan farming
Dewlap	Moderate
Weight	Males: 380–480 kg; Females: 270–350 kg
Height at Withers	122–133 cm at withers

Production & Performance

Primary Uses	Dairy, draught
Milk Yield	Moderate — 4–7 L/day; lactation 270–300 days
Milk Fat Content	4.8–5.2%
Temperament	Docile
Disease Resistance	Adapted to the Deccan plateau conditions

Special Traits

The deep red coat makes Red Kandhari one of India's most visually striking breeds. A fully dual-purpose breed that balances milk production with draught utility in the semi-arid Marathwada region.

History & Origin

The Red Kandhari breed takes its name from the Kandhar region of Nanded district. It has been maintained by farming communities of the Marathwada region for centuries and is one of Maharashtra's officially recognised indigenous breeds.

Feeding & Adaptability

Feeding Habits	Cotton seed cake, sorghum straw, green fodder; adapted to Marathwada agriculture
Climate Adaptability	Hot semi-arid Deccan plateau; black cotton soils
Conservation Status	Officially recognised; declining under crossbreeding pressure
Population	Approximately 1–2 lakh
Conservation Efforts	Maharashtra Animal Husbandry Department; NBAGR characterised

Red Sindhi

Dairy · Sindh (now Pakistan); maintained in India

Category	Dairy
Origin	Sindh (now Pakistan); maintained in India
Region	Originally Kohistan and Las Bela districts, Sindh; maintained in India at government farms and Kerala, Tamil Nadu
Alternative Names	Red Karachi, Malkani

Physical Characteristics

Coat Colour	Deep red; uniform colour; rich pigmentation
Horns	Short to medium, curved
Hump	Moderate
Body Build	Medium-sized; feminine dairy conformation; well-developed udder
Dewlap	Moderate
Weight	Males: 400–500 kg; Females: 300–380 kg
Height at Withers	120–132 cm at withers

Production & Performance

Primary Uses	Premium dairy — especially A2 milk
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Milk Yield	High — 8–15 L/day; lactation 290–320 days; up to 20 L in elite herds
Milk Fat Content	4.5–5.5% fat; exclusively A2 beta-casein
Temperament	Docile, easy to manage; popular in small farms
Disease Resistance	Excellent — highly heat and humidity tolerant; tick-resistant

Special Traits

One of the world's best Zebu dairy breeds. Red Sindhi has been exported to over 20 countries for tropical dairy improvement in Africa, the Caribbean, Southeast Asia, and South America. In India, demand for pure Red Sindhi is growing rapidly as A2 milk gains premium consumer status.

History & Origin

Red Sindhi cattle were developed by the pastoral Sindhi communities of the lower Indus valley. After Partition (1947), the breed's main breeding area fell in Pakistan, but India maintained significant herds at government livestock farms. The breed was exported globally during the 1950s–70s for tropical dairy development programmes.

Feeding & Adaptability

Feeding Habits	Coarse fodder and basic concentrate; highly efficient; suited to low-input systems
Climate Adaptability	Tropical heat; high humidity; coastal environments; adapts to varied Indian climates
Conservation Status	Stable; maintained at government farms; growing private demand for A2 milk
Population	Approximately 1–2 lakh in India (maintained at farms and A2 dairy operations)
Conservation Efforts	NBAGR gene bank; Central Cattle Breeding Farm, Sunabeda; state government farms

Sahiwal

Dairy · Sahiwal (Montgomery) district, Punjab (now Pakistan); maintained in India

Category	Dairy
Origin	Sahiwal (Montgomery) district, Punjab (now Pakistan); maintained in India
Region	Originally south-west Punjab, Pakistan; maintained in India at Hissar, NDRI Karnal, and state farms

Alternative Names	Lambi Bar, Lola, Montgomery, Multani
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Physical Characteristics

Coat Colour	Reddish-dun to yellowish-red; deeper coloured than Red Sindhi; loose, wrinkled skin
Horns	Short, often rudimentary
Hump	Large, well-developed
Body Build	Heavy, deep-bodied; loose, pendulous skin; large udder; feminine but substantial build
Dewlap	Large
Weight	Males: 450–600 kg; Females: 320–430 kg
Height at Withers	126–140 cm at withers

Production & Performance

Primary Uses	Premium dairy — the best dairy breed among all zebu cattle
Milk Yield	Very high — 10–16 L/day average; up to 25 L/day in elite animals; lactation 290–340 days
Milk Fat Content	4.5–5% fat; exclusively A2 beta-casein; high protein content
Temperament	Very docile; among the gentlest of all Indian cattle breeds
Disease Resistance	Outstanding — exceptional heat tolerance; tick-resistant; resistant to tropical diseases

Special Traits

Sahiwal is the undisputed champion dairy breed among all zebu (*Bos indicus*) cattle worldwide. It has been exported to Africa, the Caribbean, Australia, and South-East Asia where it has transformed tropical dairy farming. In Pakistan, the best Sahiwal herds produce milk yields comparable to temperate Holsteins. In India, pure Sahiwal A2 milk commands a premium of 3–5x regular milk prices. The National Dairy Research Institute at Karnal and Central Cattle Breeding Farm, Hissar maintain the largest pure herds.

History & Origin

The Sahiwal breed was developed over thousands of years in the Montgomery (now Sahiwal) district of Pakistani Punjab. It was the prize possession of the zamindars (landowners) of that region. After Partition, India maintained the breed at national livestock farms. The breed has been exported globally since the 1960s as the best heat-tolerant dairy breed for the tropics.

Feeding & Adaptability

Feeding Habits	Good-quality fodder improves yield significantly; efficient converter but responds well to supplementation
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Climate Adaptability	Tropical heat; excellent; one of the most heat and tick-resistant dairy breeds in the world
Conservation Status	Stable; growing demand from A2 milk market; maintained at national farms
Population	Approximately 1–2 lakh pure animals in India; millions globally in crossbred form
Conservation Efforts	National Dairy Research Institute Karnal; Central Cattle Breeding Farm Hissar; NBAGR gene bank; Sahiwal Cattle Breeders Association

Shweta Kapila

Dual Purpose · Udupi and Dakshina Kannada districts, Karnataka, India

Category	Dual Purpose
Origin	Udupi and Dakshina Kannada districts, Karnataka, India
Region	Udupi, Dakshina Kannada, and Uttara Kannada coastal Karnataka
Alternative Names	White Kapila, Tulu cattle

Physical Characteristics

Coat Colour	Entirely white (Shweta = white in Sanskrit); sacred appearance
Horns	Short, forward-pointing
Hump	Small to moderate
Body Build	Small to medium; compact; adapted to coastal humid conditions
Dewlap	Small to moderate
Weight	Males: 250–330 kg; Females: 170–250 kg
Height at Withers	105–118 cm at withers

Production & Performance

Primary Uses	Dairy (sacred milk), light draught, ceremonial and temple use
Milk Yield	Low — 2–4 L/day; but considered medically valuable
Milk Fat Content	5.5–6.5% fat; A2 milk
Temperament	Very gentle; kept in homes and temples

Disease Resistance	Adapted to high-humidity coastal conditions
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Special Traits

The all-white colour confers religious significance in Hinduism. These cattle are maintained in temples, used in religious ceremonies, and their milk is used in sacred rituals at major temples including the Udupi Krishna temple. A living embodiment of the intersection of Indian cattle culture and religious tradition.

History & Origin

Shweta Kapila is the white variant of the Kapila type of cattle found along the Karnataka coast. The breed has been selectively maintained for its pure white colour due to its religious and ceremonial value by the Brahmin and Bunts communities of coastal Karnataka for over a thousand years.

Feeding & Adaptability

Feeding Habits	Coconut by-products, paddy straw, coastal grasses; low-input
Climate Adaptability	Hot, humid coastal Karnataka; high annual rainfall
Conservation Status	Endangered; declining due to mechanisation and crossbreeding
Population	Under 20,000
Conservation Efforts	Karnataka government and temple trusts maintaining conservation herds; NBAGR registered

Siri

Dual Purpose · Darjeeling and Sikkim, India (also Bhutan and Nepal)

Category	Dual Purpose
Origin	Darjeeling and Sikkim, India (also Bhutan and Nepal)
Region	Darjeeling (WB), Sikkim, and parts of Bhutan and Nepal
Alternative Names	Siri cattle, Siri Gai

Physical Characteristics

Coat Colour	Variable — black, brown, or grey; long, thick coat for cold winters
Horns	Short to medium, outward-pointing
Hump	Small but present
Body Build	Medium-sized with compact, stocky build; thick coat; suited to steep terrain

Dewlap	Moderate
Weight	Males: 300–380 kg; Females: 200–270 kg
Height at Withers	112–125 cm at withers

Production & Performance

Primary Uses	Dairy, pack transport, light draught
Milk Yield	Low–Moderate — 3–6 L/day; lactation 220–260 days
Milk Fat Content	5.5–7% fat; rich mountain milk
Temperament	Docile, adaptable
Disease Resistance	Adapted to cold mountain diseases; moderate tick resistance

Special Traits

A trans-boundary mountain breed shared across the eastern Himalayan region. The high-fat milk is important for making traditional butter tea and other dairy products. Siri cattle are culturally central to the Lepcha, Bhutia, and other eastern Himalayan communities.

History & Origin

Siri cattle are one of the eastern Himalaya's primary cattle breeds, shared across political boundaries of India, Bhutan, and Nepal. The breed developed in the cool, forested mountains of the eastern Himalayas and has been the livestock backbone of mountain communities for centuries.

Feeding & Adaptability

Feeding Habits	Mountain grasses, forest browse, crop residues; seasonal grazing
Climate Adaptability	Cool to cold mountain climate (1,000–3,000 m); rugged hilly terrain
Conservation Status	Vulnerable; declining under exotic breed pressure
Population	Under 50,000 in India; more in Bhutan and Nepal
Conservation Efforts	NBAGR; Sikkim Animal Husbandry Department; Bhutan Livestock Department conservation efforts

Tharparkar

Dairy · Tharparkar district, Sindh (now Pakistan) and Barmer/Jaisalmer, Rajasthan

Category	Dairy
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Origin	Tharparkar district, Sindh (now Pakistan) and Barmer/Jaisalmer, Rajasthan
Region	Barmer, Jaisalmer, Jodhpur districts of Rajasthan; western Rajasthan
Alternative Names	White Sindhi, Cutchi, Thari

Physical Characteristics

Coat Colour	White to light grey; the coat is fine and short — reflects desert sun
Horns	Medium length, outward-pointing and slightly upward
Hump	Moderate to prominent
Body Build	Medium to large; deep-chested; good dairy lines; well-adapted to desert
Dewlap	Moderate
Weight	Males: 450–550 kg; Females: 310–410 kg
Height at Withers	128–140 cm at withers

Production & Performance

Primary Uses	Dairy in desert conditions
Milk Yield	Moderate–High for desert — 6–12 L/day; lactation 290–320 days
Milk Fat Content	4.5–5.5% fat; A2 milk
Temperament	Docile, easy to manage
Disease Resistance	Excellent — highly adapted to extreme heat, drought, and sparse water intake

Special Traits

Tharparkar is among India's most impressive livestock achievements — a breed that produces commercially significant quantities of milk in conditions of extreme heat, drought, and minimal water. It is maintained at the National Research Centre on Camel, Bikaner and at national gene banks. Particularly prized for A2 beta-casein milk.

History & Origin

The Tharparkar breed evolved in the Thar Desert over centuries, maintained by the pastoral communities of western Rajasthan and the Tharparkar district (now Pakistan). Like several Rajasthani breeds, its main breeding area was bisected by Partition. India has maintained strong herds in the western Rajasthan desert districts.

Feeding & Adaptability

Feeding Habits	Desert shrubs, dry coarse grasses, khejri pods; very low water consumption
Climate Adaptability	Extreme heat (50°C); rainfall < 300 mm; desert and semi-arid conditions; minimal water tolerance
Conservation Status	Stable; one of India's best recognised desert dairy breeds
Population	Approximately 5–7 lakh
Conservation Efforts	NBAGR gene bank; National Research Centre on Camel, Bikaner; Rajasthan Animal Husbandry Department

Thutho

Draft · Nagaland and Manipur, India

Category	Draft
Origin	Nagaland and Manipur, India
Region	Kohima, Phek, Senapati districts of Nagaland and Manipur highlands
Alternative Names	Naga cattle, North-east hill cattle

Physical Characteristics

Coat Colour	Variable — brown, black, or grey
Horns	Short to medium
Hump	Small
Body Build	Small to medium; compact; adapted to highland terrain
Dewlap	Small
Weight	Males: 200–270 kg; Females: 140–200 kg
Height at Withers	95–112 cm at withers

Production & Performance

Primary Uses	Agricultural draught, ceremonial/sacrificial in tribal culture
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Docile when raised with humans; suited to highland villages

Disease Resistance	Adapted to highland north-east Indian diseases
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Special Traits

One of India's least-documented cattle breeds, found among the tribal communities of the Naga highlands. Cattle hold deep cultural and ceremonial significance in Naga society — used in Hornbill festival ceremonies and as markers of social status and wealth.

History & Origin

Thutho (or similar local names) refer to cattle breeds maintained by the Naga and related highland communities of north-east India. These breeds evolved in isolation in the densely forested highlands and have received virtually no systematic scientific study.

Feeding & Adaptability

Feeding Habits	Forest browse, highland grasses, crop residues; minimal care
Climate Adaptability	Cool, humid, highland north-east India climate
Conservation Status	Very poorly documented; likely endangered
Population	Very small; under 10,000 pure
Conservation Efforts	Virtually none; urgent need for survey, characterisation, and conservation breeding

Umblachery

Draft · Nagapattinam and Tiruvarur districts, Tamil Nadu, India

Category	Draft
Origin	Nagapattinam and Tiruvarur districts, Tamil Nadu, India
Region	Nagapattinam, Tiruvarur, and southern Thanjavur districts — Cauvery delta
Alternative Names	Jathi Madu (local name), Mottai Madu

Physical Characteristics

Coat Colour	Dark grey to black; very distinctive
Horns	Very short, sometimes nearly absent in some animals
Hump	Small
Body Build	Small, compact, very low-slung body; short legs — ideal for narrow paddy bunds
Dewlap	Small

Weight	Males: 220–300 kg; Females: 160–220 kg
Height at Withers	100–115 cm at withers

Production & Performance

Primary Uses	Draught specifically in waterlogged paddy fields and delta land
Milk Yield	Very low — 1–2 L/day
Milk Fat Content	5%
Temperament	Docile and easy to work with
Disease Resistance	High resistance to waterlogged conditions; resistant to foot rot

Special Traits

Perfectly engineered by generations of selective breeding for working in the waterlogged Cauvery delta. The short legs allow it to walk on narrow paddy bunds without trampling crops. Its dark coat and small stature make it ideal for the intense sunlight and waterlogged conditions of the delta. Known locally as 'Jathi Madu'.

History & Origin

Umblachery cattle have been developed over centuries in the Cauvery delta region, one of Tamil Nadu's most productive agricultural zones. Farmers selectively bred for short-legged, compact animals capable of working in the delicate waterlogged paddy ecosystem. The breed is named after the Umblachery village in the delta.

Feeding & Adaptability

Feeding Habits	Paddy straw, coarse coastal grasses, aquatic plants; low-input
Climate Adaptability	Hot, humid delta; waterlogged paddy environments; high flood tolerance
Conservation Status	Vulnerable; paddy mechanisation reducing demand
Population	Approximately 1–2 lakh
Conservation Efforts	Tamil Nadu Livestock Development Agency maintains conservation herds; NBAGR registered

Vechur

Dairy · Vechur, Kottayam district, Kerala, India

Category	Dairy
Origin	Vechur, Kottayam district, Kerala, India

Region	Kottayam, Ernakulam, Alappuzha districts of Kerala
Alternative Names	Vechur dwarf, Vecchor

Physical Characteristics

Coat Colour	Variable — black, brown, grey, or white; fine, short coat
Horns	Short, slightly curved
Hump	Small
Body Build	Very small — under 90 cm at withers; smallest cattle breed in the Guinness World Records (1994–2004 until Punganur was measured); compact, fine-boned
Dewlap	Small
Weight	Males: 100–160 kg; Females: 80–130 kg
Height at Withers	80–90 cm at withers — among the world's smallest

Production & Performance

Primary Uses	Dairy (medicinal A2 milk), household milk supply, cultural preservation
Milk Yield	Very low in volume — 1–3 L/day; but extremely high in nutritional and medicinal quality
Milk Fat Content	6–8% fat; highest among Indian breeds; exclusively A2 beta-casein; rich in Omega-3 and CLA
Temperament	Extremely docile; small enough to be kept in homes; beloved as pets
Disease Resistance	Outstanding — highly resistant to mastitis, tick-borne diseases, and tropical infections

Special Traits

Formerly listed in the Guinness World Records as the world's smallest cattle breed. The Vechur cow's milk has documented therapeutic properties, including being easily digestible for lactose-sensitive individuals. The breed is central to traditional Ayurvedic medicine in Kerala, where its milk, dung, urine, and ghee are all considered medicinal. The breed nearly went extinct in the 1980s.

History & Origin

Vechur cattle are native to the Kottayam district of Kerala and have been maintained for centuries by traditional farming families of central Kerala. By the 1980s, indiscriminate crossbreeding with exotic breeds had reduced the population to a handful of animals. Dr Sosamma Iype, a professor at Kerala Agricultural University, led a heroic conservation effort

starting in 1989 that saved the breed from extinction. The Guinness Book of World Records recognised Vechur as the world's smallest cattle breed in 1994.

Feeding & Adaptability

Feeding Habits	Coconut by-products, paddy straw, kitchen waste; extremely low input; ideal for backyard farming
Climate Adaptability	Hot, humid tropical Kerala climate; high annual rainfall; backyard farming systems
Conservation Status	Critically endangered; saved from extinction by conservation efforts; recovering slowly
Population	Under 5,000 pure animals; critically low
Conservation Efforts	Kerala Agricultural University Vechur Cattle Conservation Centre (Dr Sosamma Iype's programme); NBAGR gene bank; Kerala government A2 milk promotion

Ponwar

Dual Purpose · Pilibhit and Lakhimpur Kheri districts, Uttar Pradesh, India

Category	Dual Purpose
Origin	Pilibhit and Lakhimpur Kheri districts, Uttar Pradesh, India
Region	Pilibhit, Lakhimpur Kheri, Shahjahanpur districts — Terai belt of north-west UP
Alternative Names	Ponwari

Physical Characteristics

Coat Colour	White; uniform colour; smooth
Horns	Short, slightly upward-curving
Hump	Moderate
Body Build	Medium-sized; docile build; suited to Terai farming
Dewlap	Moderate
Weight	Males: 350–430 kg; Females: 250–320 kg
Height at Withers	118–128 cm at withers

Production & Performance

Primary Uses	Dairy, draught
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Milk Yield	Moderate — 4–8 L/day; lactation 260–290 days
Milk Fat Content	4.5–5%
Temperament	Very docile
Disease Resistance	Adapted to the humid terai climate

Special Traits

A reliable dual-purpose breed adapted to the unique ecology of the Terai — the zone of marshy, forested land at the base of the Himalayan foothills. Ponwar cattle work well in both agricultural fields and the ecologically complex terai landscape.

History & Origin

Ponwar cattle developed in the Terai belt of north-west Uttar Pradesh, maintaining genetic distinctiveness despite proximity to the more widespread Haryana type. The breed has been characterised and registered by NBAGR.

Feeding & Adaptability

Feeding Habits	Green terai grasses, paddy straw, wheat fodder; moderate inputs
Climate Adaptability	Humid sub-tropical Terai; marshy conditions; seasonal flooding
Conservation Status	Officially recognised by NBAGR; declining
Population	Approximately 1–2 lakh
Conservation Efforts	NBAGR registered; UP Livestock Development Agency surveys

Beef Cow Nutrition Guide

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A good cowherd nutrition program not only meets the cow's needs, but does so at minimal cost. This emphasis on cost is essential since cost/return analyses of cowherds in Kansas indicate that feed costs represent approximately 50 to 70 percent of total costs; and, feed costs are one of the few areas in which producers can make significant changes, quickly.

Feed is not only the major cost item, but it also is the major factor influencing reproductive performance—the second most important factor in cowherd profitability. This relationship establishes what should be the primary goal of cowherd nutrition programs—maintaining a high reproductive rate. Specifically, the emphasis should be on maintaining an optimal reproductive rate, which may be somewhat below the maximum rate attainable. In other words, it may cost more in feed to move the calf crop from 92 to 94 percent than the additional calves are worth. Attaining the most profitable balance between feed costs and reproductive rate is obviously difficult, but it is the key to a profitable cowherd enterprise.

The Biological Priority for Nutrients

A key concept that producers must keep in mind when planning a cowherd nutrition program is the biological priority for nutrients as shown in Table 1. The most important concept is that before a beef female will reproduce, the other requirements (i.e., maintenance, growth and milk production) must first be met.

Meeting the nutritional requirements of beef cows at minimal cost is complicated by the fact that many variables influence cow requirements and the nutritional composition of feedstuffs. The following is a review of the factors that influence nutritional requirements of beef cows.

Factors Influencing Nutritional Requirements

1. Stage of Production

One of the most difficult aspects of feeding beef cows is that their nutritional requirements change dramatically during the year based on pregnancy and lactation status. While requirements change gradually during the year, a useful way of considering these changes is to divide the beef cow reproductive year into four periods as shown in Table 2.

Period 1. This is the most crucial period in the beef cow year in terms of production and reproduc-

tion. Not only must the cow nurse a calf, but she must rebreed within 80 to 85 days to calve at the same time next year. Consequently, her nutrient requirements are greatest during this period, with inadequate nutrition resulting in lower milk production and calf weaning weight, and poorer rebreeding performance.

Period 2. During Period 2, the cow is pregnant and lactating; however, the requirements of pregnancy during this period are relatively small. Furthermore, in a spring-calving program, high-quality forage is normally available. Consequently, Period 2 isn't considered a crucial reproductive period.

Period 3. During this post-weaning period, referred to as mid-gestation, the cow isn't nursing a calf and the requirements for the developing fetus are still rela-

tively low. Therefore, the cow's nutritional requirements are low, so Period 3 isn't considered a crucial reproductive period—it is a time to maximize the use of crop residues and other low-cost roughages. For spring-calving cows, timing the breeding season so that cows are in the middle trimester of gestation when crop residue use can be maximized will reduce annual cow cost. If cow body condition scores are low, Period 3 is an excellent time to increase the cows' weight.

Period 4. The period from approximately 60 to 90 days prior to calving is another crucial reproductive period. During this time, fetal growth is at its maximum. Additionally, the cow is laying on fat stores to assist in lactation. The consequences of inadequate nutrition during this period include:

1. Lighter calf birth weights (although calving difficulty won't be reduced).
2. Lower calf survival.
3. Lower milk production and calf growth.
4. Delayed estrus—this means a later calf next year and subsequent reduced weaning weights.

Table 3 shows the nutritional requirements for an 1,100-pound beef cow by period.

2. Weather

As any producer knows, cold weather greatly increases nutritional requirements. Thus, typical

Table 1. Biological Priority for Nutrients by Beef Cows

Priority	Function
1	Maintenance
2	Growth
3	Milk production
4	Reproduction

Table 2. The 365-Day Beef Cow Year by Periods

Period	1	2	3	4
Days	82	123	70	90
	post-calving	pregnant & lactating	mid-gestation	pre-calving

Table 3. NRC* Requirements For an 1,100-Pound Beef Cow Producing 15 pounds of Milk Per Day

	Nutritional Periods			
	1	2	3	4
TDN (lbs/day)	14.5	11.5	9.5	11.2
NEm (Mcal/day)	14.9	12.2	9.2	10.3
Protein (lbs/day)	2.3	1.9	1.4	1.6
Calcium (grams/day)	33	27	17	25
Phosphorus (grams/day)	25	22	17	20
Vitamin A (I.U./day)	39,000	36,000	25,000	
	27,000			

Table 4. Windchill Factors for Cattle with Winter Coat

wind speed (mph)	Temperature (°F)												
	-10	15	0	5	10	15	20	25	30	35	40	45	50
Calm	-10	-5	0	5	10	15	20	25	30	35	40	45	50
5	-16	-11	-6	-1	3	8	13	18	23	28	33	38	43
10	-21	-16	-11	-6	-1	3	8	13	18	23	28	33	38
15	-25	-20	-15	-10	-5	0	4	9	14	19	24	29	34
20	-30	-25	-20	-15	-10	-5	0	4	9	14	19	24	29
25	-38	-32	-27	-22	-17	-12	-7	-2	2	7	12	17	22
30	-46	-41	-36	-31	-27	-21	-16	-11	-6	-1	3	8	13
35	-60	-55	-50	-45	-40	-35	-30	-25	-20	-15	-10	-5	0
40	-78	-73	-68	-63	-58	-53	-48	-43	-38	-33	-28	-23	-18

Table 5. Estimated Lower Critical Temperatures for Beef Cattle

Coat Description	Critical Temp.
Wet or summer coat	59°F
Dry fall coat	45°F
Dry winter coat	32°F
Dry heavy winter coat	18°F

rations may be low in energy during extremely cold weather and allowances should be made for the additional requirement.

Cattle perform optimally in the thermoneutral zone where temperatures are neither too cold nor too hot. When the effective ambient temperature, an index of the heating or cooling power of the environment, is outside of the thermoneutral zone, cattle performance is depressed. The effective ambient temperature considers wind, humidity and solar radiation in addition to the actual air temperature. The most common

situation that producers face is an effective ambient temperature (wind chill index) below the lower critical temperature—the lower range of the thermoneutral zone. Table 4 shows the windchill index for varying combinations of wind and temperature. If cows have shelter from the wind, the effective ambient temperature is the same as the air temperature.

In addition to actual weather conditions, the amount of insulation on the animal influences the lower critical temperature (the temperature at which the animal's performance is adversely influenced by the environment). Table 5 shows the estimated lower critical temperatures for cattle with varying hair coats.

The only adjustment in cow rations necessitated by weather is to increase maintenance energy. Protein, mineral and vitamin requirements are not changed by weather stress. Table 6 shows the percentage increase in energy

required per degree (F) that the wind chill is below the lower critical temperature.

The general rule of thumb is to increase winter ration energy 1 percent for each degree (F) below the lower critical temperature.

As anyone who has lived in Kansas knows, there is no such thing as a typical weather pattern; however, the information in Table 7 gives monthly averages for temperature and wind speed at several locations in Kansas for the months with typical windchill indexes below the lower critical temperature. This information is provided to allow producers to consider the average wind chill or effective ambient temperature for use in adjusting rations for weather stress.

Table 8 gives examples of the amount of specific feedstuffs, i.e. grain or hay, needed to meet the additional energy requirements incurred by weather stress. In some cases simply feeding more of a low-quality feedstuff

Table 6. Increased Maintenance Energy Costs for Cattle Per Degree (F) Coldness

Coat Description	Cow Weight (lbs)			
	1000	1,100	1,200	1,300
Percentage increase per degree coldness				
Summer coat or wet	2.0	2.0	1.9	1.9
Fall coat	1.4	1.3	1.3	1.3
Winter coat	1.1	1.0	1.0	1.0
Heavy winter coat	.7	.7	.6	.6

Ames, Kansas State University.

Table 7. Average Windspeed and Temperature at Several Kansas Locations

Month/City Wichita	Concordia	Dodge City	Goodland	Topeka	
Average monthly wind speed, mph					
Jan.	12.0	13.6	12.4	10.3	12.3
Feb.	12.4	14.1	12.6	10.8	12.9
Mar.	13.8	15.8	14.2	12.6	14.4
Nov.	11.5	13.7	11.8	10.2	12.3
Dec.	11.9	13.5	12.0	10.2	12.2
Average monthly temperature					
Jan.	24.9	29.5	27.2	26.1	29.6
Feb.	30.9	35.0	31.9	32.5	35.1
Mar.	39.9	42.1	37.2	41.8	44.1
Nov.	41.1	42.5	38.1	42.7	44.3
Dec.	30.8	33.7	30.3	31.8	34.4

¹ Comparative climatic data for the United States through 1978.

² Monthly normals of temperature, precipitation, and HDD and CDD for 1951–1980 for Kansas. Sept. 1982 edition #81.

Table 8. Affect of Temperature on Energy Needs^a

Effective Temp.	Increase % in Energy	Amount of Extra Hay Needed	or	Extra Grain Needed
50°F	0	0		0
30°F	0	0		0
10°F	20	3½–4 lbs/cow		2–2½ lbs/cow
-10°F	40	7–8 lbs/cow		4–5 lbs/cow

^a Assumes a dry winter coat.

Table 9. Relationship Between Cow Weight and Nutrient Requirements—(dry, mature cows—middle third of pregnancy)

Cow Weight (lbs/day)	Requirements (lbs)		
	TDN (lbs/day)	NEm (Mcal/day)	Protein
1000	8.8	7.57	1.3
1100	9.5	8.13	1.4
1200	10.1	8.68	1.5

Nutrient Requirements of Beef Cattle, NRC, 1984.

will not meet these additional requirements, in which case the energy density of the diet must be increased by either feeding a high-quality forage or by adding a high-energy feedstuff such as grain.

3. Cow Weight

As cow weight increases, the nutritional requirement for energy and protein increases. Table 9 shows the energy and protein requirements for cows of differing weights at moderate body conditioning scores. As a general rule, each 100-pound change in weight changes the Net Energy of Maintenance (NEm) requirement by about .57 Mcal. Correspondingly, the crude protein requirement changes by about .1 pound for each 100-pound change in body weight.

4. Cow Body Condition Score

While the concept of using body condition as a tool in feeding cattle is certainly not new, it has gained considerable favor with producers and nutritionists since numerous research trials have shown that reproduction in beef cows is greatly influenced by body fatness. For example, Table 10 shows the relationship between body condition at calving, pre- and post-calving gains, and cycling activity 60 days post-calving. From a practical standpoint, the goal should be to have cows calve in good body condition and avoid significant loss in condition between calving and the start of the breeding season.

To assist producers in evaluating body condition, a nine-point cow condition scoring system has been developed with “1” being a very thin cow and “9” a very fat cow. Readers interested in more information regarding the use of this system are referred to the bulletin, C-842, *Feeding Your Cows by Body Condition Score*, available from K-State Research and Extension.

Table 10. Relationship of Body Condition and Percentage of Cows Cycling 60 Days After Calving

Condition at calving	Weight change pre-calving	Weight change post-calving	% cycling 60 days post-calving
Good	Lost	Lost	90%+
Moderate	Gained	Lost	74%
Moderate	Lost	Lost	48%
Thin	Lost	Gained	46%
Thin	Lost	Lost	25%

* Source: Whitman, Colorado State University, 1975.

Table 11. Relationship Between Milk Production and Nutrient Requirements—1,100-pound cow

Lbs. of milk per day (lbs/day)	Requirement (lbs.)		
	TDN (lbs/day)	NEm (Mcal/day)	Protein
10	12.1	11.54	2.0
15	13.3	13.24	2.3
20	14.5	14.94	2.6

Nutrient Requirements of Beef Cattle, NRC, 1984.

Table 12. Peak and Average Milk Production for Common Beef Breeds

Breed	Peak milk production lbs/day	Average milk production lbs/day
Angus	20.7	14.9
Charolais	21.6	15.1
Gelbvieh	25.3	17.8
Hereford	18.7	12.5
Limousin	20.9	14.1
Simmental	24.1	16.8

Meat Animal Research Center.

5. Milk Production

Milk production places tremendous demands for nutrients on beef cows. For example, an increase in milk production of 5 pounds per day requires about 1.7 Mcal of NEm, .3 pounds of crude protein, .012 pounds of calcium, and .006 pounds of phosphorus. Table 11 illustrates the influence of increasing milk production on cow nutrient requirements.

Not only does milk production put tremendous nutritional stress on beef cows, but peak lactation usually occurs by 60 to 80 days postpartum, which unfortunately precedes the start of the breeding

season. Also, it is important to keep in mind that milk production has a higher biological priority for nutrients than reproduction. Consequently, attempting to feed a thin cow more just prior to the breeding season to increase condition may only result in more milk production. Table 12 shows the average peak production and average total production over a 205-day lactation for the most common beef breeds. Studies of lactation curves in beef cows indicate that milk production increases rather rapidly from birth of the calf to a peak at 60–80 days and then declines slowly until weaning.

6. Age

The age of the animal influences nutritional requirements because a young cow is still growing. Again, a review of the biological priority for nutrients (Table 1) is useful since it indicates that growth is the second highest priority. The importance of this factor is clear when one considers that Kansas producers routinely indicate that most of their “open” cows are 2- and 3-year-olds. The failure of these females to rebreed means that nutrition was inadequate to meet maintenance + growth + milk production and allow for reproduction.

Table 19 shows the nutrient requirements of 2-year-old heifers during early lactation. Included in these requirements is a growth rate of .5 pounds per day. The actual growth rate may be higher for large-framed heifers or heifers that are compensating for previous restrictions in energy intake. Consequently, producers should increase the levels of energy and protein above those shown in Table 19, if these situations exist.

If possible, young females, along with old or thin cows, should be fed separately from the mature cows during the winter so that their additional nutritional requirements for growth and/or condition can be met. In short, the loss of 2- and 3-year-old cows due to reproductive failure represents a significant economic loss because the expense of heifer development has already been incurred with very limited returns.

7. Physical Activity

The maintenance requirement of beef cows is increased by activity. As a general guideline, the NEm requirement is increased by about .9 Mcal/day in cows grazing compared to cows fed in a drylot. Consequently, if cows are required to graze over wide areas, adjustments should be made to the requirements shown in Table 19.

Practical Considerations

1. Forage Intake

The beef cow, by the very fact that she has a ruminant digestive system, is designed to consume and digest forages. However, a key issue is how much she will consume when allowed free-choice access to forages either on pasture or in a drylot situation. The major factors influencing intake are:

Quality of forage available.

Intake is probably most influenced by the quality of forage with intake decreasing dramatically as quality decreases.

Quantity of forage available.

Since beef cattle selectively graze, eating the better-quality forage first, as the quantity of forage available declines, the intake also tends to decline.

Protein content of the diet.

Since protein is required by ruminal microorganisms to digest forages, if it is inadequate, voluntary intake will be substantially reduced.

Environment. Weather conditions also can impact forage intake by disrupting grazing patterns.

While cold weather generally increases intake, windy or wet weather will reduce grazing time and intake.

Table 13 illustrates the relationship between forage intake, forage quality, and supplementation with energy or protein.

One of the most important concepts illustrated in this table is that, when forage quality is low or average, forage intake is increased with protein supplementation but not with energy supplementation. This increase in intake caused by adequate protein supplementation, coupled with maintenance of forage digestibility, means that the total daily energy status of the cow is increased. Also, note that the quantity of supplement is a factor in intake since the animals may substitute the supplement for forage. This is especially true with large quantities of an energy-based supplement, such as grain. Thus, the intake guidelines for forage intake with an energy supplement shown in Table 13 may be too high when a large amount of supplement is fed.

Calculating Forage Intake

Under Grazing Situations. How much crop residue or dry, winter grass will a beef cow consume? To answer this question, using the information in Table 13, let's assume that an 1,150-pound cow is grazing a low-quality crop residue. Assuming that adequate protein supplementation is supplied, the cow will consume about 1.8 percent of her body weight in forage dry matter (DM). Taking 0.018 times 1,150 pounds, we get an intake of 20.7 pounds of dry matter. If the crop residue is assumed to be 90 percent DM, the "as-fed" intake is $20.7 / .9 = 23$ pounds of residue per day plus the protein supplement. The guidelines in Table 13 can be used in this manner to estimate the intake of any forage available free-choice.

Differences in Nutrient Content Between Clipped Forage Samples and Actual Intake by Grazing Beef Cows. It would seem logical to collect forage samples from a standing forage, such as dry-range grass in the winter, and have it analyzed. While this process will give an accurate picture of the feed value of the forage on average in the pasture, clipped samples are not representative of what cattle consume. Beef cattle, given adequate forage availability, will select the highest quality material, resulting in a diet 2 to 3 percentage points higher in protein, for instance, than clipped samples. As a general rule, the differences in nutrient composition will be greatest when forage quality is low. From a practical standpoint, it is recommended that producers add approximately 2 percentage points to the laboratory protein analysis of clipped samples before formulating rations.

2. Protein Supplementation

While energy is the most commonly deficient nutrient in beef-cow diets, protein often represents the largest "out-of-pocket" expense for cow-calf producers. Proper

Table 13. Expected Intake Levels of Forage Varying in Quality With and Without Supplementation

Roughage Type	Forage Intake (% Body Weight, DM Basis) of Beef Cows	
	Dry, bred cow	Lactating Cow
Low quality roughages:		
Unsupplemented	1.5	2.0
With protein supplement	1.8	2.2
With energy supplement	1.5	2.0
(above 4 pounds supplement, each pound supplement will decrease forage consumption by about 0.6 pound.)		
Average Quality Roughages:		
Unsupplemented	2.0	2.3
With protein supplement	2.2	2.5
With energy supplement	2.0	2.3
(above 4 pounds supplement, each pound supplement will decrease forage consumption by about 0.6 pound.)		
High Quality Roughages:		
Unsupplemented	2.5	2.7
With protein supplement	2.5	2.7
With energy supplement	2.5	2.7
(pound for pound substitution of supplement for forage)		

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Presented at 1992 National Meeting of ASAS.

protein supplementation of poor quality forages will increase forage intake. Increased forage intake meets the cow additional energy intake. Thus, to maximize profitability, it is essential to optimize protein supplementation. Inadequate dietary protein results in low forage intake and digestibility, resulting in much poorer performance. This reduction in performance may be expressed as weight loss, a decline in body condition, lower milk production, lower antibody transfer to calves through colostrum, or numerous health problems resulting from lowered resistance to disease.

Meeting the protein requirements of beef cattle is complicated by the fact that the microorganisms in the ruminant digestive system can utilize many sources of nitrogen to make protein. Additionally, the digestive characteristics of plant and animal proteins are variable. The following section is an attempt to define some of the terms used in describing types of protein and give guidelines for practical application of these concepts.

Degradable and Undegradable Intake Protein. The protein fraction of the diet can be divided into two components, degradable intake protein (DIP) and undegradable intake protein (UIP). The DIP fraction is available to the rumen microflora and, thus, can be used for their growth and digestion of dietary fiber. Supplementing low-quality forages with DIP has been shown to increase forage digestion and intake, thus increasing energy intake. The UIP is not available to the rumen microflora and, therefore, has essentially no effect on forage utilization. The UIP fraction can be a direct supply of amino acids to the host animal or it can go undigested and be expelled from the gastrointestinal tract. Oil seed byproducts (soybean meal, cottonseed meal, sunflower meal) contain a high percentage of DIP while proteins derived from ani-

mal sources contain mostly UIP. From these general statements, producers should realize that for most forage-based diets, supplementation programs should focus on the inclusion rates of DIP in the diet. K-State research has indicated that the amount of DIP required to maximize forage use appears to be about 10 to 11 percent of the digestible organic matter (which is roughly the same as TDN content). Therefore, to correctly supply the proper amount of supplement, producers need to know the source of the supplemental protein as well as an estimate of forage intake, digestibility and nutrient composition.

Non-Protein Nitrogen (NPN). From a practical viewpoint, cattle have the ability to use either “natural” protein such as that contained in feedstuffs or various other nitrogen sources. Sources other than natural protein are generally referred to as nonprotein nitrogen (NPN) sources. Common NPN sources used in protein supplements include urea, biuret and ammonia hydroxide.

Usefulness of NPN in Beef Cow Rations—The use of NPN in cow rations has been discouraged in the past because for the rumen microflora to convert NPN into protein they must also be supplied with adequate energy and carbon skeletons. Forage-based diets usually do not supply excessive amounts of either, thus, limiting the quantity of NPN that can be fully utilized. Most NPN sources are considered to be 100 percent DIP in nature. Recent research results have indicated limited amounts of the DIP in supplements can be replaced by NPN.

At present, an inclusion level of 15 percent of the total crude protein (or 20 percent of the DIP) as NPN can be used without significantly jeopardizing livestock performance. Because of the severe energy limitations of low-quality forage diets, NPN is less

potent as an energy source than say with high-quality-forage diets or high-concentrate diets. Previous research where supplement containing significant amounts of NPN indicates that, at best, 50 percent of the protein coming from NPN can be utilized by cows consuming low- to medium-quality roughages. And, as NPN inclusion rates increase, the NPN utilization can steadily fall from 50 percent to as low as 20 to 25 percent. Therefore, cattlemen need to carefully analyze their protein supplementation programs and only include those amounts of NPN that will be optimum to their operations.

Interpreting the NPN content of feedstuffs—Feed tags must show the NPN content of commercial-protein supplements. Commonly, this is done by showing the total protein concentration followed by the amount of protein coming from NPN. For example, a tag on a range cube might show the figures “20–10” for protein. This means that the total protein content of the cube is 20 percent with the equivalent of 10 percentage units or 50 percent of the protein coming from NPN.

Calculating the Protein Value of a Supplement with NPN—For use with beef cows grazing dormant range or crop residues in late winter, the 20–10 supplement should be considered, at best, a 15 percent supplement (10 percent from natural sources plus 50 percent of the 10 percent from NPN). Again, this is allowing for 50 percent utilization of the NPN which is a “best case” scenario.

If the actual rate of utilization is only 20 percent, as could be the case with crop residues in late winter, this supplement may only give performance equal to a 12 percent all natural (10 percent natural + 20 percent of the 10 percent from NPN = 12 percent). Note that the 12 or 15 percent figure for protein content should be used both for meeting protein requirements and

Table 14. Mineral Requirements and Maximum Tolerable Levels in Beef Cattle^a

Mineral	Unit	Requirement ^b :		
		Gestating	Early Lactating	Maximum Tolerable Concentration
Calcium	%	See Table 19		
Chlorine	%	—	—	1,000.00
Cobalt	ppm	0.10	0.10	10.00
Copper	ppm	10.00	10.00	100.00
Iodine	ppm	0.50	0.50	50.000
Iron	ppm	50.00	50.00	1,000.00
Magnesium	%	0.10	0.20	0.40
Manganese	ppm	20.00	40.00	1,000.00
Molybdenum	ppm	—	—	5.00
Phosphorus	%	See Table 19		
Potassium	%	0.60	0.70	3.00
Selenium	ppm	0.10	0.10	2.00
Sodium	%	0.06–0.08	0.10	—
Sulfur	%	0.15	0.15	0.40
Zinc	ppm	30.00	30.00	500.00

^a Taken from National Research Council publication Nutrient Requirements of Beef Cattle, Seventh Revised Edition, 1996.

^b Requirements for most minerals are affected by a variety of dietary and animal factors. Thus, it may be better to evaluate rations based on a range of mineral requirements and for content of interfering substances than to meet a specific dietary value.

for pricing the supplement compared to a 20 percent “all natural supplement” or other protein source.

Biuret—While urea is the most commonly used form of NPN, many supplements contain biuret which is essentially two urea molecules chemically bound together. Biuret is somewhat safer, but a review of research comparing urea and biuret indicates that the performance of cattle has been similar.

Ammonia Hydroxide—During the past 10 years, ammoniation of roughages has gained wide acceptance by beef producers in Kansas, especially as a way of making a quality feedstuff out of wheat straw and other crop residues in drought years. Feed analyses routinely show an increase in the crude protein content of wheat straw by 4 to 6 percentage points by ammoniation. Thus, a straw at 4 percent CP prior to ammoniation often tests 9 percent CP after ammoniation. However, this protein is in the form of ammonia hydrox-

ide—an NPN source. Research has shown that beef cows make some use of this NPN source, but respond very well to additional supplementation with “natural” protein. This clearly points out that the primary reason for ammoniating forages is to increase fiber and energy intake and digestibility, and not to meet all of the protein needs of the cows.

By-Pass Protein. In recent years, considerable research has focused on the concept that not all types of “natural” proteins are digested by rumen microorganisms—some of it “bypasses” or escapes the rumen to be digested directly in the intestine. Examples of high-escape protein sources include dehydrated alfalfa, blood meal, corn gluten meal, distillers grains and feather meal. There is merit in having a percentage of protein bypass ruminal digestion, and protein sources differ greatly in the proportion of protein that bypasses.

A review of the literature on bypass protein utilization by beef cows indicates that high-bypass sources rarely show any benefit compared to other “all natural” sources such as soybean meal (SBM) or cottonseed meal (CSM). However, supplements containing high bypass protein sources are better utilized in combination with NPN than are lower bypass sources such as SBM. In general, producers can price high bypass compared to low-bypass protein sources on a pound-for-pound basis for use with beef cows. It should be pointed out, however, that high-bypass sources plus NPN have shown some benefit in growing programs and may have some economic advantage.

3. Mineral Supplementation

In general, mineral requirements of beef cows can be grouped into the major minerals (calcium, phosphorus and salt) and the trace minerals which are required at much lower levels. Table 14 shows a list of the minerals for which a suggested nutrient allowance has been established.

Calcium (Ca), Phosphorus (P), and Salt. These minerals are required at significant levels by beef cows, and, thus, are major considerations in diet formulation. Since calcium is usually found in fairly high levels in forages, it is rarely deficient in diets typically fed to beef cows. Conversely, forages are relatively poor sources of phosphorus which means that it should be the first mineral considered in a supplementation program. Unfortunately, phosphorus is relatively expensive, and producers are advised to meet animal requirements but avoid over supplementation.

Cattle have a definite requirement and appetite for sodium (Na). Since most feedstuffs are deficient in this nutrient, cattle should be offered supplements containing salt (sodium chloride) at all times.

Trace Minerals. Table 14 indicates both the minimum required levels and maximum tolerable levels for trace minerals. Requirements for these minerals may be met by using trace mineralized (TM) Salt in a free-choice supplement; however, there are a few areas in Kansas where deficiencies of specific trace minerals are common. In these locations, producers must increase the concentration of specific trace minerals in the supplement to avoid deficiency symptoms. A mineral consumed at .22 pounds/day would have to contain 0.0005 percent Cobalt, 0.05 percent Copper, 0.0025 percent Iodine, 0.25 percent Iron, .25 percent Manganese, 0.0005 percent Selenium, and .2 percent Zinc to meet 50 percent of a beef cow's total daily requirement.

General Considerations in Mineral Supplementation

Do Beef Cows Have the Ability to Balance Their Mineral Requirements? Many producers believe that beef cows have the ability to consume specific minerals when a deficiency exists. However, research has shown that cows do not have the “nutritional wisdom” to balance their diets, which means that producers must do the “balancing” for their cows. However, as previously mentioned, one of the minerals that cattle do crave is salt. Thus, other minerals should be mixed with salt or other feed ingredients to promote intake. The desire for salt means that a free-choice mineral, high in TM Salt and phosphorus, will usually provide adequate mineral supplementation. Since cows don't crave minerals other than salt, it is recommended that producers avoid using a salt block separate from a free-choice mineral supplement.

Producers often comment that their cows go through periods of extremely high mineral intake and periods of little or no intake. Again, these fluctuations are prob-

ably driven by an appetite for salt. Consequently, producers should increase the salt content of their free-choice mineral when the cows consume over .2 pounds daily.

Free-Choice Mineral Supplements. Average mineral consumption per cow should range from .1 to .2 pounds daily. Intake can be adjusted by altering the salt level (higher levels reduce intake) or adding flavoring agents, such as SBM or molasses, to increase intake. Phosphorus intake and the cost of mineral supplementation can best be altered by changing the salt level in the mineral.

“Home Mineral Mixes.” In periods of high requirements, such as early lactation and the breeding season, a mixture of 50 percent Dicalcium phosphate + 50 percent TM Salt is recommended. During these periods, 3 to 5 percent SBM or molasses may be added to increase intake and prevent caking. During periods of reduced mineral requirements, a mixture of 30 percent Dicalcium phosphate + 70 percent TM Salt should be adequate.

Special Mineral Considerations.

Magnesium—Many producers must supplement with magnesium (Mg) in the early spring to prevent

Table 15. Estimated Daily Water Intake of Cattle

	Mean Temp.	Cows Nursing Calves ¹	Bred Dry Cows & Heifers	Bulls
Month	°F	Gal	Gal	Gal
January	36	11.0	6.0	7.0
March	50	12.5	6.5	8.6
May	73	17.0	9.0	12.0
July	90	16.5	14.5	19.0
Sept.	78	17.5	10.0	13.0
Nov.	52	13.0	6.5	9.0

Source: Paul Q. Guyer, University of Nebraska.

¹ Cows nursing calves during first three to four months after parturition—peak milk production period.

Table 16. Rations for the Middle Trimester of Pregnancy (October through November)

Ration	Feedstuff	Amount Per Head Per Day Lbs.	Feedstuff	Lbs.	Mineral Req.
1	Wheat Straw	17	+	Alfalfa Hay	6.5
2	Corn Stover ^a	24	—		P + TM Salt
3	Milo Stover ^a	24	—		P + TM Salt
4	Brome Hay	18 ^b	—		TM Salt
5	Prairie Hay	22 ^c	—		P + TM Salt
6	Prairie Hay	22	+	Commercial Cubes	.5 ^d

^a Both the corn and milo stovers were assumed to be 6.6 percent crude protein during this period.

^b This level of intake is only 72 percent of expected free choice consumption, which means that all cows must have access during feeding to avoid over consumption by “boss” cows.

^c Slightly low in CP

^d 20 percent CP. A practical way of delivering these cubes would be to feed 2 pounds/head every four days, or even 3 pounds/head every six days.

Note: If a high urea cube is utilized, they should be fed daily instead of periodically.

Table 17. Rations for the Last Trimester of Pregnancy (December through February)

Ration	Feedstuff	Lbs.	Amount Per Head Per Day (lbs)		Lbs.	Mineral Req.
				Feedstuff		
1 ^a	Corn Stover	18	+	Alfalfa Hay	8	P + TM Salt
2	Corn Stover	20	+	Wheat Midds	7	TM Salt
3	Milo Stover	18	+	Alfalfa Hay	7	P + TM Salt
4	Milo Stover	19	+	Wheat Midds	7	TM Salt
5 ^b	Dry Winter Grass	16	+	Alfalfa Hay	7	P + TM Salt
6 ^b	Dry Winter Grass	20	+	Commercial Cubes ^c	4	P + TM Salt
7	Dry Winter Grass	16	+	Wheat Midds	9	TM Salt
8	Brome Hay	26				TM Salt
9	Prairie Hay	16	+	Alfalfa Hay	9	TM Salt
10	Prairie Hay	20	+	Wheat Midds	5	TM Salt
11	Wheat Straw	15	+	Alfalfa Hay	10	TM Salt
12	Wheat Straw	17	+	Wheat Midds	8	Ca + TM Salt
13	Forage Sorghum Hay	25				TM Salt

^a Under the conditions described, cows consuming this ration would lose approximately 1/3 body condition score. Consequently, if this loss is unacceptable, this ration should be adjusted.

^b Under the conditions described, cows consuming these rations would lose approximately 1/2 body condition score from December 1st to March 1st. Thus, additional supplementation may be necessary.

^c 20% CP.

Table 18. Rations for Lactation (March through April)

Ration	Feedstuff	Lbs.	Amount Per Head Per Day (lbs)		Lbs.	Mineral Req.
				Feedstuff		
1	Brome Hay	28		—		TM Salt
2	Corn Stover	15	+	Alfalfa Hay	14	P + TM Salt
3	Corn Stover	22	+	Commercial Cubes ^a	8	TM Salt
4	Milo Stover	14	+	Alfalfa Hay	14	P + TM Salt
5	Milo Stover	21	+	Commercial Cubes ^a	8	TM Salt
6	Wheat Straw	15	+	Alfalfa Hay	14	P + TM Salt
7	Prairie Hay	16	+	Alfalfa Hay	11	P + TM Salt
8	Forage Sorghum Hay	20	+	Alfalfa Hay	8	TM Salt
9	Winter Native Grass	19	+	Commercial Cubes ^a	8	TM Salt
10	Winter Native Grass	14	+	Alfalfa Hay	13	TM Salt

^a 20% CP.

grass tetany, especially on cool-season forages such as brome, fescue and wheat pasture. Supplementation should be started about three weeks prior to initiation of grazing for best results. The recommended intake of magnesium oxide (MgO) is at least .03 pounds per day which means that it must make up 15 to 20 percent of a free-choice mineral supplement. A good “home mix” for this period would be 15 percent MgO, 50 percent dicalcium phosphate, 25 percent salt, and 10 percent molasses or SBM. Since MgO is unpalatable, a flavoring agent, such as molasses or

SBM, should be added to increase intake. Commercial mineral mixtures utilized during early spring should be at least 10 percent Mg. In some situations, it may be necessary to increase Mg above the normal requirement. A suggested mix for these situations, is 25 percent MgO, 25 percent dicalcium phosphate, 25 percent salt, and 25 percent molasses or SBM.

Chelated Minerals—Chelated minerals are trace minerals that are attached to an organic molecule, i.e. an amino acid or protein. Most research has shown that chelated minerals have greater

bioavailability (higher absorption potential) than some organic forms, especially oxides, used in common mineral supplements.

However, inorganic forms are much less expensive. Currently, the question of whether chelates are beneficial or not is one of the most controversial areas in beef cattle nutrition. This controversy results from the fact that even though absorption is greater, there is little, if any, research demonstrating improved cattle performance. Until further research demonstrates an economic advantage, producers are advised to use

highly available inorganic forms of trace minerals to minimize costs while assuring nutritional adequacy.

4. Troubleshooting Cow Herd Nutrition Programs

It seems to be human nature to want to blame poor reproduction or performance in a herd of cattle on exotic causes rather than the most logical causes. In troubleshooting cow herd nutrition programs, keep in mind that a very high percentage of problems, including poor reproduction or low weaning weights, can be directly attributed to inadequate energy and/or protein intake and not to a trace mineral or unknown growth factor. Only after the adequacy of energy and protein has been established should the focus turn to mineral and vitamin nutrition.

5. Vitamin Requirements

While beef cattle have requirements for several vitamins, most are produced by rumen microorganisms and, thus, are not required in the diet. However, vitamin A must be supplied in some form. Requirements for vitamin A are shown in Table 19. Research has shown that cattle are capable of storing large quantities of vitamin A in the liver during periods of high intake. This commonly occurs during the grazing season on lush, green forage, with these storage supplies being depleted during the winter months. Supplementing vitamin A is very inexpensive and can be accomplished by:

1. Feeding forages high in vitamin A such as high-quality alfalfa hay less than six to eight months old.
2. Including vitamin A in mineral mixes or in other supplements.
3. Injecting vitamin A in periods where feedstuffs are low in vitamin A content and/or requirements are high. Injecting 2 million I.U. should provide sufficient vitamin A for 80 to 100 days.

6. Water Requirements

Normally, water is easily supplied to cattle and little thought is given to this requirement. But, drought conditions are common in Kansas and hauling water an all too frequent necessity. Generally, as temperature increases water consumption increases. If water intake is limited, feed intake also will be depressed, resulting in subpar performance. Since water requirements are influenced by a number of physiological and environmental factors, Table 15 is included to provide a guideline for animal requirements.

Guidelines to Manage Beef Cow Diets

1. Feeding cattle is dynamic. Producers must consider animal, environment and diet factors to correctly feed beef cows.
2. A cow's nutrient requirements (energy, protein, minerals) will increase about 30 to 40 percent with calving. Forage intake will generally increase about 30 percent with calving.
3. Positive response to providing a supplement with high-protein concentrations is most likely when forage crude protein is less than 7 percent. The first-limiting nutrient in low-quality forages is protein. Therefore, the best approach for increasing total protein and energy supply is to supplement with DIP.
4. The NPN inclusion rate in supplements for forage-based diets must be monitored closely, as only up to 15 percent of the total dietary crude protein should be NPN in nature. Including NPN at too high of levels may result in refusal to consume supplements.
5. Starch can negatively impact forage intake and fiber digestion. Supplementing winter cow diets with corn, grain sorghum and other cereal grains can actually decrease energy intake. The key is to meet protein requirements of the rumen so that forage utilization is maximized.
6. When fed at levels exceeding 0.5 percent of body weight (e.g. 5.5 pounds of daily supplement for 1,100-pound cows) intake of low-quality forage will be reduced by 0.5 pounds for each 1 pound of alfalfa hay or low-protein concentrate fed above the 0.5 percent threshold.
7. Only minor differences in performance are evident for cattle supplemented every-other-day or three-times-weekly compared with daily.

Table 19. Nutrient Requirements of Breeding Cattle

Wt ^a (lbs)	Gain ^b (lbs)	Daily DM ^c (lbs)	Daily TDN (lbs)	NEm (Mcal)	TDN (%)	Energy in Diet DM NEm (Mcal/lbs)	Total Protein NEg (Mcal/lbs)	Daily in Diet (lbs)	Calcium in Diet DM (%)	Phos. in Diet DM (%)	Vit. A ^d Daily (1000's IU)	
Pregnant yearling heifers—Last third of pregnancy												
700	0.9	15.3	8.5	7.95	55.4	0.52	NA ^e	1.3	8.4	0.27	0.20	19
700	1.4	15.8	9.6	7.95	60.3	0.60	0.34	1.4	9.0	0.33	0.21	20
700	1.9	15.8	10.6	7.95	67.0	0.70	0.43	1.5	9.8	0.33	0.21	20
800	0.9	16.8	9.2	8.56	54.8	0.51	NA	1.4	8.2	0.28	0.20	21
800	1.4	17.4	10.4	8.56	59.6	0.59	0.33	1.5	8.8	0.33	0.21	22
800	1.9	17.5	11.6	8.56	66.1	0.69	0.42	1.6	9.3	0.35	0.21	22
900	0.9	18.3	9.9	9.15	54.3	0.51	NA	1.5	8.1	0.26	0.20	23
900	1.4	19.0	11.3	9.15	59.1	0.58	0.32	1.6	8.5	0.30	0.21	24
900	1.9	19.2	12.5	9.15	65.4	0.68	0.41	1.7	9.0	0.32	0.21	24
Dry pregnant mature cows—Middle third of pregnancy												
900	0.0	16.7	8.2	7.00	48.8	0.42	NA	1.2	7.0	0.18	0.18	21
1100	0.0	19.5	9.5	8.13	48.8	0.42	NA	1.4	7.0	0.19	0.19	25
1300	0.0	22.0	10.8	9.22	48.8	0.42	NA	1.5	6.9	0.20	0.20	28
Dry pregnant mature cows—Last third of pregnancy												
900	0.9	18.2	9.8	9.15	54.0	0.50	NA	1.5	8.0	0.27	0.21	23
1000	0.9	19.6	10.5	9.72	53.6	0.50	NA	1.6	7.9	0.26	0.20	25
1200	0.9	22.3	11.8	10.83	52.9	0.49	NA	1.7	7.8	0.26	0.21	28
1400	0.9	24.9	13.1	11.90	52.5	0.48	NA	1.9	7.6	0.26	0.21	32
Two-year-old heifers nursing calves—First 3–4 months postpartum—10 lbs milk/day												
700	0.5	15.9	10.3	9.20 ^f	65.1	0.67	0.40	1.8 ^g	11.3	0.36	0.24	28
800	0.5	17.6	11.2	9.81 ^f	63.8	0.66	0.39	1.9 ^g	10.8	0.34	0.24	31
900	0.5	19.2	12.0	10.40 ^f	62.7	0.64	0.37	2.0 ^g	10.4	0.32	0.23	34
1000	0.5	20.8	12.9	10.98 ^f	61.9	0.62	0.36	2.1 ^g	10.0	0.31	0.23	37
Cows nursing calves—Average milking ability—First 3–4 months postpartum—10 lbs milk/day												
900	0.0	18.8	10.8	10.40 ^f	57.3	0.55	NA	1.9 ^g	9.9	0.28	0.22	33
1100	0.0	21.6	12.1	11.54 ^f	56.0	0.54	NA	2.0 ^g	9.4	0.27	0.22	38
1300	0.0	24.3	13.4	12.63 ^f	55.1	0.52	NA	2.2 ^g	9.1	0.27	0.22	43
Cows nursing calves—superior milking ability—First 3–4 months postpartum—20 lb milk/day												
900	0.0	18.7	13.1	13.81 ^f	69.8	0.74	NA	2.4 ^g	12.9	0.41	0.28	33
1100	0.0	22.3	14.5	14.94 ^f	65.2	0.67	NA	2.6 ^g	11.9	0.38	0.27	40
1300	0.0	25.3	15.9	16.03 ^f	62.6	0.64	NA	2.8 ^g	11.2	0.36	0.26	45

^a Average weight for a feeding period.

^b Approximately 0.9 + 0.2 pound of weight gain/day over the last third of pregnancy is accounted for by the products of conception. Daily 2.15 Mcal of NEm and 0.1 pound of protein are provided for this requirement for a calf with a birth weight of 80 pounds.

^c Dry matter consumption should vary depending on the energy concentration of the diet and environmental conditions. These intakes are based on the energy concentration shown in the table and assuming a thermoneutral environment without snow or mud conditions. If the energy concentrations of the diet to be fed exceeds the tabular value, limit feeding may be required.

^d Vitamin A requirements per pound of diet are 1,273 IU for pregnant heifers and cows and 1,773 for lactating cows and breeding bulls.

^e Not applicable.

^f Includes 0.34 Mcal NEm/pound of milk produced.

^g Includes 0.03 pound protein/pound of milk produced.

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In each case, credit Twig T. Marston, et al, *Beef Cow Nutrition Guide*, Kansas State University, April 1998.

Kansas State University Agricultural Experiment Station and Cooperative Extension Service

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**Animal Nutrient
Requirements**

Forage Value

Evaluating Feeds

Minerals and Vitamins

Ration Balancing

Supplements

Body Condition

**Heifer Nutrition and
Development**

Beef Cattle Nutrition Workbook



Contents

The *Beef Cattle Nutrition Workbook* is an interactive workbook to be used in conjunction with Oregon State University Extension Service Winter Nutrition Programs for beef producers.

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Published December 2004



Preface

This *Beef Cattle Nutrition Workbook* is designed to be an interactive tool to help beef producers manage their herds. It contains information about beef cattle nutrient requirements, forage nutritive value, the importance of minerals and vitamins to cattle health and performance, balancing rations, economical supplementation strategies, heifer development, and the use of cow body condition score as a nutritional and reproductive management tool. Each chapter includes a worksheet to help you develop a ranch-specific plan for nutritional management of your beef herd. We hope you find this workbook helpful in the planning and execution of your nutritional programs.

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Animal Nutrient Requirements

1

Shelby Filley and
Robert Pawelek

Different classes of animals have different requirements for total pounds of feed (intake), energy, protein, vitamins, and minerals. As demonstrated in Figures 1.1 and 1.2, the specific amounts needed depend on the animal's stage of production and the level of performance desired by the livestock manager. Requirements also may vary depending on previous diet. For example, when an animal has been on a limited diet and then is fed a more nutritious diet, the diet is utilized more efficiently in a compensatory manner. Frame score (structural size) also influences requirements (see references, page 2).

Water is another crucial part of a livestock nutrition program. Livestock should always have access to adequate quantities of good-quality, clean water. Lack of water leads to decreased feed intake, lower production, and reduced revenues. Water requirements for a cow-calf pair range from 12 to 20 gallons per day. Yearling cattle require 6 to 14 gallons per day.

Certain metals and contaminants are detrimental to livestock if present in water at significant levels. Test water to make sure the water is free from harmful materials (see references, page 2).

Initial steps to evaluating livestock rations include: (1) describing the animal, and (2) determining the animal's nutrient requirements. Nutrient

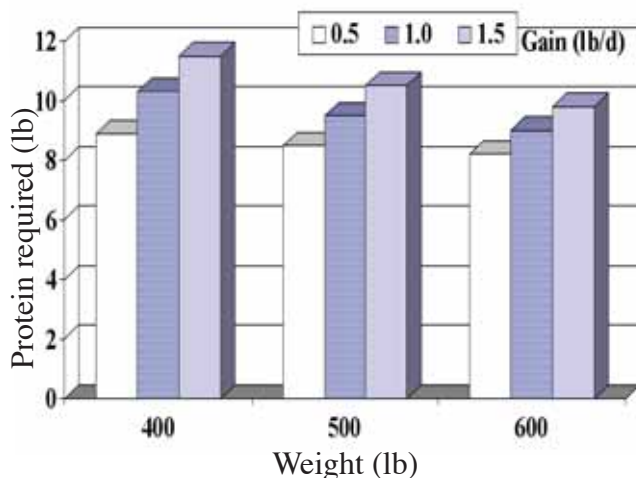


Figure 1.1—Protein requirements of medium-framed steers at various weights and rates of gain.

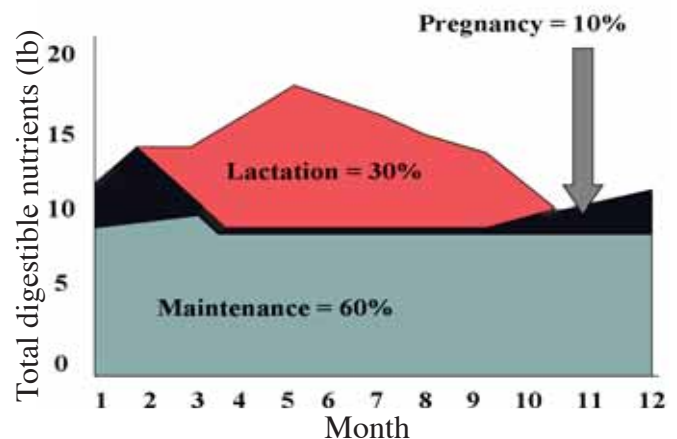


Figure 1.2—Example of annual energy inputs for a spring-calving cow. Source: Zollinger, B. and D. Hansen. 2003. *OSU Calving School Handbook*, Animal Sciences publication 110. <http://oregonstate.edu/dept/animal-sciences/cschhand.pdf>



requirements for growing and finishing cattle (Table 1.1) and for breeding cattle (Table 1.2) are included on pages 5–17.

A word of caution when using these tables: environmental factors such as mud, snow, cold, etc. increase nutrient requirements, particularly for energy. For instance, the critical temperature for most wintering cows in eastern Oregon is around 30°F. When wind chill drops below 30°F, the energy requirement increases. For each 1°F drop in wind chill, the energy requirement increases by 1 percent. If the wind chill is 0°F, the energy requirement required to maintain the cow increases by 30 percent. Also, cows in poor body condition take more energy for maintenance than cows in good body condition (see Chapter 7).

To begin an evaluation of your livestock-feeding program, fill in the following worksheets using the appropriate tables.

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Worksheet 1.1

Nutrient Requirements for Growing and Finishing Cattle

1. Animal description

Frame and sex _____

Growth characteristics (compensating or not) _____

Animal weight _____

Daily gain desired _____

Average daily gain = (final weight – current weight) ÷ days to gain

Climatic and lot conditions (temperature, precipitation, mud, snow, etc.) _____

2. Nutrient requirements (per day)—see Table 1.1

Expected dry matter intake (DMI): pounds _____

Protein: % in diet _____ pounds _____

Energy, total digestible nutrients (TDN): % in diet _____ pounds _____

Calcium (% Ca) _____

Phosphorus (% P) _____

Worksheet 1.2

Nutrient Requirements for Breeding Cattle

1. Animal description

Sex and age _____

Stage of pregnancy/lactation _____

Milking ability (average = 10 lb/day; superior = 20 lb/day) _____

Animal weight _____

Daily gain desired _____

Average daily gain = (final weight – current weight) ÷ days to gain

2. Nutrient requirements (per day)—see Table 1.2

Expected dry matter intake (DMI): pounds _____

Protein: % in diet _____ pounds _____

Energy, total digestible nutrients (TDN): % in diet _____ pounds _____

Calcium (% Ca) _____

Phosphorus (% P) _____

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Medium-frame steer calves								
300	0.5	7.8	9.6	0.75	54.0	4.2	0.31	0.20
	1.0	8.4	11.4	0.95	58.5	4.9	0.45	0.24
	1.5	8.7	13.2	1.14	63.0	5.5	0.58	0.28
	2.0	8.9	14.8	1.32	67.5	6.0	0.72	0.32
	2.5	8.9	16.7	1.48	73.5	6.5	0.87	0.37
	3.0	8.0	19.9	1.60	85.0	6.8	1.13	0.47
400	0.5	9.7	8.9	0.87	54.0	5.2	0.27	0.18
	1.0	10.4	10.3	1.06	58.5	6.1	0.38	0.21
	1.5	10.8	11.5	1.24	63.0	6.8	0.47	0.25
	2.0	11.0	12.7	1.41	67.5	7.4	0.56	0.26
	2.5	11.0	14.2	1.56	73.5	8.1	0.68	0.30
	3.0	10.0	16.6	1.65	85.0	8.5	0.86	0.37
500	0.5	11.5	8.5	0.98	54.0	6.2	0.25	0.17
	1.0	12.3	9.5	1.16	58.5	7.2	0.32	0.20
	1.5	12.8	10.5	1.33	63.0	8.1	0.40	0.22
	2.0	13.1	11.4	1.49	67.5	8.8	0.47	0.24
	2.5	13.0	12.5	1.63	73.5	9.6	0.56	0.27
	3.0	11.8	14.4	1.69	85.0	10.0	0.69	0.32
600	0.5	13.2	8.2	1.08	54.0	7.1	0.23	0.18
	1.0	14.1	9.0	1.26	58.5	8.3	0.28	0.19
	1.5	14.7	9.8	1.42	63.0	9.3	0.35	0.21
	2.0	15.0	10.5	1.57	67.5	10.1	0.40	0.22
	2.5	14.9	11.4	1.69	73.5	11.0	0.46	0.24
	3.0	13.5	12.9	1.73	85.0	11.5	0.57	0.29
700	0.5	14.8	7.9	1.18	54.0	8.0	0.22	0.18
	1.0	15.8	8.6	1.35	58.5	9.2	0.27	0.18
	1.5	16.5	9.2	1.50	63.0	10.4	0.31	0.20
	2.0	16.8	9.8	1.65	67.5	11.3	0.34	0.21
	2.5	16.7	10.5	1.75	73.5	12.3	0.40	0.22
	3.0	15.2	11.7	1.77	85.0	12.9	0.49	0.26

^aDM = dry matter; TDN = total digestible nutrients (energy); Ca = calcium; P = phosphorus

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Medium-frame steer calves (continued)								
800	0.5	16.4	7.7	1.27	54.0	8.9	0.22	0.17
	1.0	17.5	8.3	1.44	58.5	10.2	0.24	0.19
	1.5	18.2	8.8	1.58	63.0	11.5	0.28	0.19
	2.0	18.6	9.2	1.72	67.5	12.6	0.31	0.20
	2.5	18.5	9.8	1.81	73.5	13.6	0.35	0.21
	3.0	16.8	10.8	1.81	85.0	14.3	0.42	0.25
900	0.5	17.9	7.6	1.36	54.0	9.7	0.21	0.18
	1.0	19.1	8.0	1.52	58.5	11.2	0.23	0.18
	1.5	19.9	8.4	1.66	63.0	12.5	0.25	0.19
	2.0	20.3	8.8	1.79	67.5	13.7	0.28	0.20
	2.5	20.2	9.3	1.87	73.5	14.8	0.31	0.20
	3.0	18.3	10.1	1.85	85.0	15.6	0.37	0.23
1,000	0.5	19.3	7.5	1.45	54.0	10.4	0.21	0.18
	1.0	20.7	7.8	1.60	58.5	12.1	0.21	0.18
	1.5	21.5	8.1	1.74	63.0	13.5	0.24	0.18
	2.0	22.0	8.4	1.85	67.5	14.9	0.25	0.19
	2.5	21.9	8.8	1.92	73.5	16.1	0.27	0.19
	3.0	19.8	9.5	1.88	85.0	16.8	0.32	0.22
Large-frame steer calves and compensating medium-frame yearling steers								
300	0.5	8.2	9.5	0.77	52.5	4.3	0.30	0.19
	1.0	8.7	11.3	0.99	56.0	4.9	0.46	0.23
	1.5	9.1	12.9	1.19	59.5	5.4	0.58	0.27
	2.0	9.4	14.6	1.37	63.5	6.0	0.70	0.30
	2.5	9.6	16.3	1.55	67.5	6.5	0.85	0.34
	3.0	9.6	18.0	1.73	72.0	6.9	0.99	0.39
	3.5	9.3	20.3	1.88	78.5	7.3	1.16	0.45
400	0.5	10.1	8.9	0.89	52.5	5.3	0.26	0.17
	1.0	10.8	10.2	1.10	56.0	6.0	0.37	0.20
	1.5	11.3	11.4	1.30	59.5	6.7	0.47	0.23
	2.0	11.7	12.7	1.47	63.5	7.4	0.57	0.26
	2.5	11.9	13.9	1.64	67.5	8.0	0.65	0.30
	3.0	11.9	15.2	1.81	72.0	8.6	0.76	0.33
	3.5	11.5	16.9	1.94	78.5	9.0	0.90	0.36

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame steer calves and compensating medium-frame yearling steers (continued)								
500	0.5	12.0	8.5	1.00	52.5	6.3	0.24	0.17
	1.0	12.8	9.5	1.21	56.0	7.2	0.33	0.19
	1.5	13.4	10.4	1.40	59.5	8.0	0.39	0.21
	2.0	13.8	11.4	1.57	63.5	8.8	0.46	0.24
	2.5	14.0	12.4	1.73	67.5	9.5	0.55	0.25
	3.0	14.0	13.4	1.88	72.0	10.1	0.63	0.28
	3.5	13.6	14.7	2.00	78.5	10.7	0.73	0.32
600	0.5	13.8	8.2	1.11	52.5	7.2	0.22	0.18
	1.0	14.6	9.0	1.31	56.0	8.2	0.29	0.18
	1.5	15.3	9.7	1.50	59.5	9.1	0.35	0.20
	2.0	15.8	10.5	1.66	63.5	10.0	0.40	0.22
	2.5	16.1	11.3	1.81	67.5	10.9	0.47	0.23
	3.0	16.1	12.1	1.95	72.0	11.6	0.52	0.26
	3.5	15.6	13.2	2.05	78.5	12.2	0.61	0.28
700	0.5	15.4	7.9	1.21	52.5	8.1	0.21	0.17
	1.0	16.4	8.6	1.41	56.0	9.2	0.27	0.19
	1.5	17.2	9.2	1.59	59.5	10.2	0.31	0.19
	2.0	17.8	9.8	1.74	63.5	11.3	0.36	0.21
	2.5	18.0	10.5	1.88	67.5	12.2	0.40	0.22
	3.0	18.0	11.1	2.01	72.0	13.0	0.45	0.23
	3.5	17.5	12.0	2.10	78.5	13.7	0.52	0.26
800	0.5	17.1	7.7	1.31	52.5	8.9	0.21	0.18
	1.0	18.2	8.3	1.51	56.0	10.2	0.24	0.18
	1.5	19.0	8.8	1.68	59.5	11.3	0.28	0.19
	2.0	19.6	9.3	1.82	63.5	12.4	0.32	0.20
	2.5	19.9	9.8	1.96	67.5	13.4	0.35	0.21
	3.0	19.9	10.4	2.07	72.0	14.3	0.40	0.22
	3.5	19.3	11.1	2.15	78.5	15.2	0.45	0.24
900	0.5	18.6	7.6	1.40	52.5	9.8	0.20	0.18
	1.0	19.8	8.0	1.60	56.0	11.2	0.23	0.18
	1.5	20.8	8.5	1.77	59.5	12.4	0.27	0.18
	2.0	21.4	8.9	1.90	63.5	13.6	0.29	0.20
	2.5	21.8	9.3	2.03	67.5	14.7	0.31	0.20
	3.0	21.7	9.8	2.13	72.0	15.6	0.36	0.21
	3.5	21.1	10.4	2.19	78.5	16.6	0.40	0.23

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame steer calves and compensating medium-frame yearling steers (continued)								
1,000	0.5	20.2	7.5	1.49	52.5	10.6	0.20	0.17
	1.0	21.5	7.8	1.69	56.0	12.0	0.23	0.17
	1.5	22.5	8.2	1.85	59.5	13.4	0.25	0.18
	2.0	23.2	8.6	1.98	63.5	14.7	0.27	0.18
	2.5	23.6	8.9	2.09	67.5	15.9	0.29	0.19
	3.0	23.6	9.3	2.19	72.0	17.0	0.32	0.20
	3.5	22.8	9.8	2.24	78.5	17.9	0.35	0.21
1,100	0.5	21.7	7.4	1.58	52.5	11.4	0.19	0.18
	1.0	23.1	7.7	1.77	56.0	12.9	0.21	0.18
	1.5	24.1	8.0	1.93	59.5	14.3	0.23	0.18
	2.0	24.9	8.3	2.05	63.5	15.8	0.25	0.18
	2.5	25.3	8.5	2.16	67.5	17.1	0.26	0.18
	3.0	25.3	8.9	2.25	72.0	18.2	0.29	0.19
	3.5	24.5	9.3	2.28	78.5	19.2	0.32	0.21
Medium-frame bulls								
300	0.5	7.8	9.7	0.76	53.5	4.2	0.31	0.20
	1.0	8.3	11.6	0.96	57.5	4.8	0.48	0.24
	1.5	8.6	13.4	1.15	61.5	5.3	0.62	0.28
	2.0	8.8	15.2	1.34	65.5	5.8	0.75	0.33
	2.5	8.9	17.0	1.52	70.0	6.2	0.92	0.37
	3.0	8.7	19.3	1.68	76.5	6.7	1.09	0.43
400	0.5	9.6	9.0	0.87	53.5	5.1	0.28	0.18
	1.0	10.3	10.4	1.07	57.5	5.9	0.39	0.21
	1.5	10.7	11.8	1.26	61.5	6.6	0.49	0.25
	2.0	11.0	13.1	1.44	65.5	7.2	0.60	0.28
	2.5	11.1	14.4	1.60	70.0	7.8	0.70	0.32
	3.0	10.8	16.1	1.74	76.5	8.3	0.84	0.37
500	0.5	11.4	8.6	0.98	53.5	6.1	0.25	0.17
	1.0	12.1	9.7	1.17	57.5	7.0	0.35	0.20
	1.5	12.7	10.7	1.35	61.5	7.8	0.42	0.23
	2.0	13.0	11.7	1.52	65.5	8.5	0.49	0.25
	2.5	13.1	12.8	1.68	70.0	9.2	0.59	0.27
	3.0	12.8	14.1	1.81	76.5	9.8	0.69	0.31

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Medium-frame bulls (continued)								
600	0.5	13.1	8.3	1.08	53.5	7.0	0.24	0.19
	1.0	13.9	9.2	1.27	57.5	8.0	0.30	0.19
	1.5	14.5	10.0	1.44	61.5	8.9	0.36	0.21
	2.0	14.9	10.8	1.61	65.5	9.8	0.43	0.24
	2.5	15.0	11.6	1.75	70.0	10.5	0.50	0.25
	3.0	14.7	12.7	1.86	76.5	11.2	0.57	0.29
700	0.5	14.7	8.0	1.18	53.5	7.9	0.23	0.18
	1.0	15.6	8.8	1.37	57.5	9.0	0.28	0.20
	1.5	16.3	9.4	1.53	61.5	10.0	0.32	0.20
	2.0	16.7	10.1	1.69	65.5	10.9	0.38	0.22
	2.5	16.8	10.8	1.82	70.0	11.8	0.43	0.24
	3.0	16.5	11.7	1.92	76.5	12.6	0.49	0.25
800	0.5	16.2	7.8	1.27	53.5	8.7	0.22	0.19
	1.0	17.3	8.4	1.45	57.5	9.9	0.25	0.19
	1.5	18.0	9.0	1.61	61.5	11.1	0.29	0.20
	2.0	18.5	9.5	1.76	65.5	12.1	0.33	0.21
	2.5	18.6	10.1	1.89	70.0	13.0	0.38	0.23
	3.0	18.2	10.8	1.97	76.5	13.9	0.44	0.24
900	0.5	17.7	7.7	1.36	53.5	9.5	0.21	0.19
	1.0	18.9	8.2	1.54	57.5	10.9	0.25	0.19
	1.5	19.7	8.6	1.69	61.5	12.1	0.28	0.19
	2.0	20.2	9.1	1.83	65.5	13.2	0.31	0.21
	2.5	20.3	9.6	1.95	70.0	14.2	0.34	0.22
	3.0	19.9	10.2	2.02	76.5	15.2	0.39	0.23
1,000	0.5	19.2	7.5	1.45	53.5	10.3	0.21	0.18
	1.0	20.4	8.0	1.62	57.5	11.7	0.24	0.18
	1.5	21.3	8.4	1.77	61.5	13.1	0.26	0.19
	2.0	21.8	8.7	1.90	65.5	14.3	0.28	0.19
	2.5	22.0	9.1	2.01	70.0	15.4	0.31	0.20
	3.0	21.5	9.6	2.07	76.5	16.4	0.35	0.22
1,100	0.5	20.6	7.4	1.54	53.5	11.0	0.20	0.19
	1.0	21.9	7.8	1.70	57.5	12.6	0.22	0.19
	1.5	22.9	8.1	1.85	61.5	14.1	0.24	0.19
	2.0	23.4	8.4	1.97	65.5	15.3	0.26	0.19
	2.5	23.6	8.7	2.07	70.0	16.5	0.28	0.20
	3.0	23.1	9.2	2.11	76.5	17.7	0.32	0.21

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame bull calves and compensating large-frame yearling steers								
300	0.5	7.9	9.7	0.77	52.5	4.1	0.31	0.20
	1.0	8.4	11.7	0.98	56.0	4.7	0.47	0.24
	1.5	8.8	13.5	1.18	59.5	5.2	0.63	0.28
	2.5	9.2	17.0	1.56	66.5	6.1	0.91	0.36
	3.0	9.2	18.8	1.74	70.5	6.5	1.08	0.43
	3.5	9.1	20.9	1.91	75.5	6.9	1.24	0.48
	4.0	8.2	24.7	2.04	86.0	7.1	1.53	0.59
400	0.5	9.8	9.0	0.89	52.5	5.1	0.27	0.18
	1.0	10.4	10.5	1.09	56.0	5.8	0.40	0.21
	1.5	10.9	11.9	1.29	59.5	6.5	0.51	0.24
	2.0	11.2	13.1	1.48	62.5	7.0	0.61	0.28
	2.5	11.4	14.5	1.65	66.5	7.6	0.72	0.31
	3.0	11.5	15.9	1.82	70.5	8.1	0.82	0.35
	3.5	11.3	17.5	1.98	75.5	8.5	0.96	0.39
	4.0	10.2	20.3	2.08	86.0	8.8	1.19	0.48
500	0.5	11.6	8.6	1.00	52.5	6.1	0.25	0.19
	1.0	12.3	9.8	1.20	56.0	6.9	0.36	0.21
	1.5	12.9	10.9	1.39	59.5	7.7	0.43	0.22
	2.0	13.2	11.8	1.58	62.5	8.3	0.52	0.25
	2.5	13.5	12.9	1.74	66.5	9.0	0.59	0.28
	3.0	13.6	14.0	1.90	70.5	9.6	0.68	0.31
	3.5	13.4	15.3	2.05	75.5	10.1	0.77	0.35
	4.0	12.0	17.5	2.13	86.0	10.3	0.97	0.40
600	0.5	13.3	8.3	1.10	52.5	7.0	0.23	0.18
	1.0	14.1	9.2	1.30	56.0	7.9	0.31	0.20
	1.5	14.8	10.1	1.48	59.5	8.8	0.37	0.21
	2.0	15.2	10.9	1.67	62.5	9.5	0.44	0.23
	2.5	15.5	11.8	1.82	66.5	10.3	0.51	0.26
	3.0	15.5	12.7	1.97	70.5	10.9	0.58	0.27
	3.5	15.3	13.7	2.11	75.5	11.6	0.66	0.30
	4.0	13.8	15.6	2.16	86.0	11.9	0.81	0.37

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame bull calves and compensating large-frame yearling steers (continued)								
700	0.5	14.9	8.0	1.20	52.5	7.8	0.22	0.18
	1.0	15.9	8.8	1.40	56.0	8.9	0.29	0.19
	1.5	16.6	9.6	1.57	59.5	9.9	0.35	0.21
	2.0	17.0	10.2	1.75	62.5	10.6	0.39	0.22
	2.5	17.4	11.0	1.90	66.5	11.6	0.44	0.24
	3.0	17.5	11.7	2.04	70.5	12.3	0.50	0.25
	3.5	17.2	12.5	2.16	75.5	13.0	0.56	0.25
	4.0	15.5	14.1	2.20	86.0	13.3	0.70	0.33
800	0.5	16.5	7.9	1.30	52.5	8.7	0.21	0.19
	1.0	17.5	8.5	1.49	56.0	9.8	0.26	0.19
	1.5	18.3	9.1	1.66	59.5	10.9	0.31	0.20
	2.0	18.8	9.7	1.84	62.5	11.8	0.35	0.21
	2.5	19.2	10.3	1.97	66.5	12.8	0.40	0.23
	3.0	19.3	10.9	2.11	70.5	13.6	0.45	0.24
	3.5	19.0	11.6	2.22	75.5	14.3	0.50	0.26
	4.0	17.1	13.0	2.24	86.0	14.7	0.61	0.31
900	0.5	18.0	7.7	1.39	52.5	9.5	0.22	0.18
	1.0	19.2	8.3	1.58	56.0	10.8	0.25	0.18
	1.5	20.0	8.8	1.74	59.5	11.9	0.29	0.20
	2.0	20.6	9.2	1.92	62.5	12.9	0.32	0.20
	2.5	21.0	9.8	2.04	66.5	14.0	0.36	0.21
	3.0	21.1	10.3	2.17	70.5	14.9	0.40	0.23
	3.5	20.8	10.9	2.27	75.5	15.7	0.45	0.24
	4.0	18.7	12.1	2.27	86.0	16.1	0.53	0.28
1,000	0.5	19.5	7.6	1.48	52.5	10.2	0.21	0.18
	1.0	20.7	8.1	1.66	56.0	11.6	0.25	0.19
	1.5	21.7	8.5	1.83	59.5	12.9	0.27	0.19
	2.0	22.3	8.9	1.99	62.5	13.9	0.30	0.20
	2.5	22.7	9.3	2.11	66.5	15.1	0.33	0.20
	3.0	22.8	9.7	2.23	70.5	16.1	0.36	0.21
	3.5	22.5	10.3	2.32	75.5	17.0	0.40	0.24
	4.0	20.2	11.3	2.30	86.0	17.4	0.48	0.27

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame bull calves and compensating large-frame yearling steers (continued)								
1,100	0.5	20.9	7.5	1.57	52.5	11.0	0.21	0.19
	1.0	22.3	7.9	1.75	56.0	12.5	0.23	0.19
	1.5	23.3	8.3	1.91	59.5	13.9	0.26	0.19
	2.0	23.9	8.6	2.07	62.5	14.9	0.28	0.19
	2.5	24.2	9.0	2.18	66.5	16.1	0.30	0.20
	3.0	24.5	9.3	2.29	70.5	17.3	0.32	0.21
	3.5	24.1	9.8	2.37	75.5	18.2	0.36	0.22
	4.0	21.7	10.7	2.33	86.0	18.7	0.43	0.25
Medium-frame heifer calves								
300	0.5	7.5	9.6	0.73	56.0	4.2	0.29	0.21
	1.0	8.0	11.4	0.91	62.0	5.0	0.44	0.22
	1.5	8.2	13.1	1.08	68.5	5.6	0.59	0.27
	2.0	8.0	15.1	1.22	77.0	6.2	0.74	0.33
400	0.5	9.3	8.9	0.84	56.0	5.2	0.26	0.19
	1.0	9.9	10.2	1.01	62.0	6.1	0.36	0.20
	1.5	10.2	11.4	1.17	68.5	7.0	0.45	0.24
	2.0	10.0	12.9	1.29	77.0	7.7	0.57	0.29
500	0.5	11.0	8.5	0.94	56.0	6.2	0.24	0.18
	1.0	11.8	9.4	1.11	62.0	7.3	0.30	0.21
	1.5	12.1	10.3	1.25	68.5	8.4	0.38	0.22
	2.0	11.8	11.4	1.35	77.0	9.1	0.45	0.24
600	0.5	12.6	8.1	1.04	56.0	7.1	0.23	0.18
	1.0	13.5	8.8	1.19	62.0	8.4	0.28	0.20
	1.5	13.8	9.5	1.32	68.5	9.5	0.32	0.21
	2.0	13.5	10.4	1.41	77.0	10.4	0.38	0.23
700	0.5	14.1	7.9	1.13	56.0	8.0	0.22	0.19
	1.0	15.1	8.4	1.28	62.0	9.4	0.25	0.19
	1.5	15.5	9.0	1.39	68.5	10.6	0.28	0.20
	2.0	15.2	9.6	1.46	77.0	11.7	0.32	0.22
800	0.5	15.6	7.7	1.22	56.0	8.7	0.21	0.18
	1.0	16.7	8.1	1.36	62.0	10.4	0.22	0.18
	1.5	17.2	8.5	1.46	68.5	11.8	0.24	0.19
	2.0	16.8	9.0	1.51	77.0	12.9	0.28	0.20

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Medium-frame heifer calves (continued)								
900	0.5	17.1	7.5	1.31	56.0	9.6	0.21	0.18
	1.0	18.3	7.8	1.44	62.0	11.3	0.22	0.18
	1.5	18.8	8.1	1.53	68.5	12.9	0.22	0.19
	2.0	18.3	8.5	1.56	77.0	14.1	0.25	0.19
1,000	0.5	18.5	7.4	1.39	56.0	10.4	0.20	0.19
	1.0	19.8	7.6	1.51	62.0	12.3	0.20	0.18
	1.5	20.3	7.8	1.59	68.5	13.9	0.21	0.18
	2.0	19.8	8.1	1.61	77.0	15.2	0.22	0.19
Large-frame heifer calves and compensating medium-frame yearling heifers								
300	0.5	7.8	9.5	0.76	54.0	4.2	0.31	0.20
	1.0	8.4	11.3	0.95	59.0	5.0	0.45	0.24
	1.5	8.8	13.0	1.13	64.0	5.6	0.58	0.25
	2.0	8.9	14.6	1.30	69.5	6.2	0.69	0.30
	2.5	8.7	16.7	1.45	77.0	7.0	0.86	0.35
400	0.5	9.7	8.9	0.87	54.0	5.2	0.27	0.18
	1.0	10.5	10.1	1.06	59.0	6.2	0.36	0.21
	1.5	10.9	11.3	1.23	64.0	7.0	0.45	0.22
	2.0	11.1	12.6	1.38	69.5	7.7	0.54	0.26
	2.5	10.8	14.1	1.51	77.0	8.3	0.65	0.31
500	0.5	11.5	8.4	0.98	54.0	6.1	0.23	0.17
	1.0	12.4	9.4	1.16	59.0	7.3	0.30	0.20
	1.5	12.9	10.3	1.32	64.0	8.3	0.38	0.20
	2.0	13.1	11.2	1.46	69.5	9.1	0.44	0.24
	2.5	12.8	12.4	1.57	77.0	10.0	0.53	0.26
600	0.5	13.2	8.1	1.08	54.0	7.1	0.22	0.18
	1.0	14.1	8.9	1.25	59.0	8.3	0.28	0.19
	1.5	14.8	9.6	1.41	64.0	9.5	0.33	0.19
	2.0	15.0	10.3	1.54	69.5	10.4	0.38	0.22
	2.5	14.6	11.2	1.63	77.0	11.2	0.44	0.24
700	0.5	14.8	7.9	1.18	54.0	8.0	0.21	0.18
	1.0	15.9	8.5	1.34	59.0	9.4	0.25	0.18
	1.5	16.6	9.0	1.49	64.0	10.6	0.29	0.19
	2.0	16.8	9.6	1.61	69.5	11.7	0.33	0.20
	2.5	16.4	10.3	1.68	77.0	12.6	0.38	0.22

Table 1.1. Nutrient requirements for growing and finishing cattle (nutrient concentration in diet dry matter)^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Large-frame heifer calves and compensating medium-frame yearling heifers (continued)								
800	0.5	16.4	7.7	1.27	54.0	8.9	0.20	0.17
	1.0	17.6	8.2	1.43	59.0	10.4	0.24	0.18
	1.5	18.3	8.6	1.57	64.0	11.7	0.25	0.18
	2.0	18.6	9.0	1.67	69.5	12.9	0.28	0.19
	2.5	18.1	9.6	1.74	77.0	13.9	0.33	0.21
900	0.5	17.8	7.5	1.36	54.0	9.6	0.20	0.18
	1.0	19.2	7.9	1.52	59.0	11.3	0.22	0.18
	1.5	20.0	8.2	1.64	64.0	12.8	0.23	0.18
	2.0	20.3	8.6	1.74	69.5	14.1	0.26	0.18
	2.5	19.8	9.0	1.78	77.0	15.2	0.29	0.20
1,000	0.5	19.3	7.4	1.45	54.0	10.4	0.19	0.18
	1.0	20.8	7.7	1.60	59.0	12.3	0.21	0.18
	1.5	21.7	8.0	1.71	64.0	13.9	0.21	0.18
	2.0	22.0	8.2	1.80	69.5	15.3	0.23	0.18
	2.5	21.5	8.6	1.83	77.0	16.6	0.25	0.18
1,100	0.5	20.8	7.3	1.54	54.0	11.2	0.19	0.18
	1.0	22.3	7.5	1.68	59.0	13.2	0.20	0.18
	1.5	23.3	7.7	1.78	64.0	14.9	0.20	0.18
	2.0	23.6	7.9	1.86	69.5	16.4	0.21	0.18
	2.5	23.1	8.2	1.88	77.0	17.8	0.22	0.18

Source: Adapted from National Research Council. 1984. *Nutrient Requirements of Beef Cattle*. National Academies Press, Washington, DC.

Table 1.2. Nutrients requirements of breeding beef cattle (nutrient concentration in diet dry matter).^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Pregnant yearling heifers—last third of pregnancy								
700	0.9	15.3	8.4	1.3	55.4	8.5	0.27	0.20
	1.4	15.8	9.0	1.4	60.3	9.6	0.33	0.21
	1.9	15.8	9.8	1.5	67.0	10.6	0.33	0.21
750	0.9	16.1	8.3	1.3	55.1	8.9	0.27	0.19
	1.4	16.6	8.9	1.5	59.9	10.0	0.32	0.21
	1.9	16.6	9.5	1.6	66.5	11.1	0.37	0.23
800	0.9	16.8	8.2	1.4	54.8	9.2	0.28	0.20
	1.4	17.4	8.8	1.5	59.6	10.4	0.33	0.21
	1.9	17.5	9.3	1.6	66.1	11.6	0.35	0.21
850	0.9	17.6	8.2	1.4	54.5	9.6	0.26	0.20
	1.4	18.2	8.6	1.6	59.3	10.8	0.30	0.21
	1.9	18.3	9.1	1.7	65.7	12.1	0.34	0.22
900	0.9	18.3	8.1	1.5	54.3	9.9	0.26	0.20
	1.4	19.0	8.5	1.6	59.1	11.3	0.30	0.21
	1.9	19.2	9.0	1.7	65.4	12.5	0.32	0.21
950	0.9	19.0	8.0	1.5	54.1	10.3	0.27	0.20
	1.4	19.8	8.4	1.7	58.9	11.7	0.29	0.21
	1.9	20.0	8.8	1.8	65.1	13.0	0.32	0.21
Dry pregnant mature cows—middle third of pregnancy								
800	0.0	15.3	7.1	1.1	48.8	7.5	0.17	0.17
900	0.0	16.7	7.0	1.2	48.8	8.2	0.18	0.18
1,000	0.0	18.1	7.0	1.3	48.8	8.8	0.18	0.18
1,100	0.0	19.5	7.0	1.4	48.8	9.5	0.19	0.19
1,200	0.0	20.8	6.9	1.4	48.8	10.1	0.19	0.19
1,300	0.0	22.0	6.9	1.5	48.8	10.8	0.20	0.20
1,400	0.0	23.3	6.9	1.6	48.8	11.4	0.20	0.20
Dry pregnant mature cows—last third of pregnancy								
800	0.9	16.8	8.2	1.4	54.5	9.2	0.26	0.20
900	0.9	18.2	8.0	1.5	54.0	9.8	0.27	0.21
1,000	0.9	19.6	7.9	1.6	53.6	10.5	0.26	0.20
1,100	0.9	21.0	7.8	1.6	53.2	11.2	0.26	0.21
1,200	0.9	22.3	7.8	1.7	52.9	11.8	0.26	0.21
1,300	0.9	23.6	7.7	1.8	52.7	12.5	0.26	0.21
1,400	0.9	24.9	7.6	1.9	52.5	13.1	0.26	0.21

^aDM = dry matter; TDN = total digestible nutrients (energy); Ca = calcium; P = phosphorus

Table 1.2. Nutrients requirements of breeding beef cattle (nutrient concentration in diet dry matter).^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Two-year-old heifers nursing calves—first 3 to 4 months postpartum—10 lb milk/day								
700	0.5	15.9	11.3	1.8	65.1	10.3	0.36	0.24
750	0.5	16.7	11.0	1.8	64.4	10.8	0.34	0.24
800	0.5	17.6	10.8	1.9	63.8	11.2	0.34	0.24
850	0.5	18.4	10.6	1.9	63.2	11.6	0.33	0.23
900	0.5	19.2	10.4	2.0	62.7	12.0	0.33	0.23
950	0.5	20.0	10.2	2.0	62.3	12.5	0.32	0.23
1,000	0.5	20.8	10.0	2.1	61.9	12.9	0.31	0.23
Cows nursing calves—average milking ability—first 3 to 4 months postpartum—10 lb milk/day								
800	0.0	17.3	10.2	1.8	58.2	10.1	0.30	0.22
900	0.0	18.8	9.9	1.9	57.3	10.8	0.28	0.22
1,000	0.0	20.2	9.6	2.0	56.6	11.5	0.28	0.22
1,100	0.0	21.6	9.4	2.0	56.0	12.1	0.27	0.22
1,200	0.0	23.0	9.3	2.1	55.5	12.8	0.27	0.22
1,300	0.0	24.3	9.1	2.2	55.1	13.4	0.27	0.22
1,400	0.0	25.6	9.0	2.3	54.7	14.0	0.27	0.22
Cows nursing calves—superior milking ability—first 3 to 4 months postpartum—20 lb milk/day								
800	0.0	15.7	14.2	2.2	77.3	12.1	0.48	0.31
900	0.0	18.7	12.9	2.4	69.8	13.1	0.41	0.28
1,000	0.0	20.6	12.3	2.5	67.0	13.8	0.39	0.27
1,100	0.0	22.3	11.9	2.6	65.2	14.5	0.38	0.27
1,200	0.0	23.8	11.5	2.7	63.7	15.2	0.36	0.26
1,300	0.0	25.3	11.2	2.8	62.6	15.9	0.36	0.26
1,400	0.0	26.7	11.0	2.9	61.7	16.5	0.35	0.26
Bulls, maintenance and slow rate of growth—regain body condition								
<1,300	For growth and development, use Table 1.1							
1,300	1.0	25.4	7.6	1.9	55.8	14.2	0.22	0.19
	1.5	26.1	7.9	2.0	59.7	15.6	0.24	0.19
	2.0	26.2	8.2	2.2	64.0	16.8	0.26	0.20
1,400	1.0	26.8	7.5	2.0	55.8	15.0	0.21	0.19
	1.5	27.6	7.7	2.1	59.7	16.5	0.23	0.19
	2.0	27.7	8.0	2.2	64.0	17.8	0.25	0.20
1,500	0.0	25.2	6.9	1.7	48.4	12.2	0.20	0.20
	1.0	28.3	7.4	2.1	55.8	15.8	0.21	0.19
	1.5	29.0	7.6	2.2	59.7	17.3	0.22	0.19
1,600	0.0	26.5	6.9	1.8	48.4	12.8	0.19	0.20
	1.0	29.7	7.3	2.2	55.8	16.6	0.22	0.19
	1.5	30.4	7.4	2.3	59.7	18.2	0.22	0.20

Table 1.2. Nutrients requirements of breeding beef cattle (nutrient concentration in diet dry matter).^a

Weight (lb)	Daily gain (lb/day)	DM intake (lb)	Protein (%)	Protein (lb)	TDN (%)	TDN (lb)	Ca (%)	P (%)
Bulls, maintenance and slow rate of growth—regain body condition (continued)								
1,700	0.0	27.7	6.8	1.9	48.4	13.4	0.21	0.21
	0.5	29.6	7.0	2.1	52.0	15.4	0.20	0.19
1,800	0.0	28.9	6.8	2.0	48.4	14.0	0.21	0.21
	0.5	30.9	7.0	2.2	52.0	16.1	0.20	0.20
1,900	0.0	30.1	6.8	2.0	48.4	14.6	0.21	0.21
	0.5	32.2	6.9	2.2	52.0	16.8	0.20	0.20
2,000	0.0	31.3	6.8	2.1	48.4	15.2	0.21	0.21
2,100	0.0	32.5	6.8	2.2	48.4	15.7	0.22	0.22
2,200	0.0	33.6	6.8	2.3	48.4	16.3	0.22	0.22

Source: Adapted from National Research Council. 1984. *Nutrient Requirements of Beef Cattle*. National Academies Press, Washington, DC.

Forage Value

2

*Ron Hathaway and
Gene Pirelli*

Livestock producers in the western United States, including Oregon, are at an economic disadvantage relative to those in some other regions in North America because of relatively high winter feed costs. A ruminant animal consumes on average 2 to 3 percent of its body weight in dry matter each day. This equates to feeding 1.5 to 2.5 tons of conserved forage to mature cows or 0.25 to 0.5 ton to ewes during the winter feeding period. This volume of feed can represent more than 50 percent of the producer's input costs. The ability of Oregon cattle producers to compete with other regions of North America may relate to how effectively they can reduce winter feed costs while maintaining acceptable levels of beef cattle production.

In the grazing season, vegetative forage with an available height of 4 to 8 inches usually provides the nutritional needs of livestock. Outside of the grazing season, however, livestock depend on the producer to provide for their nutritional needs. It is during this period, when animals are fed conserved forage, that the nutritional management program plays an important part in maintaining the growth and condition of livestock.

In order to provide a ration that meets the nutritional needs of livestock, it is necessary to know the nutritive value of the forage. Forage value, whether hay or pasture, differs depending largely on stage of maturity, time, and weather at harvest.

Vegetative growth is the leafy growth that occurs in the early and middle parts of the growing season, and it is the most nutritious forage. Reproductive growth is the seed-producing stage of growth. It occurs later in the growing season, is a characteristic of mature plants, and is of reduced nutritive value.

Even if forage is harvested at the proper time, the nutrient content of the resulting hay or silage can vary widely. The amount and type of fertilizer used and the composition of the stand can contribute to variability.

Accurate forage analysis (see Chapter 3) is critical to profitable winter, nonpasture feeding programs. Even small deficiencies in protein and energy can lead to lower-than-desired performance levels of animals.

Once you determine the nutritive quality of the forage, you can decide how to allocate the feed. Ideally, you would formulate a complete and balanced ration, possibly including supplemental protein and/or energy (see Chapter 5). Alternatively, you could allocate the best quality feed to the animals with the highest nutrient requirements (see Chapter 1) and the lowest quality feed to those with lower requirements. The goal would be to use the feed in the most nutritionally appropriate manner.



Estimated and measured values

You can obtain a rough estimate of the value of forages through visual assessment, tables in nutrition books, and Extension publications. However, laboratory testing of your specific forage will provide the most accurate values for balancing livestock rations.

Once value has been determined, use the information to help provide balanced rations for your livestock. It can help you determine what supplements you need to add or simply the best way to use the feed you have.

Visual assessment

Several visual factors can assist in evaluating the nutrient value or quality of conserved forages. These factors are as follows.

Maturity. Maturity can be determined by observing bloom or bud stage and stem size. More abundant blooms or mature seed heads, fewer buds, and larger, woody stems indicate more mature forage.

Leafiness. Approximately two-thirds of the protein in forage is in the leaves. The ratio of leaves to stems is more important with legume forages than with grass forages.

Color. Forage that is not bright green may be damaged. Rain bleaches forage, heavy rain causes a dark brown or black color, and heat or fermentation causes browning. Yellowing can indicate overly mature forage.

Presence of foreign material. Weeds, chaff, sticks, and other matter that is not forage indicate lower quality hay.

Odor and condition. Hay that smells musty or is moldy indicates rain damage or baling at too high a moisture content. The result is lower quality hay and lower intake by animals.

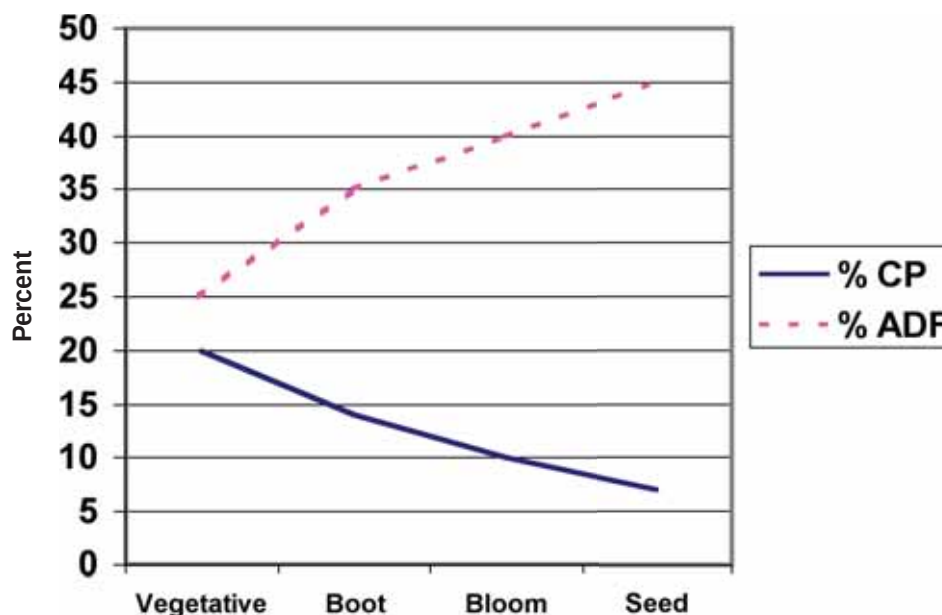


Figure 2.1—Change in crude protein (CP) and acid detergent fiber (ADF) with increasing maturity of cool-season grass forage based on laboratory tests in Oregon.

Maturity of the forage is one of the most important factors that determine proper harvest time. As forage plants or hays mature, the quality or feed value declines. As can be seen in Figure 2.1, the crude protein content decreases as the forage moves from a vegetative stage to seed head production. Additionally, the acid detergent fiber, which is an estimated measure of digestibility, increases during the same period. Increased acid detergent fiber means lower digestibility.

These changes in nutritional quality occur anytime during the growing season when plants move from a vegetative to a mature stage of production. During the most mature stage of plant growth (seed), protein and energy levels are too low to meet the nutrient requirements of most classes of livestock (see Chapter 1). Supplemental protein would be needed in order for livestock to utilize this very mature forage. Supplemental energy also might be needed, depending on animal requirements.

Table 2.1 shows data from a 6-year study in Washington, in which samples of perennial and annual ryegrass were harvested at different stages of maturity over an entire growing season. An example of the nutritional value for the boot, early boot, and vegetative growth stages is listed. Vegetative samples show much higher crude protein than the boot stage of growth. The early boot stage of grasses is considered to be the ideal cutting time to capture the highest yield and quality of a pasture or hayfield.

Stockpiled forage is pasture that was managed so as to keep the forage in the vegetative stage through the end of the growing season. As a result, the forage does not decrease in quality to the same extent as forage that is allowed to mature into the reproductive stage of growth. This standing forage is then utilized at a later date.

This practice works well in the dry climates of central and eastern Oregon, but on the west side of the Cascades the forage must be grazed before much rainfall accumulates. Rain reduces the value and desirability of the forage. Vegetative stockpiled forage is best utilized by about November 15 in western Oregon. The quality of this forage usually is intermediate relative to green, vegetative growth and dry, mature forage. Supplementation of animal diets may be required.

Table 2.1. Growth stage differences in nutritional quality (crude protein) of annual and perennial ryegrass in western Washington (average over a 6-year period).

Cutting	Crude protein (%)		Growth stage ^a
	Annual ryegrass	Perennial ryegrass	
1	9.75	10.56	Boot
2	15.56	16.94	Early boot
3	20.56	20.75	Vegetative
4	22.94	24.00	Vegetative
5	23.19	30.25	Vegetative

^aThe last three cuttings (vegetative growth) were taken during the summer and fall.

Book values and Extension publications

It has long been recognized that book value nutrient analyses for forages grown in other parts of the country do not accurately represent Oregon forages. To formulate winter beef diets, analyses for Oregon-produced forages are needed.

In order to develop a database of Oregon forages, more than 30 forages grown in Oregon were analyzed by Oregon State University researchers. These forages were evaluated for protein, energy, fat, fiber, and mineral content. In addition, the OSU research team compiled dry matter, crude protein, and acid detergent fiber results from hay surveys of more than 400 forage samples to create the Oregon Forage and Byproduct Library (see references). Mineral analyses are also included in the Library tables. There also is nutritional information on by-products as livestock feeds. This library is a valuable source of information to help producers estimate the nutritional value of locally produced or purchased forages.

Table 2.2 at the end of this chapter shows the nutrient content of common feeds used as protein and energy supplements.

Laboratory analysis

Ideally, you should have your forage tested each year for dry matter, crude protein, and acid detergent fiber (see Chapter 3). Laboratory analysis is by far the best way to develop your winter feeding program. Compare your forage analysis with the tables in the Oregon Forage and Byproduct Library and use the estimates of energy for beef cattle (TDN) and the other analyses to formulate your rations.

References

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- National Research Council. 1982. *United States-Canadian Tables of Feed Composition*. National Academies Press, Washington, DC. Web: <http://www.nap.edu/>
- Oregon Forage and Byproduct Library, Oregon State University, 112 Withycombe Hall, Corvallis, OR 97331-6702. Phone: 541-737-3431. Web: <http://osu.orst.edu/dept/animal-sciences/foragelib.htm>

Worksheet 2.1

Characterizing Your Forage Quality

What kind of conserved forage (hay, balage, or standing grass and/or legume mix) do you feed?

What was the stage of maturity when harvested? _____

What is your estimate of the crude protein and energy content of your conserved forage?

Describe your conserved forage in relation to the following factors: maturity, leafiness, color, foreign material, odor, and condition. _____

Do you have your conserved forage analyzed? _____

What is the lab analysis of your feed? CP _____ TDN _____ Other, if known _____

Table 2.2. Nutrient composition of common feedstuffs and range plants.^a

Feedstuff	Dry matter (%)	TDN (%)	CP (%)	Ca (%)	P (lb)
Grazed plants					
Brome, fresh, early vegetative	34	74	18	0.50	0.30
Dropseed, sand, fresh, stem-cured	88	59	5	0.57	0.06
Galleta, fresh, stem-cured	71	48	5.5	1.05	0.07
Needle and thread grass, fresh, stem-cured	92	49	4.4	1.09	0.06
Orchardgrass, fresh, early vegetative	23	72	18.4	0.58	0.54
Orchardgrass, fresh, midbloom	31	57	11	0.23	0.23
Redtop, fresh	29	63	11.6	0.46	0.29
Ryegrass, fresh	27	68	10.4	0.55	0.27
Sage, black, browse, fresh, stem-cured	65	49	8.5	0.81	0.17
Sagebrush, big, browse, fresh, stem-cured	65	50	9.3	0.71	0.18
Sagebrush, bud, browse, fresh, late vegetative	32	52	17.5	0.60	0.42
Saltbush, browse, fresh, stem-cured	55	36	7.2	2.21	0.12
Saltgrass, fresh, post ripe	74	53	4.2	0.23	0.07
Squirreltail, fresh, stem-cured	50	50	3.1	0.37	0.06
Summer cypress (kochia), fresh, stem-cured	85	50	9	2.36	0.12
Timothy, fresh, late vegetative	26	72	18	0.39	0.32
Timothy, fresh, midbloom	29	63	9	0.80	0.30
Wheatgrass, crested, fresh, early vegetative	28	75	21.5	0.46	0.34
Wheatgrass, crested, fresh, full bloom	45	61	9.8	0.39	0.28
Wheatgrass, crested, fresh, post ripe	80	49	3.1	0.27	0.07
Winterfat, fresh, stem-cured	80	35	10.8	1.98	0.12
Other roughage					
Alfalfa hay, early bloom	90	60	18	1.41	0.22
Alfalfa hay, midbloom	90	58	17	1.41	0.24
Alfalfa hay, late bloom	90	52	14	1.43	0.25
Alfalfa hay, mature	91	50	12.9	1.13	0.18
Barley straw	91	40	4.3	0.30	0.07
Brome hay, sun-cured, late vegetative	88	60	16	0.32	0.37
Brome hay, sun-cured, late bloom	89	55	10	0.30	0.35
Corn, silage	33	68	8.1	0.23	0.22
Fescue hay, sun-cured, early vegetative	91	61	12.4	0.51	0.36
Fescue hay, sun-cured, late vegetative	92	48	9.5	0.30	0.26
Oat hay	91	55	9.3	0.24	0.22
Oat straw	92	45	4.4	0.24	0.06
Orchardgrass, hay, sun-cured, late bloom	91	54	8.4	0.26	0.30
Redtop, hay, sun-cured, midbloom	94	57	11.7	0.63	0.35

^aTDN = total digestible nutrients; CP = crude protein; Ca = Calcium; P = Phosphorus

Table 2.2. Nutrient composition of common feedstuffs and range plants.^a

Feedstuff	Dry matter (%)	TDN (%)	CP (%)	Ca (%)	P (lb)
Other roughage (continued)					
Ryegrass, hay, sun-cured, midbloom	86	60	8.6	0.65	0.32
Sedge hay, sun-cured	89	52	9.4	—	—
Sweetclover, yellow, hay, sun-cured, midbloom	87	54	15.7	1.27	0.25
Timothy hay, sun-cured, full bloom	89	56	8.1	0.43	0.20
Wheatgrass, crested, hay, sun-cured	93	53	12.4	0.33	0.21
Wheat hay	88	58	8.5	0.15	0.39
Wheat straw	89	41	3.6	0.18	0.05
Concentrates (protein and energy)					
Barley	88	84	13.5	0.05	0.38
Beet molasses	78	79	8.5	0.17	0.03
Brewers grains	92	66	28.1	0.29	0.54
Corn	88	90	10	0.02	0.35
Cottonseed meal	91	76	45.2	0.22	1.21
Milo	89	80	12.4	0.04	0.33
Oats	89	77	13.3	0.07	0.38
Soybean meal, 44%	90	85	47.7	0.29	0.68
Wheat	89	88	16	0.04	0.42
Wheat middlings	89	69	18	0.18	0.99

Source: Adapted from National Research Council. 1984. *Nutrient Requirements of Beef Cattle*. National Academies Press, Washington, DC.

Evaluating Feeds

3

Robert Pawelek and
Shelby Filley

As discussed in the previous chapter, nutrient content of feeds varies not only among different feed types, but also among batches of the same type of feed (hay lots, grain shipments, growth stage of pasture, etc.). The objective of sampling forages and concentrates is to obtain representative samples for laboratory analysis to estimate the value of feed for livestock.

Rapid feeding and turnover of grains and supplements may reduce the value of analyses compared to analyzing larger batches fed over a longer period of time. However, analyses completed even after the ration is consumed can detect seasonal differences that may influence future use.

Sampling feeds

Forages may be sampled as hay or standing pasture. Sampling methods include mechanical coring of bales with a hay probe, pulling hand-grab samples from bales or windrows, pulling hand-grab samples from standing forage, and clipping standing forage samples.

Accuracy depends largely on sampling method and lab technique. Having dependable samples can prevent unwelcome surprises. It is important to identify the sample by date, cutting, forage or concentrate type, pasture location, and owner before shipping it to the lab.

Remember: the key to accurate sampling of feed is to sample consistently and mix the samples thoroughly, especially if you plan to divide them.

Baled hay

There are several methods for sampling baled hay. The best technique is to use a mechanical coring probe made specifically for this purpose. Place the serrated edge on the side of the hay bale that is most resistant to puncture (usually the round side of a round bale or the small end of a square bale) and obtain a sample by drilling with a brace.

The sample should be as representative of the composition of the bales as possible. Repeat the process on several hay bales within the sampling lot. Sample hay from different fields or cutting times separately.



An older traditional method of obtaining forage samples is the hand-grab technique. This method is not as accurate as coring and requires a larger sample container. Hand samples do not provide consistently reliable results.

Pasture forage

Standing forage may be sampled by the hand-grab or clipping technique. Collect 20 subsamples and combine a sufficient amount to fill a 1-gallon reclosable plastic bag. In a pasture having a pure stand of forage, take samples randomly from different locations. If the pasture is a mixed stand of forages, it is important to sample only plants that animals will graze. Clip forage no shorter than 1 inch to avoid contaminating samples with soil. You can use scissors or a knife. Unlike hay, pasture forage is not normally dry when sampled. To avoid spoiling, air dry moist samples before shipping.

Grain

Grade of grain, along with protein and energy content, can vary widely. Periodically analyze the major feed grains for lactating and finishing diets. You also can request a lab analysis from the supplier. To sample, use a grain probe if available; otherwise, hand sampling is acceptable. It is best to take grain samples while the material is being off-loaded from the delivery truck. However, you can take five samples from different locations in the pile, bunk, bin, or truck. Combine all samples and ship in a clean, dry container.

Other feeds

Bagged or block products require several separate samples. A general guideline is to take samples from 10 percent of the bags or blocks in question. Take at least one handful per bag. Sample blocks by slicing or chiseling a chunk from each block.

Choosing a laboratory

A List of Analytical Laboratories Serving Oregon (EM 8677) is available in your county Extension office or on the Internet (<http://eesc.oregonstate.edu/agcomwebfile/edmat/html/em/em8677/em8677.html>). It contains contact information for labs that analyze forage samples. Prices and sample submission guidelines may differ among laboratories. Call the lab prior to taking the sample. Lab fees start at around \$30.00 per sample and increase depending on analyses requested.

Analyses (lab tests to request)

There are many possible tests to perform on forages. Results for moisture, protein, and energy are the most important. Ask the lab to run the required analyses to give you this basic information. Standard analyses include percent dry matter (DM), percent crude protein (CP), percent acid detergent fiber (ADF), and percent neutral detergent fiber (NDF). Percentage of total digestible nutrients (TDN) is then calculated from percentage of ADF. Periodic analysis for basic mineral content also is helpful. Other analyses are nitrate nitrogen and trace mineral composition. Because non-standard analyses can be costly, request them only when needed.

Laboratory method

Wet chemistry uses chemical solutions to directly measure plant components in feed. This method is a well recognized and accepted way to measure the components of feed.

Near Infrared Reflectance Spectroscopy (NIRS) uses light transmitted through the sample to estimate the components of the feed. The estimates are made using mathematical equations based on previous wet chemistry data. Large amounts of data from many reference samples are required in order to create accurate equations for estimating feed components in this manner. While some forages lend themselves well to NIRS, and some laboratories have compiled the necessary data for accuracy, this is not always the case.

Be sure to discuss the use of wet chemistry versus NIRS with your county Extension faculty, nutrition consultant, or analytical lab before you decide which method to use. NIRS usually is less expensive and may be available overnight. Some labs can run wet chemistry samples overnight, while others take longer.

Lab report definitions

Acid detergent fiber (ADF)—A measure of cellulose and lignin. ADF is negatively correlated with overall digestibility (high ADF = low digestibility).

Adjusted crude protein—The crude protein adjusted for availability to the animal. Some proteins can be bound with fiber and are unavailable, especially in heat-damaged forages.

As-fed basis—A way to express percentage of nutrients in a feed, including moisture. As-fed will equal as-sampled or as-received, if the feed is not altered between sampling, testing, and feeding time.

Crude fat—Fat and other ether extractable compounds. Fat is an energy-dense nutrient and contains 2.25 times the energy found in carbohydrates and proteins.

Crude protein (CP)—An estimate of the protein content of the feed.

Laboratories measure the nitrogen (N) content of the forage and then calculate crude protein using the formula $CP = \%N \times 6.25$.

Dry matter (DM)—The dry portion of the forage (not including the water).

Dry matter basis—A way to express percentage of nutrients in a feed, without the moisture. At some point in the calculations, DM must be converted to as-fed basis for the final mixing and feeding (see Chapter 5).

Minerals and vitamins—See Chapter 4.

Moisture—The water portion of a sample.

Net energy (NE)—An estimate of the energy in a feed that is available to the animal after allowing for energy lost during digestion and metabolism. Estimates for NE are divided into NE for maintenance (NE_m) and NE for gain (NE_g) and are more precise than TDN.

NE_m is the energy value of a feed to maintain animal tissue without gain or loss of weight.

NE_g is the energy value of a feed used for body weight gain above that required for maintenance.

Neutral detergent fiber (NDF)—A measure of hemicellulose, cellulose, and lignin, representing the fibrous bulk of the forage. NDF is negatively correlated with intake (high NDF = low intake).

Nonstructural carbohydrates (NSC)—Starches and sugars inside the cell that serve as energy sources for the animal.

Relative feed value (RFV)—A way to rank feed based on digestibility (ADF) and intake (NDF) potential. An RFV of 100 is considered the average score and represents an alfalfa hay containing 41 percent ADF and 53 percent NDF on a dry matter basis. The higher the RFV, the better the forage quality. RFV is used in feed marketing, not in balancing a ration for animals.

Total digestible nutrients (TDN)—A rough estimate of the feed energy available to the animal. It often is calculated from ADF.

References

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- Stokes, S.R. and E.P. Prostko. 1998. *Understanding Forage Quality Analysis*. Texas A & M University.

Worksheet 3.1

Sampling Plan for Your Forages

Each year, analyze new lots of hay or batches of grain (or other supplements) prior to the intended feeding period. You then can make and follow a plan for using this feed in an efficient manner. List the feeds you need to have analyzed.

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

Minerals and Vitamins

Important Little Things in Life

4

Ron Hathaway and
James Oldfield

Two things that characterize Oregon’s agriculture are its abundance of forage on range and pasture lands and its diversity, which makes possible the production of numerous crop and animal products. Diversity is a two-edged sword, however, because it means that different soil and weather conditions affect the composition of forages or other crops, and the effects sometimes are undesirable.

Suboptimal animal performance has been observed when cattle consume forage-based diets, despite an adequate supply of protein and energy. An insufficient supply of micronutrients, including minerals and vitamins, may be involved because cattle must receive all of the essential nutrients in proper quantities to maintain good health, grow, and reproduce at their maximum potential.

Minerals

A good example of the effect of diverse growing conditions is the concentration of mineral elements in forage, particularly the trace minerals, which normally are present in very small (“trace”) quantities. Sometimes, these minerals are present at levels that are too low for good animal health, and so-called “deficiency diseases” result. On the other hand, elevated levels of these minerals sometimes cause toxicity.

Table 4.1 identifies the essential minerals as macrominerals or microminerals (trace). Macrominerals are required in relatively large amounts, and the requirements are expressed in percentages. Trace minerals, or microminerals, are required in smaller amounts, and the requirements are expressed in mg/kg (parts per million or ppm) of livestock diets. Many of the essential minerals usually are available in livestock feed in adequate

Table 4.1. Nutritionally essential minerals.

Macrominerals		Microminerals (trace)	
Calcium (Ca)	Potassium (K)	Cobalt (Co)	Manganese (Mn)
Phosphorus (P)	Chlorine (Cl)	Copper (Cu)	Molybdenum (Mo)
Magnesium (Mg)		Chromium (Cr)	Selenium (Se)
Sodium (Na)		Iodine (I)	Zinc (Zn)
Sulfur (S)		Iron (Fe)	Nickel (Ni)



amounts. However, in some cases feeds do not contain sufficient amounts, and supplementation is necessary to optimize animal performance.

Minimum requirements and maximum levels of minerals are shown in Table 4.2.

Macrominerals

Calcium (Ca) and phosphorus (P) are the minerals required in the largest amount by livestock. Calcium is the most abundant cation in an animal's body. Approximately 99 percent of the calcium is found in bones and teeth, with the remaining 1 percent distributed in soft tissues. Phosphorus is a major constituent of bones and teeth and is an essential component of organic compounds involved in almost every aspect of metabolism.

For optimum performance, Ca and P need to be present in the diet at required levels. In addition, the Ca to P (Ca:P) ratio is important. The ideal Ca:P ratio is approximately 1:1 to 2:1. With adequate amounts of P in the diet, a higher Ca:P ratio can be tolerated. In beef cattle, diets with a ratio as high as 7:1 have performed satisfactorily if levels of P are well above

Table 4.2. Mineral requirements and maximum tolerable level of minerals in beef rations.^a

Mineral	Minimum requirement			Maximum tolerable level
	Growing and finishing cattle	Gestating cows	Early lactation cows	
Calcium (%) ^b	—	—	—	—
Cobalt (ppm)	0.10	0.10	0.10	10.00
Copper (ppm)	10.00	10.00	10.00	100.00
Iodine (ppm)	0.50	0.50	0.50	50.00
Iron (ppm)	50.00	50.00	50.00	1,000.00
Magnesium (%)	0.10	0.12	0.20	0.40
Manganese (ppm)	20.00	40.00	40.00	1,000.00
Molybdenum (ppm)	—	—	—	5.00
Nickel (ppm)	—	—	—	50.00
Phosphorus (%) ^b	—	—	—	—
Potassium (%)	0.60	0.60	0.70	3.00
Selenium (ppm)	0.10	0.10	0.10	2.00
Sodium (%) ^c	0.06–0.08	0.06–0.08	0.1	10.00
Sulfur (%)	0.15	0.15	0.15	0.40
Zinc (ppm)	30.00	30.00	30.00	500.00

^appm = parts per million (1 ppm = 0.0001%; 1% = 10,000 ppm; 1 mg/kg = 1 ppm)

^bCalcium (Ca) and phosphorus (P) requirements vary with age, weight, type of animal, and production level (pounds/day growth, stage of gestation/lactation, etc.). Young animals have high requirements because of bone growth. Also, high rates of gain or milk production and pregnancy increase Ca requirements. See Chapter 1 tables for amounts required.

^cPercent sodium chloride (NaCl = salt)

Source: National Research Council. 1996. *Nutrient Requirements of Beef Cattle*. National Academies Press, Washington, DC. Web: <http://www.nap.edu/>

the required level and vitamin D level is high. Depressed performance has been observed with ratios below 1:1. Therefore, the ratio should not exceed 7:1 nor be less than 1:1.

Excess P in relationship to Ca can result in a very detrimental situation, even when Ca is at or above required levels. High levels of P can cause increased calcium resorption from the bone in adult animals. Urinary calculi (urolithiasis) are also caused by excess P. Stones in the kidney or bladder block urine, which can lead to death from uremia.

Calcium also has an interrelationship with other nutrients, and feeding higher levels of Ca for extended periods can have a detrimental effect on performance. Excess calcium can cause a deficiency of other essential elements, i.e., phosphorus, magnesium, iron, iodine, zinc, and manganese.

Consuming forage low in magnesium (Mg) leads to low blood levels of Mg (hypomagnesemia), which can cause grass tetany. Cattle in a state of severe magnesium deficiency become hypersensitive to tactile or sound stimuli and experience tremors. Convulsions and death usually follow.

Calcium and potassium (K) also can affect Mg blood levels by influencing Mg absorption. Thus, Mg concentration, as well as Ca and K levels, is important. By calculating a tetany ratio, $K \div (Ca + Mg)$, you can estimate the risk of grass tetany.¹ A tetany ratio of greater than 2.2 represents a tetany-prone diet.

Microminerals (trace)

The levels of microminerals in forages or other feed crops may make the difference between profit and loss in an animal operation. For example, a deficiency of selenium may cause animals to suffer white muscle disease, which can interfere with growth and reproduction and may cause death.

An excess of molybdenum results in toxic molybdenosis and, because molybdenum ties up copper in the diet, may cause copper deficiency. Molybdenum levels of 5 to 6 ppm produce signs of molybdenosis. Some areas in Baker County have high levels of molybdenum in the soil, and animals show symptoms of copper deficiency, even though the forage contains normal levels of copper.

Deficiency symptoms often are similar for several microminerals, making diagnosis difficult. One specific symptom is goiter (a neck swelling), which indicates iodine deficiency.

The importance of trace minerals has been known for years. The need for copper and cobalt was discovered in the 1930s; for selenium in the 1950s. Yet, many livestock operators fail to ensure adequate quantities in rations for their animals. A 1996 audit of cow/calf feeds in 18 states showed deficiencies of various trace elements ranging from about 5 percent to nearly 50 percent of the 352 forage samples studied (Corah and Dargatz, 1996).

¹The tetany ratio is calculated on an equivalent weight basis (i.e., it does not use the percentage or ppm of the mineral as shown in the forage analysis report, but rather uses the amount of each element corrected for molecular weight.) Thus, the formula actually is:

$$(K \text{ concentration} \div 39) \div [(Ca \text{ concentration} \div 20) + (Mg \text{ concentration} \div 12.1)]$$

The concentration of the elements must be on the same basis—percentage, ppm, or mg/kg.

To overcome a deficiency is fairly easy and generally not costly. One simply locates a source of the deficient mineral, usually a salt-mineral supplement, and adds it to the diet at a level to meet the animals' needs.

Preventing toxicity is more difficult, but it can be accomplished by eliminating the toxic feed from the diet (changing grazing sites, for example) or by adding a mineral that counteracts the effect of the toxic element, such as copper in cases of molybdenum toxicity.

Supplements seldom contain an element as such, but rather a salt of the needed element, such as sodium selenite in the case of selenium. Recently, interest has been shown in chelated minerals, which are thought to be more available to animals. A chelate is a combination of a mineral element and an organic substance, such as a protein or an amino acid. Chelated forms are more expensive, so consider using a blend of chelated and simple salt forms.

When supplementing feeds with essential trace elements, you'll need to calculate the amount of supplement to provide, based on the percentage of the needed element in the salt (see Worksheet 4.1).

Vitamins

Cattle require most of the vitamins needed by other mammals. However, ruminants are unique because rumen microorganisms can synthesize some essential vitamins. Synthesis by rumen microorganisms, supplies in natural feedstuffs, and synthesis in tissues meet most of the requirements.

Calves from adequately fed mothers have minimal stores of vitamins at birth. Colostrum is rich in vitamins, providing an immediate source of vitamins to the newborn calf. The ability to synthesize B vitamins and vitamin K in the rumen develops rapidly when solid feed is introduced into the diet. High-quality forages contain large amounts of vitamin A precursors and vitamin E. Vitamin D is synthesized by animal exposure to sunlight and is found in large amounts in sun-cured forages.

In ruminants, five classes of vitamins are important. Vitamin A likely is the most important. Vitamins D, E, K, and some of the B vitamins also are known to be essential. Supplemental vitamins can be added to the trace mineral salt mixture. Information on requirements, metabolic role, and signs of deficiency and toxicity is available in the references listed at the end of this chapter.



Vitamin A

When feed sources are likely to be low in vitamin A, supplementation is recommended to prevent potential problems with productive efficiency of cattle. Supplementing brood cows with vitamin A before (16,000 IU/day) and after (40,000 IU/day) the calving season can increase conception rates by 10 percent and decrease calf morbidity by as much as 50 percent as compared to a deficient state.

It is not uncommon for liver stores of vitamin A to be reduced when cattle consume a diet of low-quality, mature forage. Typically, 2 to 4 months of protection can be expected from stored vitamin A. Increasing dietary vitamin A by 5,000 IU per day can double the amount of vitamin A stored in the liver. Supplemental vitamin A may not always be necessary when cattle are grazing dormant rangelands, or if the cattle are fed properly cured and stored hay.

Vitamin E

Supplementation is recommended for brood cows consuming poor-quality forage during late pregnancy through early lactation. During the late stages of pregnancy, vitamin E concentration in the blood tends to decrease below marginal levels. Feeding supplemental vitamin E is an effective method of maintaining proper levels during this period.

Neonatal calves must receive dietary vitamin E because its placental transfer is very limited. However, it may not be necessary to supply supplemental vitamin E directly to the calf. Feeding an additional 1,000 IU of α -tocopherol (vitamin E) to cattle during the last trimester of pregnancy can increase antibody production and sequestration in the colostrum and increase vitamin E concentration in the colostrum by up to 30 percent. You can maintain plasma concentrations of vitamin E in the calf above the minimal level by continuing maternal supplementation into the early stages of lactation.

Vitamin D

Sunlight typically stimulates sufficient production of vitamin D in grazing cattle. Vitamin D status is improved more effectively by increasing the animal's exposure to sunlight than by dietary supplementation.

Vitamin K

Ruminal microorganisms generally synthesize and supply vitamin K in amounts sufficient to meet the ruminant animal's requirements. Supplemental vitamin K usually is justified only when cattle are suspected to have been exposed to vitamin K antagonists.

Consumption of a natural vitamin K antagonist (dicumarol) can be a problem if cattle ingest moldy sweetclover forage. Because dicumarol antagonizes vitamin K, an essential component of the blood clotting process, feeding moldy sweetclover hay to cattle may cause hemorrhage and uncontrollable bleeding.

Response to vitamin K therapy depends somewhat on differences in the dicumarol source and dosage. Therefore, prevention (avoid feeding moldy sweetclover hay) is a better alternative to supplemental treatments.

B vitamins

Microbial synthesis of the B vitamins in the rumen is thought to satisfy the ruminant's requirements. Supplementation with B vitamins should not be necessary under normal production situations, especially if

cattle consume high-quality forage or are fed supplemental protein and/or energy that improves ruminal digestibility and subsequent microbial synthesis. Injections of B vitamins seem to be beneficial only to combat overt stress and disease.

Conclusions

Minerals and vitamins, whether essential or nonessential, can adversely affect an animal if included in the diet at excessively high levels. Feeding excessive amounts is costly in terms of dollars and animal performance. Having your forage sampled and analyzed for mineral content is the most practical way to assure that you are feeding an adequate, but not excessive, level.

Because of the availability of low-cost sources of vitamins A and E, consider regular supplementation of these vitamins to cattle fed low-quality forage. Supplementing other vitamins is not recommended, except under special circumstances.

It is in keeping with good husbandry and nutritional practice to maintain the intake of these important dietary constituents at required levels, but below the maximum tolerable levels. The benefits of such programs are seen in increased production efficiency. Aside from direct effects on reproduction, producers should benefit from cattle in better health.

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Worksheet 4.1

Practice Questions for Mineral Supplementation

1. A herd of cattle shows signs of copper (Cu) deficiency (scouring, light coat color), but when the forage they are grazing is analyzed, it shows 10 ppm Cu, which should be adequate. What do you suppose is causing the problem?
2. You wish to supplement a diet with copper, using copper sulfate as the source. Copper sulfate contains 39.81 percent Cu. How much copper sulfate must you add to supply 7 ppm Cu to a ton of feed?
1 ton = 908,000 grams (see example below)
3. Young calves are born with large swellings in the upper throat area. Can you relate this symptom to a micromineral deficiency? What is the common name for this deficiency disease?

Example for calculating amount of mineral compound to add to a desired mineral mix (and why you should consider a salt-mineral supplement that is formulated and mixed by a reputable dealer)\
You are losing some calves to white muscle disease and decide to supplement their diet with 0.1 part per million (ppm) of selenium (Se). You obtain some sodium selenite. How much should you add to the feed to get 0.1 ppm of selenium? Sodium selenite contains 45.65 percent Se.

When calculating small quantities, such as parts per million, it is easier to use small units (grams), rather than pounds. There are 454 grams in a pound, and 908,000 grams in a ton (2,000 pounds in a ton x 454 grams in a pound = 908,000 grams in a ton). A paper clip weighs about 1 gram.

1. You want 0.1 ppm Se in the 1 ton of feed ($0.1 \div 1,000,000$) x 908,000 grams per ton = 0.0908 gram Se per ton of feed.
2. Sodium selenite is 45.65 percent Se, so you need 2.19 grams of sodium selenite to get 1 gram of Se ($100 \div 45.65 = 2.19$).
3. You need 0.1989 (approximately 0.2) gram of sodium selenite to get 0.0908 gram of selenium per ton ($2.19 \times 0.0908 = 0.1989$).

This is a very small amount to disperse evenly in a ton of feed, so it's best to make a premix. Add 0.2 gram of sodium selenite to 100 pounds of feed, mix thoroughly, and then add the other 1,900 pounds and continuing mixing. It may be best to have a feed dealer do the mixing since dealers are familiar with premixes and have efficient mixing equipment.

Resources

You can find information for calculating supplement values from the following resources.

- *The Merck Veterinary Manual* provides brief descriptions and percentage composition of all mineral supplements.
- *The Mineral Nutrition of Livestock* is a good source of information on animal symptoms of various deficiencies and supplementary levels of minerals needed to correct them.
- *Nutrient Requirements of Beef Cattle*

Ration Balancing

5

David Bohnert and
David Chamberlain

Feed costs are a major component of the total operating expense for most beef operations. By formulating and feeding balanced rations, you can conserve feed dollars while allowing for the most efficient level of production. Ration balancing depends on having accurate and reliable nutrient analyses of feedstuffs in addition to knowing the animals' nutrient requirements. Terms commonly used in ration balancing include the following.

Feedstuff—An ingredient used in formulating a ration.

As-fed—Refers to the moisture and nutrient content of feedstuffs normally fed to animals. This value must be corrected to account for moisture content (determination of dry matter) when balancing rations.

Dry matter—The portion of feed that remains after all the water has been removed. It contains the nutrients other than water.

Nutrients—The chemical substances found in feedstuffs that can be used, and are necessary, for the maintenance, production, and health of animals. The chief classes of nutrients are carbohydrates, fats, proteins, minerals, vitamins, and water.

Ration—The amount of feed an animal receives in a 24-hour period.

Balanced ration—A ration that supplies nutrients in the proper amount and proportion for an animal's maintenance, growth, lactation, and/or gestation. Information necessary to compose a balanced ration includes the nutrient composition of feedstuffs and the animal's nutrient requirements.

Nutrient composition—The amount of specific nutrients contained in a ration or feedstuff. Normally expressed as a percentage of dry matter.

Nutrient requirement—The amount of a specific nutrient that is required to meet an animal's minimum need for maintenance, growth, reproduction, lactation, and work. Nutritional requirements depend on the type, size, and physiological status of the animal.

TDN—Total digestible nutrients. A term used to express energy in feeds.



There are several ways to balance daily nutrient intake with daily nutrient requirements. All of these methods rely primarily on mathematical computations, which can be carried out by hand or with computer software. However, in order to formulate a balanced ration, you first must know the animal's nutrient requirements (see Chapter 1) and the nutrient composition of the feedstuffs to be included in the ration (see Chapters 2 and 3).

Send samples of feedstuffs to a certified analytical laboratory to determine nutrient composition (see Chapter 3). If this is not possible, ask your local Extension office for average values. However, there is no substitute for a lab analysis of your particular feedstuff. Feedstuff nutrient composition can be greatly influenced by stage of maturity, harvesting, processing, storage conditions, etc.

Methods of ration balancing

As mentioned previously, you can balance rations by hand or by using specifically designed computer software. We will describe one of the more common and useful methods of hand balancing a diet (Pearson Square) and review the positive aspects of computer programs.

Pearson Square

Use of the Pearson Square is relatively easy. You can use this technique to determine the proportions of two feedstuffs that will yield a ration containing a desired nutrient concentration. It can be used only for two feed materials; however, one or both can be a mixture. An example is provided here.

In this example, a ration is developed for a 500-pound heifer calf having a desired gain of 1.5 pounds/day. Her daily requirements are as follows (from page 12, Table 1.1, Chapter 1):

12.1 pounds dry matter intake
10.3 percent crude protein
68.5 percent TDN

The feedstuffs to be used in developing the balanced ration are listed in Table 5.1. Step by step, the procedure is as follows.

1. Balance for TDN. Draw a square and place 68.5 (the desired TDN level) in the center (Figure 5.1).
2. At the upper left corner of the square, write "meadow hay = 50," and at the lower left corner write "ground barley = 75." These numbers represent the TDN percentage in each feedstuff.
3. Subtract diagonally, smaller from larger ($68.5 - 50 = 18.5$; $75 - 68.5 = 6.5$) and write the numbers on the right side of the square as shown in Figure 5.1.
4. Add the numbers on the right side of the square ($6.5 + 18.5 = 25$). These numbers indicate that a ration of 6.5 parts meadow hay and 18.5 parts ground barley will give a 68.5 percent TDN ration. This is a total of 25 parts.

Table 5.1. Feedstuffs used in Pearson Square example.

Feedstuff	Dry matter (%)	Total digestible nutrients (%)	Crude protein (%)
Meadow hay	92	50	6
Ground barley	88	75	11
Cottonseed meal	90	65	41

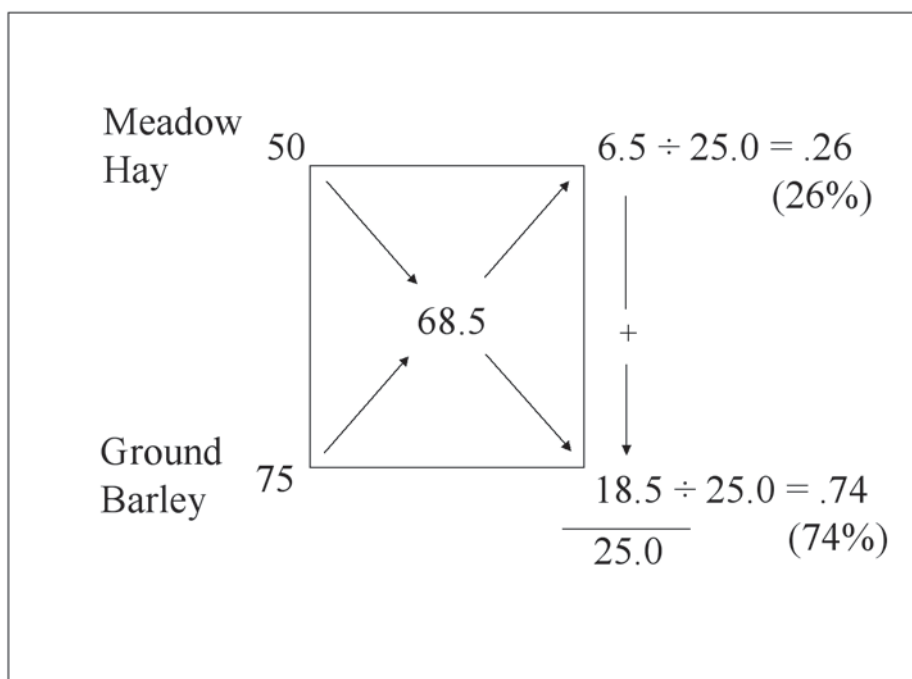


Figure 5.1—Balancing for TDN using a Pearson Square.

- Divide the meadow hay and ground barley parts by 25 to get the preliminary percentages of hay ($6.5 \div 25 = 26\%$) and barley ($18.5 \div 25 = 74\%$).
- Determine the crude protein concentration in the meadow hay and ground barley mixture. Multiply the percentage of each feedstuff in the mix by its crude protein content. Meadow hay is 26 percent of the mix and contains 6 percent crude protein. Ground barley is 74 percent of the mix and contains 11 percent crude protein. Add the results. Therefore, the crude protein concentration in the mix is:

Meadow hay	0.26×6	$= 1.56\%$
Ground barley	0.74×11	$= 8.14\%$
		<u>9.70%</u>
- Determine whether crude protein is adequate. The concentration in the meadow hay/ground barley mix is 9.7 percent. The heifer requires 10.3 percent crude protein. Therefore, the crude protein content needs to be increased by adding a protein supplement (cottonseed meal in this example).

8. Use the Pearson Square method again to balance for crude protein. Draw a square and put 10.3 in the center (Figure 5.2).
9. Write “meadow hay/ground barley mix = 9.7” in the upper left corner, and “cottonseed meal = 41” in the lower left corner; these numbers indicate the crude protein percentage in each feedstuff.
10. Subtract diagonally, smaller from larger ($10.3 - 9.7 = 0.6$; $41 - 10.3 = 30.7$), and write the numbers on the right side of the square.
11. Add the numbers on the right side of the square ($30.7 + 0.6 = 31.3$). These numbers indicate that a ration of 30.7 parts meadow hay/ground barley mix and 0.6 part cottonseed meal will give a 10.3 percent crude protein ration. This is a total of 31.3 parts.
12. Divide the meadow hay/ground barley mix and cottonseed meal parts by 31.3 to get the preliminary percentages of meadow hay/ground barley ($30.7 \div 31.3 = 98\%$) and cottonseed meal ($0.6 \div 31.3 = 2\%$).
13. Determine the pounds of dry matter that each feedstuff contributes to the total. Multiply pounds of dry matter required daily by the heifer (12.1) by the percentage for cottonseed meal (0.02, or 2 percent). Thus, the dry matter component made up by cottonseed meal is $12.1 \times 0.02 = 0.24$ lb.

Subtract this amount (0.24) from the total dry matter intake (12.1) to determine how much dry matter will come from the meadow hay/ground barley mix ($12.1 - 0.24 = 11.86$ lb). There should be 11.86 lb of meadow hay/ground barley on a dry matter basis.

To determine the amount of dry matter for meadow hay and ground barley, multiply 11.86 by the percentages of meadow hay and ground barley obtained in the first square (step 5): 26 percent meadow hay and 74 percent ground barley. $11.86 \times 0.26 = 3.08$ lb meadow hay and $11.86 \times 0.74 = 8.78$ lb ground barley.

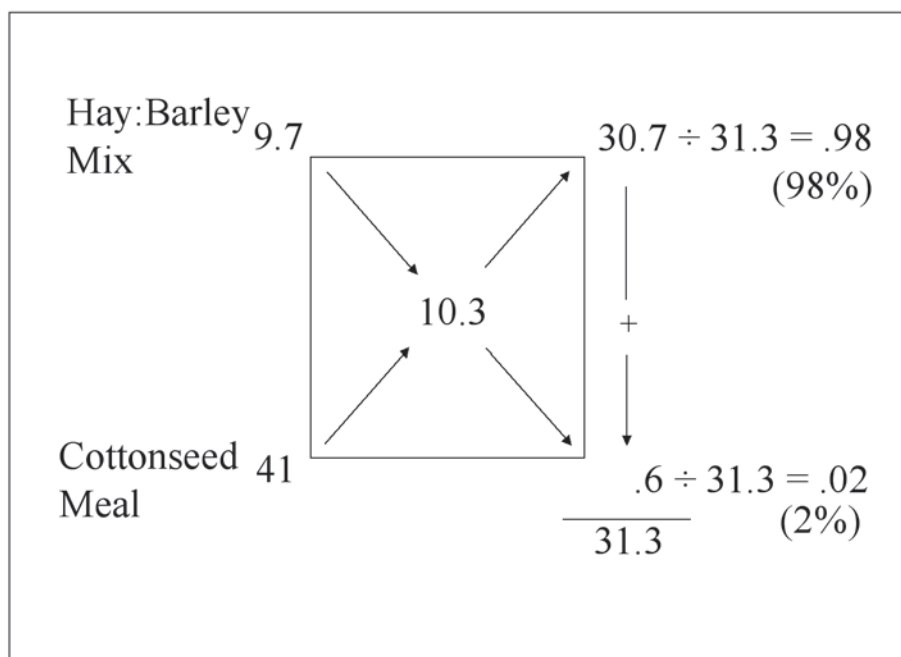


Figure 5.2—Balancing for crude protein using a Pearson Square.

14. Change each amount from a dry matter basis to an “as-fed” basis so that you know how much to feed. To do so, divide the pounds of dry matter for each feedstuff by the percentage of dry matter in each feed (see Table 5.1).

Meadow hay = $3.08 \text{ lb} \div 0.92$ (92% dry matter) = 3.35 lb

Ground barley = $8.78 \text{ lb} \div 0.88$ (88% dry matter) = 9.98 lb

Cottonseed meal = $0.24 \text{ lb} \div 0.90$ (90% dry matter) = 0.27 lb

A blank Pearson Square is included for your use as Worksheet 5.1.

Computer software

Computers provide immediate access to vast amounts of information concerning feedstuff nutrient content and animal nutrient requirements. As a result, computers and associated software can be powerful management tools to help lower feed costs and improve animal nutrition.

Computer programs designed for ration balancing include databases containing information on nutrient content and prices of hundreds of feedstuffs. Also, you can update databases with your current nutrient analyses and/or locally available feedstuffs. Most programs use management variables, such as class of beef cattle and physiological stage, to calculate nutrient requirements.

In addition, nutrition programs often provide an estimate of the quantity of feed consumed. Intake greatly influences the concentration of nutrients in a balanced ration, but it can be difficult to estimate. Intake is important because the total quantity of nutrients obtained by an animal in a given day depends on both the total intake and concentration of nutrients in the ration.

Ration balancing software also can allow you to store and change the price of individual feedstuffs. This enables you to obtain economic information about a particular ration in addition to nutrient analysis.

The ability to have feedstuff nutrient and price information available in a computer database is a major improvement over having to obtain information from tables and/or memory. Computers can evaluate numerous rations in the time it takes to formulate one ration by hand. In addition, you can evaluate and compare various feedstuffs to determine which are most cost-effective.

Exercise extreme caution when using computer programs to balance or evaluate rations and/or feedstuffs. Ration balancing software alone cannot determine a nutritional management program. Software is only one of many tools used in developing a complete nutritional program. There is no substitute for personal experience and common sense. Ration balancing programs can't determine whether a particular diet is prudent or practical. For example, nutritional software may suggest a diet consisting of 100 percent barley because it meets the animal's nutrient requirements. You need a knowledge of feeds and ruminant nutrition to realize that rations such as this are not practical.

Some commonly used computer programs from educational services include the following.

- SPARTAN, from Michigan State University (<http://www.msu.edu/user/ssl/index.htm>). Available for \$100 from the MSU Software Distribution Center, 517-353-6740.
- TAURUS, from University of California–Davis (<http://animalscience.ucdavis.edu/extension/software.htm>). Available for \$400 from the Department of Animal Science at the University of California Davis, 530-752-5886.
- AUTONRCAF, from Oklahoma State University (<http://www.ansi.okstate.edu/software/>). Available free of charge on the Web from the Department of Animal Sciences at Oklahoma State University.

Contact suppliers for updated cost of programs.

Important points to remember

Successful ration balancing and/or evaluation requires accurate information regarding feedstuff nutrient composition and beef cattle nutrient requirements. You need to know the dry matter percentage of feedstuffs and understand how to convert as-fed values to dry-matter values.

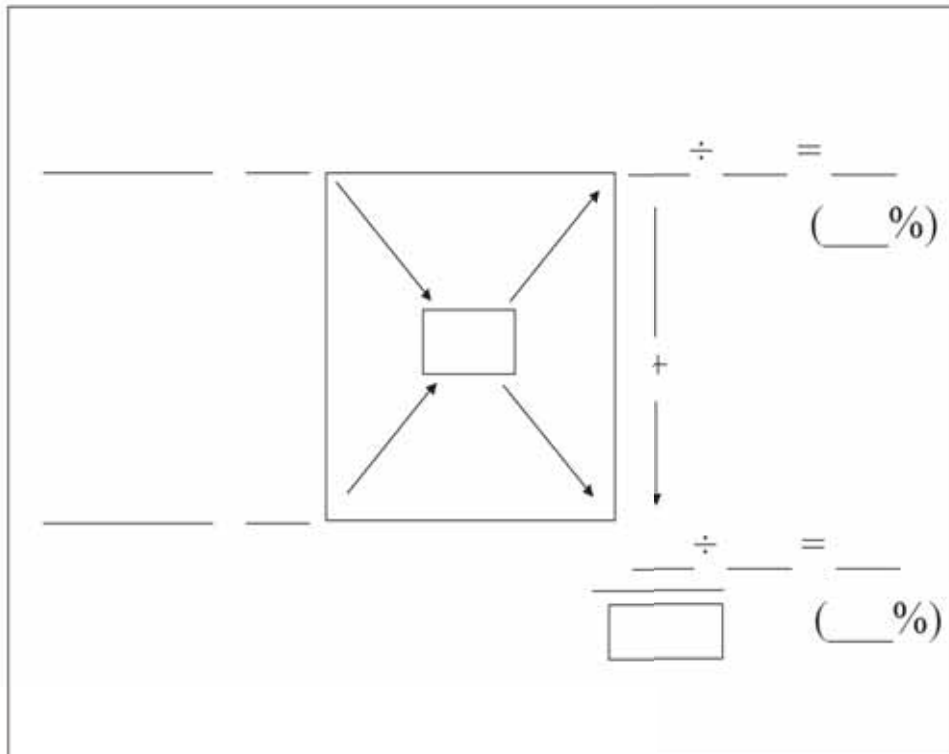
In addition, a fundamental understanding of mathematics will improve your ability to formulate rations and interpret hand- and computer-generated diets. Your ability to balance and evaluate rations is only as good as the information used to develop them.

Efficient use of feedstuffs means providing a diet that meets an animal's nutritional requirements. Overfeeding, underfeeding, and/or feeding nutritionally unbalanced diets are inefficient management practices that increase feed costs and reduce profitability.

Worksheet 5.1

Pearson Square Diagram for Balancing Rations

Note: The Pearson Square can be used to balance rations containing two feed ingredients. The feed ingredients can be single feedstuffs or mixtures of feedstuffs.



Supplements and Supplementation Strategies

Used to Keep Livestock Productive During the Winter Feeding Period

Having learned about nutrient requirements, feed value, and forage evaluation in previous chapters, you probably have discovered that the winter feed resources available on your operation don't always meet the requirements of your cows and heifers at every physiological state. Late-gestation and early-lactation cows have increased nutritional requirements. Bred heifers have these same requirements plus the elevated nutritional demands of body growth. These increased demands, combined with the lower quality feed that often is fed during the winter, commonly create a need for nutrient supplementation.

Determining how much and what kind of supplementation you need

There are three nutritional components that may need to be supplemented: protein, energy, and minerals. Supplements are feedstuffs that are added to the base forage to provide the nutrients required to support the desired level of production. The goal of supplementation is to supply the difference between the animal's requirements and the nutrients available through forage. Supplementation of protein and energy is discussed in this chapter. Chapter 4 discusses supplementation of minerals.

To develop a supplementation plan, begin by answering the following questions.

1. What is the physiological state of the animal?
2. What nutrients (and amounts of those nutrients) are required for the desired level of production? Keep in mind that cattle require quantities of nutrients, not percentages of nutrients. Percentages apply only when the animal's intake is the same as the predicted intake. It is best to break percentages down into quantities when evaluating rations and feeds (see Chapter 1).
3. What is the nutrient content of the forage available to you?
4. How much forage is available?

6

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After you answer these questions, you should have a pretty clear idea of what you need to supplement. A common mistake is to “blanket supplement,” or to buy a supplement that includes protein, minerals, and a little energy. These supplements are fine if they provide what you need, but often they cause producers to waste money supplementing something that isn’t deficient.

A worksheet to help you determine what supplement is best for you is included at the end of this chapter (Worksheet 6.1).

Protein supplementation

Protein is a crucial part of the diet that often is deficient in winter-feeding situations. Both the animal itself and ruminal microorganisms need protein; hence, protein deficiency can severely depress animal performance. Research shows that protein deficiency also can suppress appetite and limit intake. Limited intake leads to an increased potential for other deficiencies.

If you determine that your forage is deficient in protein, you will want to consider: (1) what kind of protein supplementation is available, (2) the nutritional value of those supplements, and (3) the costs associated with using them.

Nonprotein nitrogen vs. natural protein

An important consideration in choosing a protein supplement is whether the supplement is nonprotein nitrogen (NPN) or natural protein. Nonprotein nitrogen is a good source of nitrogen, but it does not contain preformed amino acids. Natural proteins, on the other hand, are made up of amino acids.

Ruminal microorganisms can use NPN effectively as a nitrogen source in the production of microbial protein; however, NPN is not utilized as efficiently as natural protein. While utilization of NPN varies greatly (Table 6.1), you might expect 50 to 60 percent efficiency when feeding NPN with high-forage diets. The actual utilization depends on the source and the base diet of the animal.

Consider the above factors when pricing and purchasing a protein supplement. In addition, be aware that an NPN protein source should not exceed one-third of the animal’s total protein requirements.

Supplement options

Oilseed by-products such as cottonseed meal and soybean meal are common sources of protein supplements. These sources of natural protein contain more than 40 percent crude protein (CP) and a high density of energy. They work especially well when energy supplementation also is needed. The drawback to oilseed by-products is that they are relatively expensive.

Table 6.1. Comparison of the utilization rate of non-protein nitrogen (NPN) in liquid and dry forms when supplementing various feeds.

Study	NPN efficiency (as a % of natural protein)
Oklahoma—dry range	
Dry urea (6 studies)	92
Liquid urea (3 studies)	84
Maryland—corn stalks	
Liquid supplement	55
Southern Illinois—corn stalks	
Liquid supplement	14
Oklahoma—dry range	
Dry urea	37
Nebraska—dry range	
50% NPN/50% natural protein	74

Source: National Research Council. 1976. *Urea and Other Nonprotein Nitrogen Compounds in Animal Nutrition*. National Academies Press, Washington, DC.

Alfalfa or alfalfa cubes are another common protein supplement, especially in areas where they are readily available. Alfalfa is a good source of natural protein that provides the same benefits as other protein supplements when fed on an equal crude protein basis.

One potential drawback to alfalfa is that it doesn't have the high caloric density that some oilseed meals do. Thus, it may not be as effective if an energy source is also needed. One advantage is that it does not require special equipment or facilities to deliver it to the herd.

Commonly used sources of nonprotein nitrogen are urea and biuret. These products may be available as a liquid (typically with molasses as the carrier) or in a dry form (usually with a grain as the carrier). While these supplements may seem to be the most economical, keep in mind that NPN is not used as efficiently as natural protein.

Finally, high-quality grass hays may be a potential supplement for low-quality forages. They typically have a lower crude protein content

Table 6.2. Chemical composition of some potential feed ingredients used as sources of supplemental protein for low-quality forages.^a

Protein source	CP (%)	% of CP		TDN (%)	ME (Mcal/kg)
		DIP	UIP		
Brewers grain	26	41	49	7	2.53
Canola meal	41	68	32	69	2.49
Coconut meal	22	62	38	64	2.31
Corn gluten meal	47	38	62	84	3.04
Cottonseed meal					
Mech	44	57	43	78	2.82
Sol-41% CP	46	57	43	75	2.71
Sol-43% CP	49	57	43	75	2.71
Distillers grain	30	45	55	90	3.25
Soybean meal-44	53	80	20	84	3.04
Soybean meal-49	50	65	35	87	3.15
Soybean, whole	40	65	35	94	3.40
Sunflower meal	26	38	62	65	2.35
Urea	291	100	0	0	0
Alfalfa hay					
Vegetative	22	86	14	64	2.31
Early bloom	20	84	16	62	2.24
Midbloom	17	82	18	60	2.17
Full bloom	13	77	23	56	2.02
Wheat middlings	18	77	23	83	3.00
Tall fescue hay	9	67	33	56	2.02
Meadow hay	13	77	23	60	2.17

^aCP = crude protein; DIP = degradable intake protein; UIP = undegradable intake protein; TDN = total digestible nutrients; ME = metabolizable energy

Source: Torell, R. and J. Ballietto. 1994. Pricing Protein and Energy Supplements. CL313 in *Cow-Calf Management Guide and Producer's Library*. Agricultural Communications, College of Agricultural and Life Sciences, University of Idaho, Moscow, ID.

than do protein supplements, but, depending on the base diet and their availability, they might be a viable option.

Cost analysis

Once you have determined how much protein you need to supplement, and what supplements are available, you need to do a cost analysis. It is essential that you compare the cost per ton of utilizable protein, not the cost per ton of supplement. Looking at the cost per ton of supplement can be very misleading.

Keep in mind that you are buying a specific nutrient, either protein or energy. Mineral requirements typically can be met with a less expensive free choice salt/mineral program. You do not want to pay for something you don't need!

The following steps will guide you through the cost analysis process.

1. Determine the total pounds of nutrients in 1 ton of supplement. To do so, multiply 2,000 pounds by the percentage of the nutrient in the supplement.

Example:

$2,000 \text{ lb} \times 0.17 \text{ (17\% CP alfalfa hay)} = 340 \text{ lb of actual protein}$

2. Determine the cost per pound of nutrients. Divide the per-ton price of feed by the pounds of actual nutrient contained in a ton.

Example:

$\$100/\text{ton} \div 340 \text{ lb CP} = \$0.29/\text{lb of protein}$

Therefore, the cost of protein in the alfalfa hay listed above is \$0.29/pound (\$580/ton). Use this value in your cost comparison.

If a protein supplement consists partially of NPN, you need to account for the lower utilization of NPN when figuring the cost per ton of protein. You want to figure the cost per ton of utilizable protein. Following is an example of how to make this adjustment.

1. Determine the amount of utilizable protein in the supplement.
 $\% \text{ NPN} \times \text{utilization rate} = \% \text{ utilizable protein (from NPN)}$

Example—liquid supplement:

Crude protein, not less than 25%

Includes not more than 18.5% equivalent protein from nonprotein nitrogen

$25\% - 18.5\% = 6.5\% \text{ natural protein}$

$18.5\% \times 0.6 \text{ utilization} = 11.1\% \text{ utilizable protein (from NPN)}$

$6.5\% \text{ natural protein} + 11.1\% \text{ utilizable NPN} = 17.6\% \text{ total utilizable protein}$

When calculating the cost per pound of protein in this supplement, you would use 17.6 percent as the protein content.

Energy supplementation

Energy is another nutrient that is commonly supplemented into low-quality forage diets during critical periods of the biological cycle. Energy supplementation leads to problems not encountered when supplementing for protein. High-energy feeds have been shown to decrease forage utilization. In other words, cattle tend to use energy supplements as a replacement or substitute for lower-quality forage instead of as a

supplement. The degree to which they use energy as a replacement depends on the forage quality, the amount of energy in the supplement, the protein in the diet, and the energy source.

Supplement options

There are two general types of energy supplements: starch-based supplements and fermentable fiber sources. Starch-based supplements include grains such as corn, sorghum-grain, barley, oats, and wheat. Sources of fermentable fiber include soybean hulls, wheat middlings, beet pulp, and corn gluten feed.

Starch-based supplements can depress forage intake and digestibility by increasing the proportion of starch-digesting bacteria and decreasing the number of fiber-digesting bacteria within the rumen. While results vary, supplementation of grain at 0.4 percent of body weight or less typically does not depress the intake and digestibility of low-quality forage by beef cattle. At supplementation levels greater than 0.8 percent of body weight, forage intake and digestibility can be greatly depressed.

Supplementation with fermentable fiber sources generally does not decrease forage intake and/or digestibility as much as grain-based supplements. Fermentable fiber supplemented at rates of 0.2 to 0.8 percent of body weight has yielded favorable results.

Cost analysis

As in the case of protein, cost analysis of energy is an important step in determining which energy source is best for your operation. A number of different measures of energy are used in evaluating feedstuffs (see Chapter 3, page 30). Regardless of the measure you use, it is important that you be consistent. Cost analysis of energy follows the same steps as cost analysis of protein.



Frequency of supplementation

The frequency with which you supplement depends on whether you are supplementing protein or energy. Energy should be supplemented on a daily basis, as infrequent and irregular supplementation of energy can cause digestive problems.

Frequency of protein supplementation is more flexible. Ruminants have the ability to recycle absorbed nitrogen back to the rumen. Therefore, research has shown that infrequent protein supplementation is acceptable and safe as long as it results in the same total amount of supplement. Thus, you can save labor and time by supplementing every 2 to 7 days rather than every day.

Additionally, studies show less variation in weight change and supplement intake within a herd when supplementation is less frequent. There probably is less competition for the supplement when greater quantities are provided in a single feeding.

When practicing infrequent supplementation, be aware that you still must provide the same total amount of supplement, just in a larger quantity less frequently. Infrequent supplementation works best with natural protein supplements.

In the case of either protein or energy supplementation, be sure to provide adequate space for cattle to access the supplement. This will help prevent excessive variation in consumption within the herd.

Supplement placement

In many parts of the Northwest, stockpiled dormant forages are used as a winter feed resource. In this case, it is advantageous to feed supplements in a location that encourages more efficient use of the stockpiled forage. You might move the supplement site to increase use of the forage in specific areas and discourage the tendency of animals to remain at a supplement site where feed is not available. While the supplement may be necessary to meet the animals' nutrient requirements, it also is a potential tool for achieving management goals.

Other considerations

As in any management situation, the selection of protein and energy supplements is not always straightforward. Factors beyond price must be considered. The convenience, or feedability, of the product is important. Labor and time are valuable. It might be worth extra cost to drop off a feed tub or block once a week instead of having to load and feed one or several loads of alfalfa once or twice a week. Product availability also is critical.

Finally, consider what other nutrients you must supplement to balance the ration. If your ration is deficient in protein, energy, and minerals, and you can find one product that satisfies all of the deficiencies, it might be worth the extra cost. While vitamins and minerals can be supplemented in a free-choice salt/mineral supplement, the convenience of an all-in-one product may be worth the extra dollars. In this respect, each operation differs. It is up to you as the manager to use cost analysis and consider other factors to make the best choice for your operation.

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Worksheet 6.1

Determining Which Supplement Is Best for You

1. What supplements are available and what is their cost?

(Figure cost based on ton of nutrient, not on ton of supplement. Examples are included below to help you work through the cost comparison.)

Example supplements—nutrient content and cost^a

Supplement	Energy content (% TDN)	Protein content (% CP)	NPN (%)	Cost/ton of supplement	Cost/ ton of nutrient
Alfalfa	60	19	0	\$100	\$526 (CP)
Liquid		25	18	\$280	\$1,573 (CP)
Barley	73	11	0	\$200	\$273 (TDN)
Tub		18	18	\$250	\$1,389 (CP)

^aTDN = total digestible nutrients; CP = crude protein; NPN = nonprotein nitrogen

Alfalfa

0.19 (% crude protein) x 2,000 lb = 380 lb of protein/ton of alfalfa

\$100/ton of alfalfa ÷ 380 lb of protein = \$0.26/lb of protein x 2,000 lb/ton = \$526.32/ton of protein

Liquid

25% – 18% = 7% natural protein

18% x 0.60 (NPN utilization) = 10.8% utilizable protein from NPN

7% natural protein + 10.8% utilizable NPN = 17.8% total utilizable protein

0.178 (% crude protein) x 2,000 lb = 356 lb of protein/ton of liquid

\$280/ton of liquid ÷ 356 lb of protein = \$0.79/lb of protein x 2,000 lb/ton = \$1,573/ton of protein

Barley

0.73 (% TDN) x 2,000 lb = 1,460 lb of protein/ton of barley

\$200/ton of barley ÷ 1,460 lb of TDN = \$0.14/lb of TDN x 2,000 lb/ton = \$273/ton of TDN

Protein tub

0.18 (% CP) x 2,000 lb = 360 lb of protein/ton of tub

\$250/ton of tub ÷ 360 lb of protein = \$0.69/lb of protein x 2,000 lb/ton = \$1,389/ton of protein

continued on next page

Worksheet 6.1 (continued)

Determining Which Supplement Is Best for You

2. Other factors that are very important when choosing a supplement are: (1) how much of the supplement cows will need to eat to balance the ration, and (2) whether it is physically realistic for them to eat that amount of that particular supplement.

Review the amount of protein (pounds) needed to supplement the forage you are feeding (see Chapter 5) and enter below:

Pounds of supplemental protein required to meet the cows' protein requirement _____ lb

What type of protein supplement do you prefer to use (based on convenience and price/nutrient needed—see above)? _____

What is the percentage of protein in the supplement? _____

How much will each animal need to consume to meet its requirement? _____

Example

Your mature cows are grazing on dry winter range. The grass they are eating is about 4.5% CP. Daily consumption is about 21 lb DM, and their protein requirement is 7.8%, or 1.6 lb of CP. You are planning to supplement with a baked protein tub containing 18% CP (all natural).

Pounds of supplemental protein required to meet the cows' protein requirement: 0.655 lb
(21 lb DM x 4.5% CP = 0.945 and 1.6 lb CP – 0.945 = 0.655 lb)

What type of protein supplement do you prefer to use (based on convenience and price/nutrient needed—see above)? Protein tub

What is the percentage of protein in the supplement? 18%

How much will each animal need to consume to meet its requirement? 3.64 lb
(See below for calculations.)

Calculations

0.18 (protein content of baked tub) x _____ (amount of tub consumed) = 0.655 lb (protein needed to balance ration)

$0.655 \div 0.18 = 3.64$ lb of supplement consumed to balance ration

Many of the protein tubs suggest consumption levels of 1.5 to 2 pounds/animal. If each cow needs to eat 3.5 pounds of supplement from the tub, you might look for a different source of protein, where consumption likely would be greater.

Body Condition

The Beef Cow's Energy Gauge

7

Bruce Nisley and
Cory Parsons

Monitoring body condition is a management tool for evaluating the nutritional status and energy reserves of beef cattle. Nutritional requirements of cattle must be met in order to attain an optimal level of performance and efficient use of feed resources. Proper feeding strategies and ration balancing techniques have been discussed earlier in this publication. By following these techniques and strategies, you can provide the proper nutrients required to maintain or improve body condition, or nutrient reserves.

Cow body condition scoring (BCS) is a method of categorizing breeding animals by their degree of body nutrient reserves. This concept is not new. BCS simply puts a quantitative score on a procedure many cow/calf producers have followed for years to determine the body fat reserves of their herds.

BCS allows you to be more exact in describing your cows. It also provides a standardized tool for the beef industry to use when monitoring the energy reserves of the cow herd. Body condition scores range from 1 (severely emaciated) to 9 (very obese); see Table 7.1.

Monitoring body condition is an effective way to evaluate the herd's nutritional status. It also allows you to monitor the effectiveness of your feeding strategies and ration balancing. By assigning body condition scores, you will be better able to sort cows according to their nutritional needs, thereby improving the efficiency of your nutrition programs.

Table 7.1. Body condition score.

1	↑	Severely emaciated. Bone structure of shoulders, ribs, back, hooks, and pins is sharp to the touch and easily visible. Little evidence of fat deposits or muscling.
2		Emaciated. Little evidence of fat deposition, but some muscling in the hindquarters. The backbone feels sharp to the touch.
3	Caution	Very thin, no fat on ribs or brisket and some muscle still visible. Backbone easily visible.
4		Thin, with ribs easily visible but shoulders and hindquarters still show fair muscling. Backbone visible.
5		Moderate to thin. Last two or three ribs cannot be seen unless animal has been shrunk (held off feed and water for 12 to 18 hours). Little evidence of fat in brisket, over ribs, or around tail head.
6		Good, smooth appearance throughout. Some fat deposits in brisket and over tail head. Ribs covered and back appears rounded.
7	Caution	Very good flesh, brisket full. Fat cover is thick and spongy and patchiness is likely. Ribs very smooth.
8		Obese, back very square, brisket distended, heavy fat pockets around tail head. Square appearance.
9	↓	Rarely observed. Very obese. Animal's mobility may be impaired by excessive fat.

Source: Spitzer, J.C. 1986. *Influences of Nutrition on Reproduction in Beef Cattle*. W.B. Saunders Co., Philadelphia.

There is a strong link between body condition and weight change. As BCS increases or decreases, corresponding weight changes occur. Body condition of beef cows affects herd performance and profitability.

How to determine BCS

Body condition scoring can be done by physically palpating individual cows, visually appraising each cow, or scoring a representative sample of the herd. Palpation likely is more accurate, but it is not always practical due to large numbers of cattle or inadequate facilities.

Once properly trained, you can inspect cattle visually and accurately assign a BCS. When visually inspecting cattle and assigning BCS, it is important to know where to look on the cow and what to look for (Figure 7.1). Be careful to adjust for fill and hair coat, particularly in cold climates.

Always assign a BCS when animals have the same amount of rumen fill as the last time a BCS was assigned. Some producers prefer to assign BCS when the animals are shrunk (held off feed and water for 12 to 18 hours). This removes the rumen fill and makes it easier to see the ribs; thin cows with full rumens may seem to have more body condition than they actually do.

It may be advantageous to compare BCS to live weight. It has been reported that for every 1 BCS you can expect a 75- to 95-pound change in weight, depending on the frame size of the cow.

The ideal BCS

The BCS of beef cows varies depending on breed, environmental conditions, physiological conditions, and time of year. It is recommended that BCS of all cattle be taken and recorded at weaning and again at calving time. By doing so, you can sort thin cows from more fleshy ones and provide the thin cows with a nutritional supplement to increase their body condition prior to calving and rebreeding.

It is recommended that mature cows enter the calving season at a BCS of at least 5 but not more than 7. First-calf, 2-year-old heifers should

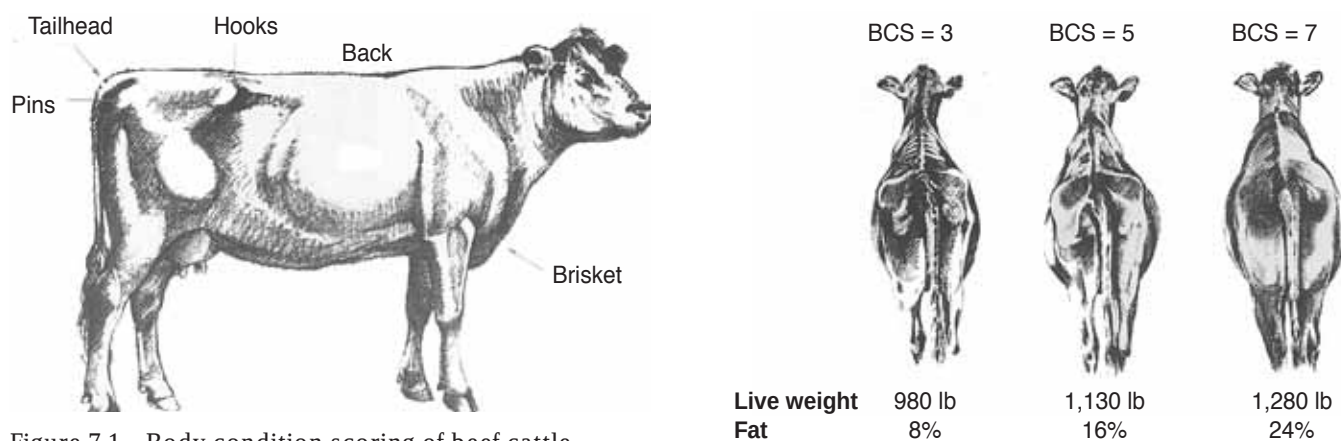


Figure 7.1—Body condition scoring of beef cattle.

enter the calving season at a BCS of at least 6 but not more than 7. A higher BCS is recommended for first-calf heifers because they are still growing. Consequently, they require more nutrients for continued growth, milk production, and repair of their reproductive tract for rebreeding. It also is recommended that you calve first-calf heifers 30 days prior to the main herd. This gives heifers an additional 30 days to rebreed and stay in the herd.

The consequences of thin cows

The cow is an amazing animal, with a nearly unequaled ability to survive on poor-quality forages. Yet even with her incredible ability to survive, she must prioritize nutrient allocation. When a cow doesn't have enough nutrients in her diet to meet all her needs, her body prioritizes how the limited nutrients will be used. The priorities are as follows (highest priority first):

1. Maintenance
2. Lactation
3. Growth in young cows
4. Reproduction

Thus, when a cow has inadequate feed and lacks body reserves, reproduction is the first thing to suffer and the last to recover. Reproductive losses can be devastating to ranch profitability.

Information collected from Texas producers involved in the Standardized Performance Analysis program (1991–1999) showed that the key measure of reproductive and productive performance was pounds of calf weaned per exposed female. This number is a function of both conception rate and days from calving to conception (postpartum interval). Cows that calve early in the calving season wean heavier calves. Thus, it is important not only to get cattle bred, but also to get them bred early in the breeding season.

Numerous researchers have shown that body condition has a significant impact on pregnancy rate and postpartum interval. It has been reported that only 50 percent of cows having a BCS of 4 were pregnant after a 90-day breeding season, while 90 percent of those with a BCS of 7 were settled in the same period (Figure 7.2). Others found a difference of 58 days in calving to conception between cows with condition scores 3 and 7 (Figure 7.2).

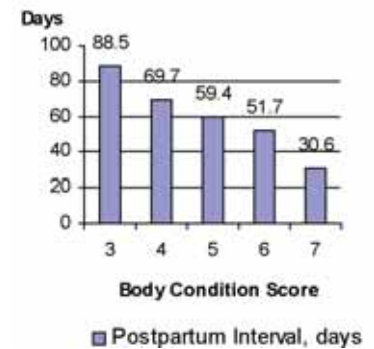
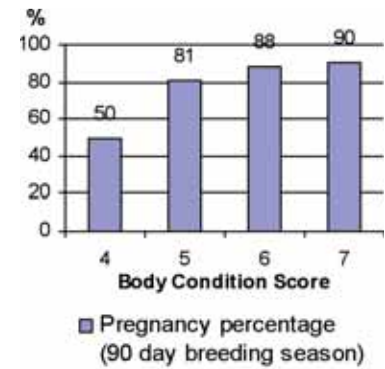


Figure 7.2—Effect of body condition score on reproductive performance. As BCS increases, the postpartum interval decreases and the pregnancy percentage increases.

Table 7.2. The relationship of body condition score to beef cow performance.

BCS	Pregnancy rate (%)	Calving interval (days)	Calf average daily gain (lb)
3	43	414	1.6
4	61	381	1.75
5	86	364	1.85
6	93	364	1.85

Low BCS has significant impacts on calf health and growth, and hence on profits. Researchers have reported an increased calf death loss of 5 percent for thin cows (92 percent survival in thin cows compared to 97 percent for cows in good condition). Furthermore, studies have shown reduced gains of 0.25 pound per day in calves from cows with a BCS of 3 as compared to those from cows with a BCS of 5 or greater (Table 7.2).

The cost of poor condition

We have discussed the production loss associated with underconditioned cows, but it is more powerful to look at economic losses and costs. Using an example herd of 100 cows, we have calculated the cost of mismanagement in Table 7.3. In this example, we compare BCS 3 to 5.

This example is not all-inclusive, but it helps to show some potential costs of thin cows. You would not expect to see an entire herd in such poor condition. However, it is not uncommon to see part of a herd as thin as BCS 3. Often, the thin cows are old cows or 2-year-olds weaning their first calf. Thus, in reality you would (hopefully) not lose \$141 per exposed cow, but possibly a percentage of this amount.

How and when to improve body condition score

There is no doubt that thin cows have lower production and produce less income. Thus, you must adopt the most economically efficient management. The ideal is to optimize production at minimum expense. Thus, you need to put condition on the cows at the most efficient times and minimize condition loss when economically viable.

Table 7.3. Economic comparison of BCS 3 to BCS 5 cow herds.

	BCS 3 herd (100 bred cows)	BCS 5 herd (100 bred cows)
Cows exposed to produce 100 bred cows	132 cows exposed 76% conception	106 cows exposed 94% conception
Calf survival	92 head of calves weaned (92%)	97 head of calves weaned (97%)
Age at weaning, adjusted for postpartum interval (PPI)	Calves 180 days at weaning (88.5 days PPI)	Calves 210 days at weaning (59.4 days PPI)
Calf weaning weight (using 85-lb average birth weight and adjusted weaning age)	373 lb Gain 1.6 lb/day	474 lb Gain 1.85 lb/day
Pounds of weaned calf per herd	34,316 lb total	45,978 lb total
Value of calves at weaning	\$105/cwt = \$36,032	\$95/cwt = \$43,679
Income per cow calved	\$360	\$437
Income per cow exposed	\$273	\$412

First, you must decide which cattle really need more condition. If cattle are already in good flesh, you don't want to expend additional resources. However, you may have some cows with BCS ranging from 3 to 7.

There is no value in overfeeding already well-conditioned cows, so sorting into feeding groups likely is the best option. Three groups often are recommended, but in most operations two groups are more practical.

If the herd is sorted into two groups, the first likely would include thin mature cows with BCS of 4 or less and 3-year-old and younger cows with BCS of 5 or less. Feed this group to gain condition. Feed the remainder of the herd (those with an adequate BCS) a maintenance ration.

To improve a cow's BCS by 1 point, she will need to gain 75 to 95 pounds. To achieve this gain, you must be aware of what is happening at each phase of the cow production cycle (Table 7.4.), as well as what nutritional and feed resources are available. For example, in a spring calving herd, calves are weaned in late fall, when the fetus in the pregnant cow is developing slowly. From a production sense, this is a practical time to consider increasing body condition. Yet, it may be a time of limited-quality feed.

Evaluate what will work best to maintain and add condition to your herd. It is important to estimate the number of pounds to be gained in order to reach the desired condition score and to formulate the ration accordingly. Balancing rations is discussed in Chapter 5.

There are no simple management solutions to optimize production. However, by making good decisions, you can improve your herd performance and pay dividends for the time spent making decisions.



Table 7.4. Estimated weight gain from weaning to calving to achieve desired BCS.

Current BCS	Desired BCS	Approximate amount of body weight gain
1	5	Needs to gain more than 350 lb Economics are questionable
2	5	Needs to gain 300 to 350 lb Economics are questionable
3	5	Needs to gain 200 to 300 lb
4	5	Needs to gain 150 to 200 lb
5	5–7	Needs to gain weight of fetus (100 lb)
6	5–7	Needs to gain weight of fetus (100 lb)
7	5–7	No weight gain needed
8	5–7	Probably can lose from 50 to 150 lb (during midpregnancy)
9	5–7	Probably can lose from 100 to 200 lb (during midpregnancy)

Worksheet 7.1

Application of Body Condition Scoring

1. What would you estimate your cows' BCS to be at calving?

1 2 3 4 5 6 7 8 9

2. What would you estimate your cows' BCS to be at weaning?

1 2 3 4 5 6 7 8 9

3. Approximately how many pounds on average (if needed) would each cow need to gain to reach "ideal" condition? _____ (1 BCS = 75 to 95 pounds)

4. When would be the best time to improve the BCS of underconditioned cows in your herd?

5. Assume the difference in postpartum interval between BCS 4 and BCS 5 cattle is about 10 days and your herd averages BCS 4. How many pounds might you add at weaning by having calves born 10 days earlier? Calves usually gain from 1.6 to 2.2 pounds per day.

Estimated gain (in pounds per day) _____ x 10 days = _____ pounds per head
 x _____ number of calves sold at weaning = _____ pounds of lost opportunity at weaning

It is important to remember that weaning weight is only one area of lost performance for thin cows.

Heifer Nutrition and Development

8

Peter Schreder

Adequate heifer development is critical to the long-term sustainability of a productive cow herd. Replacement heifers are the foundation on which your herd is built. Whether heifers are purchased or self-raised, they represent your long-term herd productivity.

The improved profitability resulting from calving heifers for the first time as 2-year-olds rather than as 3-year-olds has been long proven. Also, research indicates that well-developed heifers that calve early in their first calving season continue to calve early in subsequent calving seasons. Consequently, they wean heavier calves throughout their lifetimes, compared to heifers that calve later in their first calving season.

Although aspects such as breed characteristics and genetics are important, meeting animals' nutritional requirements is a key part of developing productive heifers. The nutritional program for developing heifers should not require expensive purchased feeds. Target gains often can be met with little energy and protein supplementation.

Divide the heifer nutritional program into stages: weaning, weaning to breeding, breeding to first calving, first calving to second breeding, and the second postcalving period. Each stage is discussed below.

Weaning

Weaning is a time of stress, and it is a time when heifer calves' development can be retarded. A good weaning program typically lasts 4 weeks and should overcome the weight loss and nutritional stress of the weaning process. Prior to weaning, contact your veterinarian to set up a vaccination and preconditioning schedule.

A practice that is proving to be effective is fence line weaning. In a recent study, researchers weaned 248 calves and placed them in two feed yard pens. The dams of the calves in one pen were allowed to stay in the area just outside the pen. The other dams were removed from the area. After 10 days, there was no significant difference in weight gain, immune function, or physiological indicators of stress. But there was much less fence walking and bawling in calves weaned next to their dams. Fence-line weaning seemed to be a calmer process.



When weaning, get the calves started on high-quality hay and a palatable grain mix, and provide the recommended level of trace minerals (see Chapter 4, Table 4.2). The grain mix should contain a protein supplement (soybean meal, cottonseed meal, etc.) and some molasses to increase palatability.

It is important to provide a palatable, high-quality grain mix because dry matter intake normally is depressed during weaning and, consequently, nutrient intake is decreased. The grain mix should allow the heifers to obtain a significant amount of their required nutrients from a small amount of supplement.

You can get calves to the feed bunk quickly by feeding a familiar ingredient (such as high-quality hay or alfalfa). You might begin by providing high-quality hay in the feed troughs, in addition to about 1 pound/head/day of ground barley or corn oats barley (COB) and 1 pound/head/day of soybean meal.

Once the animals have gotten used to the grain mix (usually 5 to 7 days), you can increase the quantity to 2 pounds/head/day grain and 2 pounds/head/day soybean meal. After about 4 weeks (to monitor for health disorders and allow adequate time for adaptation to life without “mom”), you can turn the calves out on high-quality pasture and initiate the growing program.

Postweaning

The postweaning period lasts from the end of the weaning phase to the time the heifers are bred. The growth rate during this phase should allow heifers to reach their target weight for breeding, or approximately 60 to 65 percent of mature weight. At this weight, most heifers reach puberty before the breeding season. Since the pubertal estrus is less fertile than later cycles, it is wise to plan for heifers to reach their target weight at least 21 days prior to the breeding season.

If the target weight is not reached, breeding will be delayed. Conversely, with excessive growth and condition, the developing udder fills with fat, retarding mammary development and resulting in poor milking ability.

The daily gain of heifers typically needs to be about 1.25 pounds during this stage. You can accelerate the growth rate so that heifers lagging in development can catch up, but the gain should not exceed 2 pounds/day.

Adequate growth often can be achieved with high-quality pasture alone. During wintering in western Oregon, grazing on small grain, ryegrass, or fescue pastures can result in adequate gain.

Average-quality hay or pasture rations require supplementation, as determined by a forage analysis. The supplement may need to provide protein, energy, or both. A mineral mix containing an ionophore can stimulate performance if forage quality is marginal compared to the heifer’s requirements. Chapter 1 discusses the protein, energy, and mineral requirements of heifers at different stages of development.

Worksheet 8.1 (at the end of this chapter) will help you calculate required heifer gains.

Breeding to calving

It is recommended that first-calf heifers weigh 75 to 85 percent of their mature weight at first calving. Heifers need to gain approximately 1 pound/day from the time they are bred until calving. In western Oregon, this gain usually is achieved on pasture alone.

While it might seem easy to achieve this performance goal, do not turn heifers out to pasture and ignore them before calving. Take steps to assure adequate body condition of heifers precalving, and pay extra attention to mineral and vitamin requirements during the last trimester.

Heifers should have a body condition score of at least 6 (see Chapter 7) entering the calving season. If heifers are in poor condition, place them on a higher level of nutrition. It is difficult to improve their condition as they approach calving, and it is especially difficult after calving. Improving heifer body condition score will increase the quantity and quality of colostrum, decrease the time from calving to estrus, decrease death loss in calves, and increase calf vigor.



Calving to rebreeding

If a heifer that calves in good body condition loses condition rapidly after calving, her reproductive rate may be reduced. Heavy-milking heifers are especially prone to rapid weight loss, resulting in delayed cycling.

Plan to provide the lactating first-calf heifer the highest quality hay or pasture available, and be prepared to provide a grain supplement sufficient to maintain body condition above 5. Provide a good, palatable mineral supplement balanced specifically for early lactation (see Chapter 4).

It is important to understand a cow's order, or hierarchy, of nutrient allocation. Nutrients are utilized in the following order: (1) maintenance, (2) lactation, (3) growth, and (4) reproduction (see Chapter 7). It is easy to see that a cow in a negative energy and protein state will experience decreased reproduction and conception. Reproduction is a luxury (not a necessity) for the heifer; however, it is essential for producer profitability.

Second calving

Second-calf cows often are the most difficult to get bred; many producers notice the highest rate of open cows in this age bracket. When the young cow is preparing for her second calf, she should be about 90 percent of her mature body weight. She should maintain a body condition of 5 or better after calving.

This cow is still growing, especially in the case of later maturing breeds, so she needs a slightly higher level of nutrition than do mature cows. In general, however, if milk production is moderate and forage

quality is high, 3-year-old cows can be managed with mature cows. Some producers find it easier to group 3-year-olds with 2-year-olds, but in either case it is difficult to target nutrition specifically to this group.

Summary

The development program for heifers is divided into stages. Providing heifers with nutrition to meet the target gains and weights indicated will result in a high level of fertility and calf health. Providing heifers with a good start also improves the overall profitability of the beef operation.

Many producers develop their own heifers. It is important to examine the possibility of purchasing 2-year-old replacement heifers from a producer who specializes in developing quality heifers. Worksheet 8.2 will help you compare the cost of raising versus purchasing replacement heifers.

Worksheet 8.1

Calculating Required Heifer Gains

	Example	Your herd	Notes
Mature cow size	1,100 lb		
Target weight at breeding (65% of mature weight)	715 lb		$1,100 \times 0.65 = 715$
Current weight	450 lb		
Total gain needed	265 lb		$715 - 450 = 265$
Current date	October 15		
Start of breeding season	June 1		
Length of feeding period	225 days		

Worksheet 8.2—Sample Change in Net Income if Replacement Heifer Is Purchased

Positive effects

\$ per head

Added returns

1. Net returns from sale of raised heifer calf (535 lb x \$0.80/lb)\$428.00
2. Interest on net returns from heifer calf sale
\$428 [line 1] x 0.11 (interest rate) x 15 months^a ÷ 12\$58.85
3. Total added returns [line 1 + line 2]\$486.85

Reduced cost^b

4. Value of hay fed to raised heifer calf (1.82 tons x \$75/ton)\$136.50
5. Value of pasture grazed by raised heifer calf\$55.14
6. Value of salt and minerals for raised heifer calf\$15.97
7. Other feed costs for raised heifer calf\$11.59
8. Veterinary and medicine expenses for raised heifer calf\$5.00
9. Value of labor and management for raised heifer calf (8 hours x \$10/hr)\$80.00
10. Raised heifer calf's share of bull cost
\$840 (annual bull cost) ÷ 25 females per bull x 15 months^a ÷ 12\$42.00
11. Other nonfeed costs for raising heifer calf^c\$50.00
12. Interest on feed and nonfeed costs for raised heifer calf
396.20 [sum of lines 4–11] ÷ 2 x 0.11 (interest rate) x 15 months^a ÷ 12\$27.24
13. Total reduced cost [sum of lines 4–12]\$423.44
14. Total positive effects [line 3 + line 13]\$910.29

Negative effects

Added cost

15. Cost of purchased replacement heifer\$750.00
16. Other costs for purchased replacement heifer^b\$0
17. Total added cost [line 15 + line 16]\$750.00

Reduced returns

18. Reduction in returns experienced if replacement heifer is purchased^b\$0
19. Total negative effects [line 17 + line 18]\$750.00

Financial analysis

20. Change in net income per heifer replacement [line 14 – line 19]\$160.29
21. Change in annual net income for herd
\$160.29 [line 20] x 20 (number of heifer replacements required per year)
x (12 ÷ 15 months^a)\$2,564.64
22. Average annual rate of return
\$160.29 [line 20] ÷ \$750.00 [line 15] x (12 ÷ 15 months^a) x 10017.1%

^aThe numerator (lines 2, 10, 12) and denominator (lines 21, 22) should equal the number of months between the sale of the heifer calf and the purchase of the replacement heifers.

^bEnter only the effects occurring during the period between the sale of the heifer calf and the purchase of the replacement heifer.

^cMay include repairs, utilities, fuel, insurance, etc.

Source: Nelson, D.D. and G.S. Willett. 1992. *Analyzing the Economics of Raising versus Buying Beef Replacement Heifers*. EB17. Washington State University Cooperative Extension.

Worksheet 8.2—Blank

Change in Net Income if Replacement Heifer Is Purchased

Positive effects

\$ per head

Added returns

1. Net returns from sale of raised heifer calf (_____ lb x \$_____/lb)\$_____
2. Interest on net returns from heifer calf sale
\$_____ [line 1] x _____ (interest rate) x _____ months^a ÷ 12\$_____
3. Total added returns [line 1 + line 2]\$_____

Reduced cost^b

4. Value of hay fed to raised heifer calf (_____ tons x \$_____/ton)\$_____
5. Value of pasture grazed by raised heifer calf\$_____
6. Value of salt and minerals for raised heifer calf\$_____
7. Other feed costs for raised heifer calf\$_____
8. Veterinary and medicine expenses for raised heifer calf\$_____
9. Value of labor and management for raised heifer calf (_____ hours x \$_____/hr)\$_____
10. Raised heifer calf's share of bull cost
\$_____ (annual bull cost) ÷ _____ females per bull x _____ months^a ÷ 12\$_____
11. Other nonfeed costs for raising heifer calf^c\$_____
12. Interest on feed and nonfeed costs for raised heifer calf
_____ [sum of lines 4–11] ÷ 2 x _____ (interest rate) x _____ months^a ÷ 12\$_____
13. Total reduced cost [sum of lines 4–12]\$_____
14. Total positive effects [line 3 + line 13]\$_____

Negative effects

Added cost

15. Cost of purchased replacement heifer\$_____
16. Other costs for purchased replacement heifer^b\$_____
17. Total added cost [line 15 + line 16]\$_____

Reduced returns

18. Reduction in returns experienced if replacement heifer is purchased^b\$_____
19. Total negative effects [line 17 + line 18]\$_____

Financial analysis

20. Change in net income per heifer replacement (line 14 – line 19)\$_____
21. Change in annual net income for herd
\$_____ [line 20] x _____ (number of heifer replacements required per year)
x (12 ÷ _____ months^a)\$_____
22. Average annual rate of return
\$_____ [line 20] ÷ \$_____ [line 15] x (12 ÷ _____ months^a) x 100 %

^aThe numerator (lines 2, 10, 12) and denominator (lines 21, 22) should equal the number of months between the sale of the heifer calf and the purchase of the replacement heifers.

^bEnter only the effects occurring during the period between the sale of the heifer calf and the purchase of the replacement heifer.

^cMay include repairs, utilities, fuel, insurance, etc.

Source: Nelson, D.D. and G.S. Willett. 1992. *Analyzing the Economics of Raising versus Buying Beef Replacement Heifers*. EB17. Washington State University Cooperative Extension, Pullman, WA.

Dairy Cattle Feeding and Nutrition management

Training Package for Dairy Extension Workers



August, 2017



SNV



DAIRY TRAINING CENTRE
Developed with
Dairy Training Centre

Dairy Cattle Feeding and Nutrition management

Part I Training manual

Part II Training Guideline

Preface

SNV Ethiopia, through EDGET project (Enhancing Dairy Sector Growth in Ethiopia, 2013-2018), engages in the capacity building, extension services and innovative support to the Ethiopian dairy sector particularly working with smallholder dairy farmers. The aim of the project is to increase milk production and productivity in order to double the income of the smallholder dairy farmers. EDGET is operational in the regions Oromia, Amhara and SNNP, and working with 65,000 dairy farming households.

The project works closely together with livestock regional bureaus and their respective zonal, woreda and kebele staff in delivering extension and other supports. One area of collaboration is the development of practical training and coaching tools and materials for extension workers based on a need assessment.

SNV has engaged the Netherlands based Dairy Training Centre (DTC) for the development of the Training package for extension workers. The documents were more elaborated and validated with the utmost contribution of high level experts from regional Livestock and Fisheries resources Development Bureaus/Agencies and Research Centers from the three operational regions of EDGET.

Overall nine training packages were developed on Breed Improvement and Fertility Management; Dairy Cattle Feeding and Nutrition Management; Dairy Cattle Health Management; Dairy Farm Management; Dairy Housing and Manure Management; Farm Economics; Forage Production and Management; Hygienic and Quality Milk Production; Young Stock Management.

*This training package is on **Dairy Cattle Feeding and Nutrition Management.***

SNV, also on behalf of the experts that contributed and DTC, would hope to see the materials widely used outside the project areas by all interested dairy development practitioners. The materials will be available in hard copies and soft copies including on SNV website www.snvworld.org and other relevant websites.

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Abbreviations

ADF	Acid Detergent Fiber
BCS	Body Condition Scoring
CF	Crude Fiber
CP	Crude Protein
DCP	Digestible Crude Protein
DM	Dry Matter
EE	Ether Extract
IOM	Inorganic Matter
NDF	Neutral Detergent Fiber
NFC	Non Fiber Carbohydrates
NFE	Nitrogen Free Extract
NPN	Non-Protein Nitrogen
NRC	National Research Council
OM	Organic Matter
TDN	Total Digestible Nutrients

I. Dairy Cattle Feeding and Nutrition Management Training manual

1. Introduction

Over the last decades an enormous amount of research has been done into the ways in which livestock use feed to maintain life, to grow, to reproduce and to produce all of the animal products which man uses, especially milk.

In Ethiopia animal production is often a secondary activity within the subsistence farming system. However, commercial dairy farming in Ethiopia is expanding due to income increase, rapid urbanization and changing dietary habits.

Dairy cows (and all other ruminants) have the capacity to convert roughages and other by-products from the human food industry, otherwise useless waste products, into valuable products like milk and meat. In this capacity as “waste product utilizers” ruminants contribute to the environment. However also negative impact from managing livestock is there. In Ethiopia overgrazing has caused land erosion and in 2007 the FAO calculated that the livestock sector contributes 18 % of all greenhouse gas emissions which lead to climate change.

Another aspect is the cost of feeding. In virtually all commercial livestock farming systems the purchase of feed and/or the cost of growing fodder is the most expensive item of the production costs. Because of these high costs it is therefore very important that feedstuffs are utilized in the most optimal way. To do so it is very important that one has excellent knowledge and skills with regard to feeding management.

In some cases, due to the “green revolution”, farmers are able to produce more and better quality feed and fodder. However, due to the absence of proper skills in feeding management, the extra feed and fodder resources are not utilized in the best way.

Farmers always have to work against limitations which are usually not of their own making or choosing. For example shortages of land, shortages of high quality feedstuffs to offset an abundance of poor quality roughages, climate change, etc. Each of these situations brings different, farmer specific, practical problems, but the nutritional challenge and dairy cattle nutrition management objective remains the same: An attempt to produce a properly balanced diet at the least cost to the farmer without affecting the environment giving the farmer the optimum income from his/her dairy enterprise.

Although this manual will not provide the answer for each and every “specific situation” it does provide the (theoretical) knowledge and guiding principles to help farmers in their feeding management.

2. Dairy Cow Behavior and Impact on Dairy Cattle Nutrition Management

2.1 Dairy Cow Behavior

Dairy cows have evolved from originally ruminating prey animals. Rumination and the fermentation which takes place in the rumen allows dairy cows/ruminants to turn low quality forages and human food by-products into high quality food: milk and meat. Ruminating prey animals forage in open plains with plenty of grazing grass and quickly

return to hiding places in the bush/forest where they lie down at a comfortable and dry place and ruminate protected from predators.

They are social animals and always will graze, lie down and ruminate together. This behavior is still present in the dairy cows today. Dairy cows also use eating and drinking to confirm relationships and dominance in the herd. Dominant cows make it clear that they are the boss and cows lower in the hierarchy indicate that they know their place at eating and drinking places. It is essential behavior which will be expressed.

Prey animals always keep an eye on each other and their surroundings and react immediately to any threats, other animal's reaction and unfamiliar situations. A nervous cow eats more quickly resulting in a lower feed intake. Cows will also stand more in such a situation instead of lying down. Nervousness can be caused by many things such as conflicts with other cows, fear of people, not enough space to lie down or cubicles which are not comfortable (wet/dirty, incorrect dimensions, hard surface, etc.), not enough eating or drinking space causing dominant cows to chase away lower ranking animals, different feeding times (cows need "routine" !), etc.

2.2 Impact of behavior on Feeding Management

The above explained natural behavior has the following implications on Feeding Management:

- The feeding manger should be large enough to allow all cows to eat at the same time! The feeding manger should have feed, preferably of the same quality, 24 hours per day to avoid dominant cows to eat the best. Preferably feed mangers should be 5-10 cm higher than the standing place of the cow (a position similar to the situation of grazing animals). In this position cows will produce more saliva during feeding which improves rumen function. Make sure that the feed fence is comfortable for cows to access, see also the Dairy housing and manure management manual.
- Feed cows always at the same time. Make sure that feed is easily accessible. Cows need/want routine. Feed enough so that the feeding manger will never be empty and accept feed leftovers. Clean the feeding manger at least once a day (feeding mangers with a smooth easy to clean surface also leads to reduced heating up of feed). In this situation cows will eat in small portions throughout the day ensuring that the rumen is always full and functioning well, see also chapter 3.
- Make sure that cows have access to clean and fresh water 24 hours per day. Cows drink 6 to 14 times per day. Usually after milking and after every meal. Preferred temperature should be between 17 – 27 oC. Cows prefer to drink together from an open low trough (50 cm). Make sure that at least 10 % of cows can drink at the same time.
- Make sure that there are enough comfortable cubicles/lying down areas for the cows (cows should lie down at least 12 hours per day). They will spend 8 to 10 hours ruminating.
- Treat cows gently (no shouting, no beating, etc.). Avoid stressful situations.



Figure 1: Enough feed and feeding space ensures a high feed intake

3. The digestive system of ruminants.

3.1 The foregut of the ruminant.

Digestion means the breaking down of the different components of a feedstuff in such a way that they can be taken up by the blood stream and transported to the places in the body where they are needed. The process of digestion in non-ruminants as well as in ruminants takes place under the influence of enzymes. Enzymes are substances which stimulate or facilitate certain chemical processes.

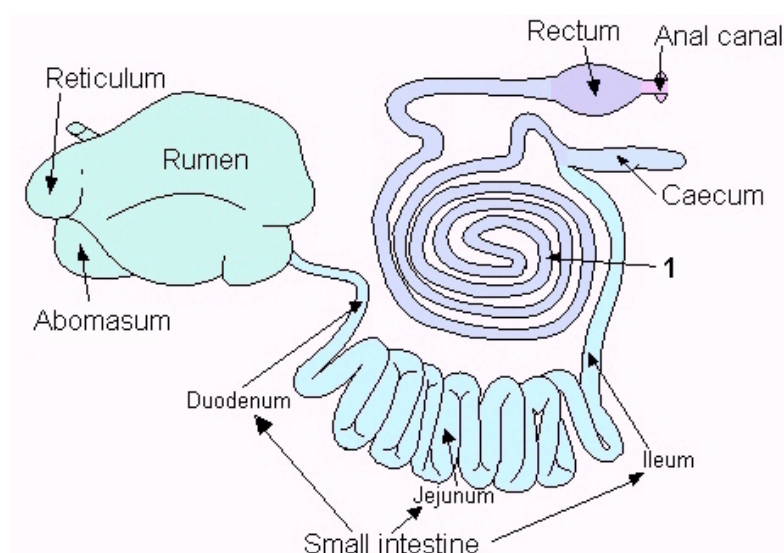


Figure 2: Schematic representation of ruminant's digestive system

The abomasum is comparable to the stomach of non-ruminants; the other stomachs (the so called fore-stomachs) are specific for the ruminant.

Due to the relative large volume and specific action of the fore stomachs, the cow is able to take in quite a lot of bulky vegetable feed. Through breeding the exotic breeds (Holstein Frisian and the Jersey for example) have evolved into cattle breeds having relatively big digestive systems.

A common descriptive term for the rumen is the fermentation vat, as it serves as a reservoir of feed and supports an active microbial fermentation.



Figure 3: Rumen Papillae

The inside of the rumen wall is lined with papillae, small finger-like projections, which increase the rumen's absorptive area (see figure 3). Volatile fatty acids, ammonia and water move through the rumen wall directly to the blood stream. Increasing the rumen available energy content of the diet (starch) will stimulate papillae growth. This in turn improves rumen absorption of volatile fatty acids into the blood stream. The number and size of rumen papillae of cows not receiving concentrates (like dry cows) are therefore less. Just after birth the fore stomachs of the calf are still not developed. The milk which the calf drinks goes directly to the abomasum through the esophageal groove. However, the fore stomachs develop quite rapidly and at the age of 6 weeks the volume of the fore stomachs is about the same size of that of the abomasum. This development is (and should be) stimulated through the feeding of roughage and especially concentrate.

In adult Holstein Frisian cows the volume of the fore stomachs is about 15 times of that of the abomasum. The rumen has then a volume of 100 – 150 liters.

The rumen content is made up of three layers. At the bottom is the liquid layer. On this layer floats a solid mass of roughage particles and in the space above this layer gasses will be present.

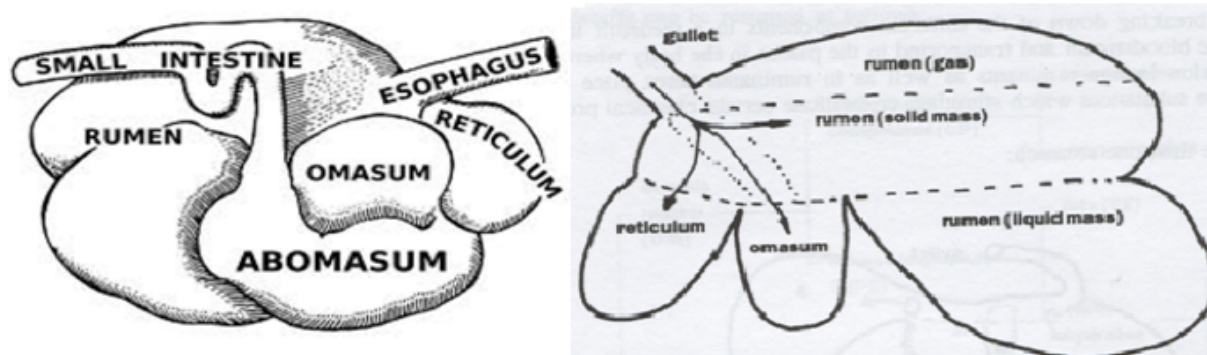


Figure 4: Schematic representation of the fore stomachs.

The rumen will have over two contractions per minute. This mixes the rumen contents, brings bacteria and feed into contact, moves material out of the rumen if it is dense and short, and brings long particles to the top surface for rumination to occur.

The cow spends 8 -10 hours per day ruminating (chewing her cud). The rumination cycle consists of four functions. **Regurgitation** occurs when an area around the esophagus is stimulated with coarse material (scratch effect of fibrous roughages). A handful of coarse material is regurgitated to the mouth by a reverse peristalsis movement.

Once the coarse feed is in the mouth, the second phase (**re-chewing**) occurs, grinding the feed to a smaller particle size which will allow it to pass out of the rumen.

The third phase is **re-salivation**. Saliva contains buffers, which mixed with the regurgitated material will stabilize the rumen pH. A cow produces over a 150 liters of saliva every day when proper rumination occurs.

The fourth phase is when the cow **re-swallows** the cud. If the coarse material has been mechanically reduced through chewing, the material should sink in the rumen and moves to the reticulum. Feed leaves the rumen when the particle size is dense (heavy) and short. Long particles will float in the rumen, forming the forage or hay raft in the top of the rumen and stimulate additional rumination time. If material is too coarse, such as straw, too much time is needed to re-chew the feed, and this reduces overall dry matter intake. When feed is fermenting in the rumen large amounts of methane, carbon dioxide, and other gases are formed and must be expelled (from 30 to 50 liters per hour). Methane and carbon dioxide are important greenhouse gases.

Under normal conditions, distension from gases will stimulate the cow to have a rumen contraction which will clear the esophagus region and allow gas to be belched or eructated. If this area can't be cleared or gas forms a froth, the cow may bloat. If the animal takes liquid feed it will pass directly into the abomasum through the esophageal groove. Digestion continues in the abomasum in the same way as with animals having one stomach (non-ruminants).

The above makes it clear that ruminants are able to digest feedstuffs which are not digestible for animals having only one stomach. These feedstuffs are often waste products from the production of food for human consumption and/or grown on land which is unsuitable for anything else but fodder crops/grass production. As such ruminants are essential in each farming system for optimum utilization of natural resources and waste products and are therefore playing a very positive role in protecting the environment. The greenhouse gas emissions however pose a threat. Increasing milk production per cow is the shortest route to a reduction of these greenhouse gases per liter of milk!

3.2 Digestion of carbohydrates, protein and fat in the fore stomachs.

The feed mixed with some saliva enters the fore stomachs. In the rumen the micro flora (billions of bacteria and protozoa) starts with the digestion process. What happens in the fore stomachs during this process is the following:

a) The breaking down of part of the carbohydrates, mainly into:

- **acetic acid**, which is needed in the butter fat making process; especially cellulose contributes to the formation of acetic acid.
- **propionic acid**, which provides energy to the animal and which is needed for the formation of milk sugar; especially starches and sugars in the feed contribute.
- **butyric acid**.

These volatile fatty acids are for the largest part absorbed through the rumen wall into the blood stream of the cow and transported to the liver, mammary gland, fatty tissue deposits and other tissues. These volatile fatty acids provide 60 – 80 % of the energy requirements of the cow. Fat, carbohydrates not degraded in the rumen and protein provide the remaining energy. The remaining volatile fatty acids produced in the rumen are utilized by the microbes for the production of microbial protein.

Under special circumstances also lactic acid can be produced in the rumen. Since this acid is not volatile it can play a negative role in the rumen fermentation, causing acidification of the rumen content and rumen acidosis. Rumen acidosis will cause damaging of rumen papillae (see figure 5).



Figure 5: Damaged rumen papillae due to rumen acidosis

b) The breaking down of part of the protein of the feed into amino acids. These amino acids are either used in the formation of protein made by the micro flora in the rumen or broken down still further into ammonia, carbon dioxide and volatile fatty acids.

On average around 60 - 70 % of the dietary protein is transformed/ "digested" by the rumen microbes in the rumen. The microbes in the rumen not only use the amino acids from the feed, they are also able to make all necessary amino acids out of non-protein nitrogen (NPN), for example out of ammonia. The non-ruminant animal does not have this capability. The microbes in the rumen will only "produce" amino acids out of urea if certain conditions are met:

- There must be sufficient easily digestible carbohydrates available in the rumen to supply the energy to the micro flora to "produce" amino acids.
- There should be a clear shortage of "natural" protein in the feed.
- The urea should be supplied in small quantities and at regular intervals.

There is a limit to what the dairy cow (in fact her rumen) can “handle” in the form of NPN. Under normal circumstances around 130 g of microbial protein is produced for every 1000 g of digestible organic matter which enters the rumen! In order to achieve this level the rumen fluid should contain 150 g of ammonia per liter.

- c) Fat is present at modest levels in diets commonly fed to dairy cattle. Dietary fat and oil are consumed either as triglycerides (three fatty acids attached to a glycerol molecule) or as free fatty acids.

Rumen microbes hydrolyze triglycerides to fatty acids and glycerol (used by microbes as a minor energy source). The fatty acids can be classified as saturated and unsaturated. Rumen microbes partially hydrogenate unsaturated fatty acids, forming more saturated fatty acids with similar carbon length. Fatty acids (such as those in soybeans or fish meal) can interfere negatively with rumen fermentation and lower fiber digestibility.

Unsaturated fatty acid can be toxic to fiber digesting bacteria and coat fiber particles, thereby reducing bacterial attachment and fiber digestion. Rumen inert fats do not affect rumen fermentation. In general fat content in the ration should be limited to a maximum of 5% whereby feeding of fish oil should be minimal. Feed oil seeds whole or coarsely processed to minimize exposing rumen microbes to high levels of unsaturated fatty acids.

Digestibility coefficient.

Not all the feed an animal eats is digested. The part that is not digested will leave the body as feces and is not used by the animal. The digested part of the feed can be expressed as a percentage of the total feed intake. This percentage is called the digestibility coefficient.

Example :

A cow eats 12 kg dry matter per day, of which 3 kg is later found in the feces. This means that $12 - 3 = 9$ kg dry matter has been digested. The digestibility coefficient of the dry matter is then $9/12 \times 100 = 75 \%$. In other words 75 % of the dry matter has been digested.

Depending on the feedstuff the digestibility may vary a lot. Some products such as most feeds of animal origin, young grass and other green forages are very easily digested (high digestibility coefficient) while other products such as straw and chaff are difficult to digest (low digestibility coefficient).

Other factors which might influence the digestibility of a feedstuff are:

- The kind of animal (goat, sheep, cow; local or “exotic” breed, etc.)
- The level of feeding (the more feed the animal consumes, the lower the digestion will be)
- The structure of the feed (“fine” feed is digested more easily than “coarse” feed; it is for example not sensible to feed whole grains to cattle).

- The age of the animal; very young and very old animals digest the feed somewhat less efficiently.
- Maturity stage of the plant, early harvested and 50% blooming stage is more digestible and nutritious than matured plants.
- Physical composition of the plant, i.e. leaf to stem ratio of the feed

4.The composition of feedstuffs.

4.1 Water and dry matter.

All feedstuffs consist of water and dry matter. For example if the water content of a feedstuff is 75%, the dry matter (DM) content is 25 %. Although water is very important for cattle, it plays a vital role in functioning of the body, milk production, and the total intake of food. A shortage or uneven supply of drinking water has a direct impact on the dry matter intake and milk production. It is the dry matter which contains the nutrients cattle need.

4.2 Carbohydrates.

Carbohydrates in feedstuffs comprise of sugars, starches (e.g. derived from cereals, tubers and roots) and substances which are part of the plant cell walls, vessels and woody tissues, mainly hemi-cellulose, pectin, cellulose and lignin. Pectin is found in high levels in beet pulp, citrus pulp and legumes and is the most rapidly fermented portion of the cell wall.

Part of the carbohydrates in feedstuffs is called "crude fiber" (CF) and the rest "non fiber carbohydrates" (NFC). The latter consists of sugars, starches and sugar like substances. Sugars and starches are much easier to digest than crude fiber by the rumen microbes.

In the animal body carbohydrates are mainly used to provide the energy covering the direct needs of the animal. If more energy is provided than immediately needed, the surplus is stored as body fat.

Crude fiber is very important for rumination and for the proper functioning of the rumen. The ration for cattle therefore should contain a good quantity of crude fiber. The minimum amount is considered 20%. Because the old system for determining the CF content (boiling of feed with dilute acid and alkali) is considered not so accurate other chemical tests are now also in use:

- **Acid detergent fiber** (ADF). ADF consists of cellulose, lignin and lignified nitrogen components (also called heat damaged protein, or ADF-nitrogen).
- **Neutral detergent fiber** (NDF) consists of ADF (cellulose and lignin) and hemi-cellulose). Another term for NDF is cell wall content. NDF represents the total fiber content of the feed.
- **Effective NDF** (eNDF). eNDF measures the physical form or length of the feed. Values for eNDF are determined by sieving feeds or using screens to separate forage particle size.

4.3 Fats.

Fat or “ether extract” also provide energy. In fact a certain amount of fat provides much more energy ($\pm 3\times$) than the same amount of carbohydrates. Some vitamins (for instance A, D, E and K) are found in the fat fraction of a feedstuff.

Roughages have a low fat content. Concentrates which are by products from the vegetable oil industry (like soybean, sunflower, etc.) have a relatively high fat content. Some fat should be present in the ration given to dairy cattle to ensure the supply of the above mentioned vitamins.

Too much fat in the ration however lowers the feed intake of the ruminant and disturbs the functioning of the rumen. If whole cotton seed cake for example is fed one should not provide more than 2-3 kgs to dairy cows per day. The maximum percentage of fat what a ration can contain in general is 5 %, but depends a lot on the kind of fats.

4.4 Crude protein.

Protein is needed for growth, maintenance, reproduction and production. In general every animal needs a certain minimum amount of protein daily in order to stay healthy and productive. In the case of dairy cows a shortage of protein will eventually result in:

- Small calves at birth
- Low milk production
- Lower protein content in the milk
- Loss of body weight
- Increased risk of infectious and metabolic diseases
- Fertility problems

Crude protein is made up of “true protein” (chains of 20 different amino acids) and of inorganic nitrogen salts, ammonia, urea, amino acids and other substances. These “non protein nitrogen” (NPN) are summarized under the name of “amides”

Ruminants, such as dairy cows, can very well utilize non-protein nitrogen. Hence, instead of feeding dairy cows expensive (true) protein, cheaper sources of nitrogen can be used as well. Urea, which is a relatively cheap chemical product, is such a non-protein nitrogen. However one must be very cautious with adding urea to rations for dairy cattle.

4.5 Inorganic matter or “ash”.

Inorganic matter (IOM) is also called ash because the IOM content is determined by burning samples of feedstuffs until they are free of carbon. A high level of ash in a sample often indicates contamination with soil. In the ash are the minerals.

4.6 Summary of the composition of feedstuffs.

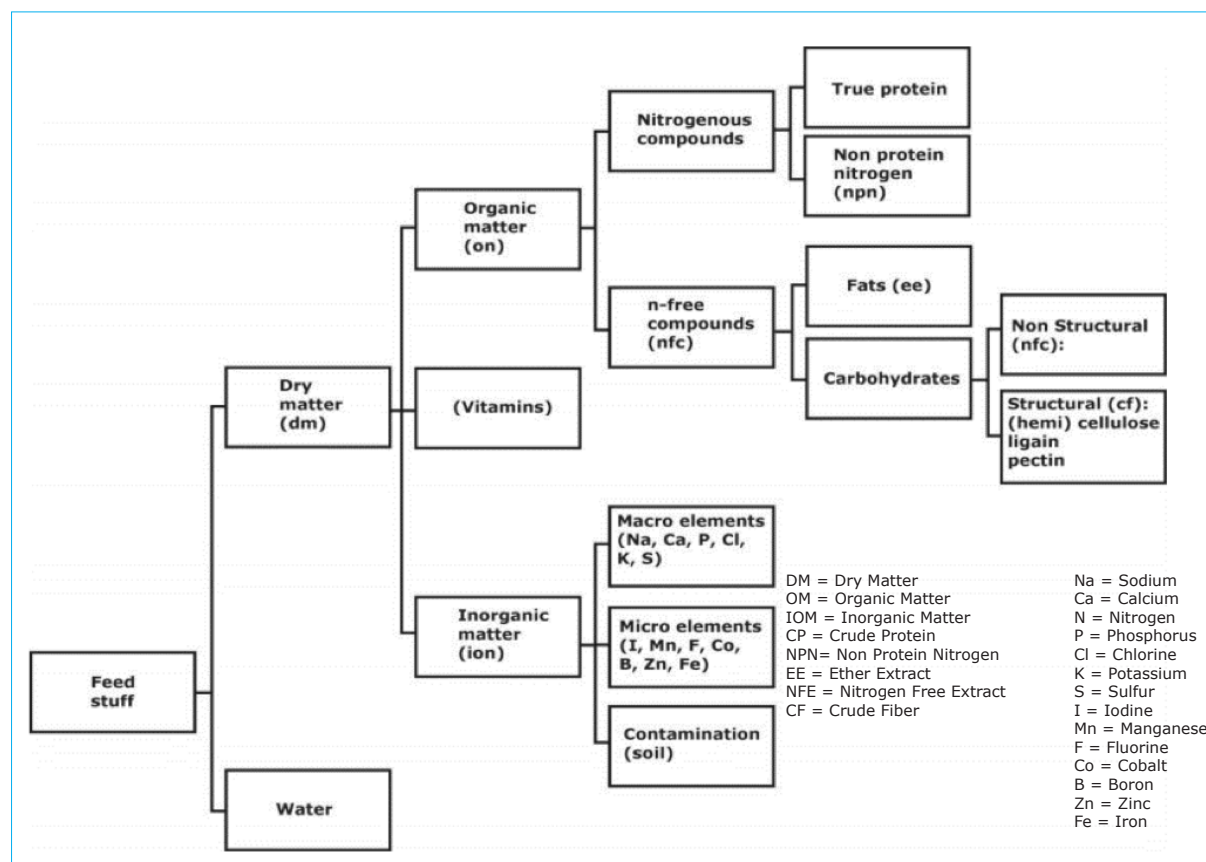


Figure 6: Composition of feedstuff

5. Determination of the feeding value of feedstuffs.

5.1 Methods used to analyze the feeding value of feedstuffs.

The feeding value of a feedstuff is determined mainly by its energy and protein content. To a lesser extent also minerals and vitamins content is important, but are not taken into consideration in this chapter.

To measure the energy and protein content of a feedstuff various methods are used:

The chemical analysis.

In the laboratory the following parts of a feedstuff are determined:

- Dry matter (DM)
- Crude protein (CP)
- Crude fiber (CF)
- Fats (EE)
- Inorganic matter or the ash

The chemical analysis used to determine the above parts is called the "Weende Analysis" or "Proximate Analysis". It is the oldest system used to determine the feeding value of feedstuffs.

In vitro analysis.

In this system feedstuffs are incubated in rumen fluid or in a laboratory solution containing various enzymes for specific periods of time in order to determine the digestibility. For calculation of exact energy and protein contents also the DM, CP and inorganic matter is determined. This method takes more time and labor, but gives a better estimation of the digestibility.

The near infra red spectroscopy (NIRS).

NIRS is a method which is fast, cheap and accurate under well controlled conditions involving only a few feedstuffs. It is based on the extent to which the various feedstuffs absorb certain wavelengths of infra red light. The accuracy is depending a lot on the calibration with standard samples (the data base) of the same feedstuff.

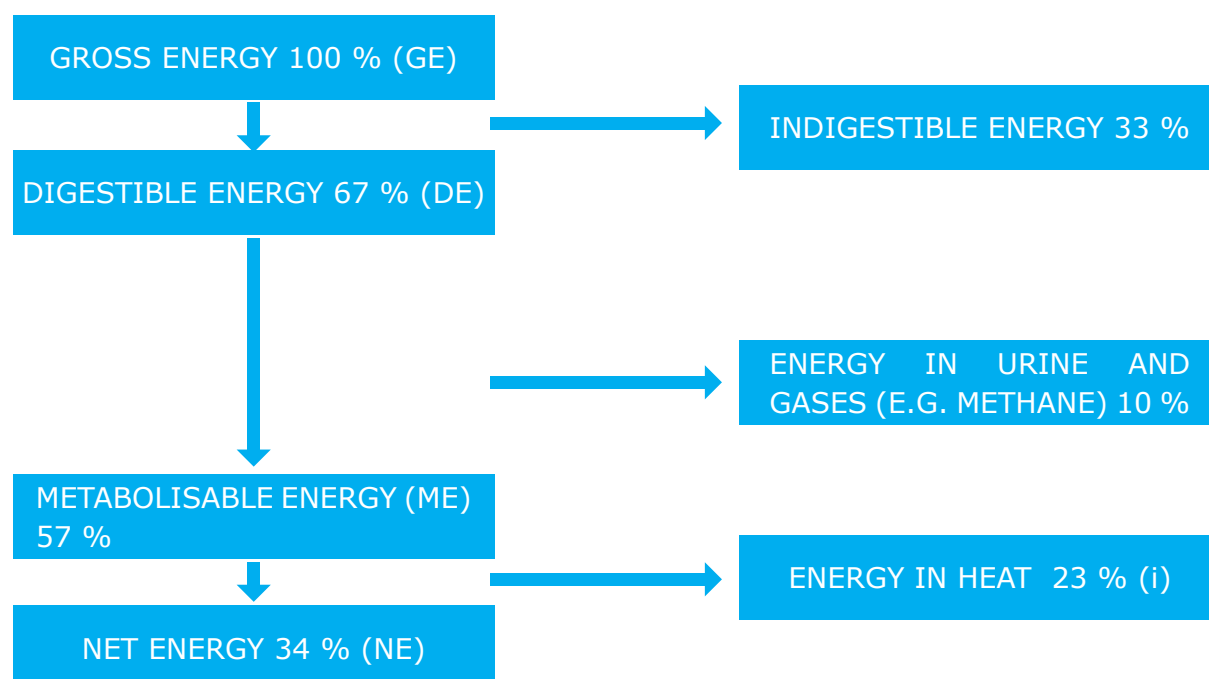
In vivo analysis.

Through incubation of feedstuffs in live animals (the so called "nylon bag method" in which nylon bags with a feedstuff are placed in the rumen through a "whole") research is being done into the degradability of various components of a feedstuff. This is a time consuming expensive method and therefore mostly used by research stations.

5.2 The utilization of energy by a ruminant animal.

The gross energy of a feedstuff can not be used for the full 100 % by the animal for its maintenance and production requirements. It can be compared with the use of petrol by a car. During combustion losses occur and only 20 – 30 % of the gross energy contained in the petrol is effectively used for moving the car.

In the following schedule a general description is shown of the utilization of energy by a ruminant:



Heat losses are due to the energy spent in chewing, rumination, absorption, etc. The losses occurring in the above schedule depend among other things upon the feedstuff, the ruminant category and their production target. In “climate cells” these losses can be measured exactly. This kind of research is however very expensive. For most feedstuffs therefore, especially feedstuffs used in developing countries, hardly any information with regard to “net energy levels” is available.

5.3 Energy evaluation systems.

Each country has often its own system for the determination of feeding values. It is not within the scope of this manual to explain each and every system and therefore energy evaluation systems available will be described in general terms.

The energy evaluation systems are categorized according to the kind of energy (NE, ME or DE) and for which production purpose. In this manual we will only focus on energy evaluation systems for milk production. In this manual we will use the most common system which is used universally, the total digestible nutrients (TDN) system.

In the TDN system the following calculating rules are used:

- 1 kg digestible carbohydrates (made up of digestible crude fiber and digestible non fiber carbohydrates) is assumed to liberate 4000 Kcal when digested.
- The same applies to 1 kg digestible crude protein.
- Digestible crude fat, however, is assumed to liberate 9000 Kcal per kg (we all know that fat is “energy rich” !).

The TDN value of a feedstuff is calculated as follows :

TDN = amount of digestible CP + amount of digestible (CF + NFC) + 2.25
x amount of digestible EE in the feedstuff

or

TDN = amount of digestible OM + 1.25 x amount of digestible EE

Example :

100 kg wheat bran has 11 kg dCP, 6 kg dCF, 42 kg dNFC and 2 kg dEE. The TDN value is then : $11 + 6 + 42 + (2.25 \times 2) = 63,5$ kg TDN per 100 kg of wheat bran.

The digestibility coefficients are determined through analyzing the composition of the feedstuff and the composition of the feces.

5.4 Protein evaluation systems.

Through the Proximate analysis the nitrogen (N) content of a feedstuff is determined, protein is composed of chains of amino acids. Research has shown that protein contains a fairly constant proportion of 16% of N. Once the N content has been determined the protein content can therefore be calculated as follows:

$100 / 16 (\%) = 6.25 \times \text{gN} = \text{protein content.}$

Based on the above principles the following protein evaluation systems are commonly used:

A. Crude protein system.

The N content of a feedstuff is determined and multiplied by 6.25 as described above. The term “crude” protein is used because the outcome of the calculation is influenced by other feed components which contain N, but are not proteins. For example amino acids, nucleic acids, nitrates, ammonia, urea, etc. These components are called amides and form the non protein nitrogen (NPN). This NPN can be used by the rumen flora and fauna and will result to a certain extent into microbial protein, see also chapter 3.2.

In short crude protein = true protein + amides

In this manual we will work with grams crude protein (g CP) of a feedstuff since this system is also used internationally.

B. Digestible crude protein.

No matter how much protein a food contains it will have no nutritional value for an animal if it is not digestible. Having determined the N content of a feedstuff the figure is corrected for the N content found in the feces. This results in the “digestible crude protein content” or dCP content of a feedstuff.

6. Feed resources in Ethiopia and their characteristics.

6.1 Roughages.

In general roughages have the following characteristics:

- Low TDN content (+ below 65% on DM basis)
- High crude fibre content (30% or above)
- Comprise of long particles which can stimulate rumen function also called “effective fibre”.

Roughages in Ethiopia can be grouped as follows :

1. Grass, including range grasses, and grassland products (hay and silage)
2. Legumes (vetch, lucern, desmodium, leucaena, clover, etc.)
3. Straw (wheat, rice, soya, bean, etc.)
4. Stover (Maize and sorghum)
5. Other (in general by products from processing of crops like sugar cane tops and tree leaves)

A. Grazing and browsing of grasses and legumes.

The feeding value of grasses is depending for a large part on the growth stage of the grass. During “aging” the CF content increases rapidly and energy content is consequently decreasing. The protein content is decreasing even more quickly during the aging of the grass. Often the soil on which grasses and legumes grows is lacking in nutrients for proper growth. Application of a fertilizer (usually containing nitrogen and phosphorus) will not only increase the quantity, but it will also improve the quality of the grasses and legumes.

Ethiopian highlands are rich in pasture species, particularly indigenous legumes. At lower altitudes native legumes are less abundant and commonly have a climbing or sprawling growth habit which leads to higher grazing losses. The area available for grazing is determined by the intensity of annual cropping. The importance of natural pasture is gradually declining due to the expansion of crop farming into grazing lands, redistribution of common lands to the landless and land degradation.



Figure 7: Naivasha Star grass pasture

Legumes usually contribute a lot to the feeding value of a ration, especially the protein content. They are particularly valuable to grazing cattle in the dry season when the protein content of grass is very low. They are also especially valuable in supplementing high energy/low protein forages like maize and maize silage. During aging legumes in general maintain a good protein level, much better compared to grasses. Also hay made of legumes still has a relatively high protein content and can function as protein supplementation in rations comprising mainly out of grass hay and/or silage. For roughage rations it is generally recommended to feed 75% of the roughage dry matter from grasses and 25% from legumes in order to balance energy and protein content.

B. Crop residues.

Cereal straw from teff, barley and wheat is the largest component of livestock diet in the intermediate and highland areas. Straw is stacked after threshing and fed to animals during the dry season, as are pulse-crop residues (e.g. horse beans, field peas and lentils). At lower altitudes maize, sorghum and millet stovers occur to a greater extent. In general cereal and rice straw have very poor energy



Figure 8: Maize silage

values and hardly contain protein. Straw from legume crops still have a reasonable protein content. Maize stover has a higher energy content compared to cereal straws.

C. Maize and sorghum stovers. Maize and sorghum are rather exceptional roughages because of their high TDN value, even when conserved as silage. They have a low protein content, however, and are best supplemented with legumes or medium high protein concentrates.

D. Cultivated Forages

Fodder crops are commonly grown for feeding dairy cattle; Oats, Desho Grass and Vetch mixtures, Alfalfa, Rhodes grass and fodder beet being the most common. Introduced fodder trees (Leucaena, Sesbania spp., Tree Lucerne) have been used for soil and water conservation programs, animal feed source, wind break, for improving soil fertility and around farmers' homesteads.



Figure 9: Young, dark green (well fertilized) leafy Napier

E. Other

There are many other crop by products used for feeding of livestock like sugar cane tops and leaves, bagasse, tree leaves, etc. Usually these products have a poor feeding value and are only used in times of feed shortages.

Judging the quality of locally produced/ available roughages is a very important skill in feeding management. Giving the above information the following criteria are important in assessing the quality/feeding value of a roughage:

- Origin of the roughage, growth stage, when harvested
- Colour, smell, palatability
- DM content
- Price

6.2 Concentrates.

Concentrates are feedstuffs that are easily digested in the rumen and have high energy and protein values compared with roughages. They usually have a low crude fibre content and comprise of very small particles. Their "structure value" is therefore very low (from 0.0 up to 0.2). Concentrates can be categorized as follows:

Energy rich concentrates.

- Grains (maize, sorghum, wheat, barley, teff, oats, etc.)
- Grain processing by-products (wheat bran, rice bran, maize gluten, etc.)

The various milling by-products obtained through processing wheat are of great interest as livestock feed for state farms, city dairy holders, and to a lesser extent for some dairy co-operatives. Wheat grain is processed in big mills, whereas in the case of teff, barley, maize and sorghum the whole grains are processed and used for food.

- Root crops (cassava, potato, etc.)
- Left over bread, by-products from the baking industry
- Sugar industry by-products (molasses, beet pulp)
- Beer industry by-product (brewer's grain). Brewer's grains are traditionally valued for lactating cows because of their palatability and milk-producing property. In addition to commercial beer production small-scale home brewing is also practiced.
- Apples, carrots, onions, fodder beet, pumpkin, banana, citrus pulp, other fruit by-products, etc.

Protein rich concentrates.

- Industrial by-products (mainly oilseed meals, cotton seed cake, etc.) Oil cakes are an excellent concentrate feed for ruminant livestock and have a high level of by-pass protein. The processing of oil seed is widely practiced on a family basis or in small village mills.
- Products of animal origin (blood meal, etc.)
- Mixed compound feeds

For more details see also annex 1 (List of common feedstuffs and their nutritive value for dairy cattle).

Concentrate mixtures in combination with suitable roughages are balanced in composition to meet the requirements of the type of animal they are designed for. They are often sold as a meal. Such a meal should not contain more than 13% of water or otherwise they can not be stored for a long period. The concentrate mix which is best for dairy cattle provides the nutrients lacking in the roughage part of the ration so that production is as high as economically possible. If the roughage is high in TDN content but low in protein content (e.g. maize silage), a concentrate may be used which has a medium TDN content and a high protein content. If the roughage has only a medium TDN value but is high in protein (e.g. young grass/legume pasture), then a high TDN/low protein concentrate may be used.

Concentrates may be bought from milling companies or made on the farm from mainly farm grown ingredients. On farm mixing is often cheaper, but it may not be advisable if the farmer does not have all the ingredients required or knowledge how to mix them.

In practice three kinds of concentrates are used for dairy cattle:

- | | |
|-----------------------------------|----------------------|
| • Low protein content mixtures | 100 – 150 g CP/kg DM |
| • Medium protein content mixtures | 150 – 180 g CP/kg DM |
| • High protein content mixtures | 180 – 250 g CP/kg DM |

6.3 Comparing the quality of roughages and concentrates on dry matter basis.

The nutritive value of a feedstuff can be expressed per kg product or per kg DM. In practice both will be used. Since the DM content of feedstuffs varies a lot comparison of the nutritive value can only be done if one takes the feeding value per kg DM.

Maize silage with 40% DM has a feeding value of 280 gTDN. The feeding value per kg DM is:

$$100/40 \times 280 = 2.5 \times 280 = 700 \text{ gTDN}$$

Wet brewer's grain with 23% dry matter has a feeding value of 736 gTDN. In the dry matter one kg product contains:

$$100/23 \times 736 = 4.35 \times 736 = 3200 \text{ gTDN per kg product.}$$

7. Feed requirements for various categories of dairy animals.

7.1 Energy and protein requirement for maintenance and lactation.

Feed requirements of dairy cattle vary according to their production stage and age categories. Cows produce approximately half of their total milk yield during the first 100 days of lactation. For this reason it is essential to feed cows properly during the early days of lactation. Feed intake is poor at the beginning of lactation but improves as lactation increases. As feed intake is not proportional with milk production requirements the cow possesses the unique ability to utilize her body reserves for milk production.

The lactating cow usually losses weight at the beginning of lactation as a result of withdrawal of her body reserves until a point when she reaches her peak. Cows can even under good feeding conditions lose as much as 66 kg in body mass during the first three months of lactation. From 120 days after calving the body mass gradually increases until calving. Adult cow, which is not producing still, needs nutrients to survive. These nutrients are required for vital body functions, like respiration, blood circulation, maintaining the body temperature, etc. and also for movement. The nutrients needed for this purpose are called the maintenance requirement of the cow. If a cow receives less than its maintenance requirement, it reduces its reserves of body fat and it starts to lose weight. The following table shows the maintenance requirements of adult cows of different body weight.

Table 1: Maintenance requirements of adult cows per day.

BODY WEIGHT	PROTEIN G CP	TDN (G)	ME MJ
200	233	1900	34.5
250	256	2200	38.7
300	276	2500	42.9
350	299	2800	47.2
400	321	3100	51.4
450	344	3400	55.7
500	366	3700	60.0
550	389	4000	64.2
600	411	4300	68.4
650	434	4600	72.7
700	456	4900	76.9
750	479	5200	81.2
800	501	5500	85.4

Source NRC

When a cow is producing (giving milk), extra nutrients are needed above the maintenance level. These extra nutrients needed for production make up the production requirements of the cow. In the next table the amount of CP and TDN is shown which a cow needs to produce one (1) kg of milk. The production requirements are related to the fat content of the milk.

Table 2: Production requirements per kg of milk

% FAT IN THE MILK	PROTEIN (G CP)	TDN (G)	ME MJ
2.5	72	255	4.19
3.0	78	280	4.53
3.5	84	305	4.87
4.0	90	320	5.21
4.5	96	340	5.55
5.0	102	365	5.89
5.5	108	390	6.23
6.0	114	415	6.57
6.5	120	440	6.91
7.0	126	465	7.25
7.5	132	490	7.59
8.0	128	510	7.93

Source : NRC

Example :

A cow weighs 500 kg and produces 10 kg milk/day. The milk contains 4 % fat. What are the total TDN and CP requirements of this cow ?

For maintenance the cow needs: 366 g CP
 3700 g TDN
 For production the cow needs: 10 x 90 = 900 g CP
 10 x 320 = 3200 g TDN
 The cow's requirements are: 366 + 900 = 1266 g CP 3700 + 3200 = 6900 g TDN

7.2 Feeding of dry and pregnant animals.

The aim of feeding the pregnant cow is to:

- Get the cow into good condition at calving
- Produce a good calf and
- Have reserves to produce milk during early lactation.

If pregnant animals are underfed, they will be in poor body condition when they give birth and begin lactating. Milk production will be significantly reduced. They will also be

slow to start cycling again. Underfed animals may also give birth to light weight and/or weak offspring. Overfeeding pregnant animals can also cause undesirable outcomes:

- Dams may have fat deposits and poor muscle condition that interfere with giving birth.
- Over conditioning also puts animals at risk of pregnancy ketosis and/or fatty liver syndrome.
- Fat young stock may have reduced milk production due to the deposition of fat in the udder.

The dry, pregnant cow should be fed with good quality roughage such as good pasture, grass, and good hay or Napier grass. During the eighth month of pregnancy, the cow should be fed as if she was producing 5 kg of milk and towards the time of calving, as if she was producing 10 kg of milk. During the eighth month, the cow can be maintained on good pasture or good Napier grass. On poor pasture/Napier grass, 1-2 kg concentrate (e.g. 50% noug cake+50% wheat bran) is necessary. During the last month of pregnancy the cow should get 1-2 kg concentrate with good quality roughage and 3-4 kg concentrate with poor quality roughage. Remember that salt and other minerals are needed. The feeds fed during the dry period should be the same as those fed during early lactation. A poor nutritional program during the pregnancy/transition period decreases reproductive efficiency and increases incidences of metabolic disorders.

Dry matter intake of the dry pregnant cow varies according to age, pregnancy status and time relative to calving. A substantial decline in intake occurs within the last 2–3 weeks of gestation for all age and pregnancy status groups supporting the need for a two-diet dry cow program. Concentrates fed to a drying cow should be stopped followed by stopping milking. It should not require any concentrates until she is “steamed up” two weeks before calving. Observation is the key though. If the animal is not in good condition it should receive improved fodder, concentrates, and minerals during her dry period to enable her to give birth to a strong healthy calf, withstand post-calving problems and have reserves to enable her to produce higher milk yield. Preferably the cow should already have the right body condition (a body condition score of around 3 to 3.5) when she is dried off. The dry cow should not gain excessive body weight. Cows that become too fat are more likely to have problems at calving time and over-feeding dry cows is wasteful and does not make economic sense.

Transition cows should not have to walk long distances to obtain water. Water is the most important nutrient. Watering devices must be kept clean. A dry period of approximately 60 days between lactations is essential for fair milk production. In short, the dry period is the time from the end of lactation until the cow calves again. Many farmers have a tendency to neglect their cows during the dry period. The cow should be dried off about two months before the expected calving date.

The reasons for drying off are:

- The nutrient requirement of the fetus are high during the last months of gestation
- To replenish body with nutrients depleted during lactation and good body condition at calving.

- To repair and regenerate the alveolar system, the milk secretor cell.
- To gain new stimulation for following lactation as a result of parturition
- To produce colostrum, this is indispensable for the calf during the first days after calving.

If the cow does not have optimum dry period between lactations its subsequent milk yield will be reduced. The dry period allows the udder to rest and the cow to become strong again, to build up a body reserve ready for the next lactation. During the dry period the cow udder gets the opportunity to recover after the lactation period. If the cow is not allowed this rest period the next lactation could result in a loss of at least 30% in milk production. It is estimated that the calf (foetus) gains 60% of its birth weight during the last six (6) weeks before calving. For an overview of energy and protein requirements of dry pregnant cows see the table below.

Table 3: Maintenance and pregnancy requirements per day (last 2 months of gestation)

BODY WEIGHT	PROTEIN (G CP)	TDN (G)	ME MJ
250	611	3020	57.2
300	700	3440	61.4
350	786	3850	65.7
400	875	4270	69.9
450	963	4690	74.2
500	1052	5100	78.4
550	1126	5520	82.7
600	1223	5910	86.9
650	1298	6290	91.2
700	1372	6680	95.4
750	1447	7070	99.65
800	1521	7450	103.9

Source : NRC

7.3 Feeding of milking cows in different stages of lactation.

Several changes occur in cows as they progress through different stages of lactation. As well as variations in milk production, there are changes in feed intake and body condition, and stage of pregnancy.

7.3.1 Feeding of a cow during early lactation

Immediately after calving, the cow needs many nutrients to recondition the body as well as to support milk production. During the first and second month after calving, the cow produces a high amount of milk per day. Adjusting the cow to the milking ration after calving is therefore an important management issue during early lactation. Feed intake will not keep pace with nutrient needs for milk production, especially for energy, and body fat will be mobilized to meet energy requirements. Cows at early lactation should be provided with all the necessary nutrients like energy and protein in order to help the cow to produce milk and allowing her to achieve peak production. Feeding

protein according to the requirements helps to stimulate feed intake and permits efficient use of mobilized body fat for milk production.

Feeding of good quality roughage and a high rate of supplementation with concentrates (noug cake and wheat bran for example) is needed to achieve a high milk production. Increasing concentrate supplementation in a gradual way after calving will increase nutrient intake and make sure the cow will continue to eat enough roughages. With high quality roughage 8-10 litres of milk per day can be obtained while it is only 2-3 litres per day on poor quality hay and straw.

Poor feeding in the early lactation results in:

- Low lactation yield,
- Poor condition
- Delayed heat or no heat signs at all (leads to poor fertility) and
- Low fat test and ketosis

Therefore, improve the feeding system of the cow at early lactation through providing good quality roughages.

7.3.2 Feeding of a cow during mid and late lactation.

After the cow has reached peak production, the milk yield will drop gradually. As a result, the energy requirement is less demanding during this stage. Sharp increase in feeding in mid lactation will result in an increase in body mass (fattening) rather than in an increase in production especially in poor productive cows. If a cow is not fed well during early lactation, maximum production will not be obtained.

Milk production requires a lot of energy. If not enough energy is provided to milking cows it will:

- Loose body condition, become thin and weak.
- Milk yield will drop.
- Pregnant cows may become ill after calving
- The calf is likely to be small

If too much energy in the diet of milking cows the animal will become fat which lead to increased incidence of:

- Difficult birth
- Retained placenta
- Displaced abomasum
- Milk fever and ketosis

Cows cannot store much protein in their bodies and so it must be supplied in the daily ration in order to maintain high milk production.

Too little protein in the ration/imbalanced rations can lead to:

- A rapid drop in milk production if the amount of protein in the ration is suddenly reduced.
- Rations providing far too little protein will also cause excessive weight loss in milking cows, in case of sufficient energy levels and low protein levels cows might also fatten

- Rations for dry and pregnant cows having too low protein levels will result in small calves being born.

Protein-rich feeds are expensive. Feeding too much protein to dairy cows is wasteful because the surplus is broken down by microorganisms in the rumen and then excreted from the body. It even takes energy from the cow to get rid of the excess also having a negative impact on milk production.

7.4 Mineral and vitamin requirements.

7.4.1 Which minerals are needed?

Two categories of minerals are needed, Micro/trace minerals and Macro minerals. There are seven macro minerals that need to be analyzed and balanced within a cow's diet. These are calcium (Ca), phosphorus (P), magnesium (Mg), sulfur (S), sodium (Na), chlorine (Cl) and potassium (K). Some of these minerals work together, while others work independently. Macro minerals are supplemented as grams per cow per day and expressed as a percentage of the ration dry matter in the NRC feeding value tables 2001

Trace or micro minerals include cobalt, copper, iodine, iron, manganese, selenium and zinc. These minerals are added as milligrams per day and expressed as parts per million (PPM) in the ration dry matter. Trace minerals or microminerals are needed for blood synthesis, hormone structure, normal reproduction, vitamin synthesis, enzyme formation, and immune system integrity. A major challenge for nutritionists and dairy managers is deficiencies do not immediately impact milk yield or growth. Reduced reproductive performance or impaired health can take several months before a deficiency appears.

Table 4: Function and deficiency signs of macro and micro elements.

a. Macro minerals	Function	Deficiency signs
Calcium (Ca)	Bone & teeth formation, blood clotting, smooth muscle contraction	Rickets, slow growth, bone fractures, lower milk yield, milk fever (hypocalcaemia)
Phosphorus (P)	Bone & teeth formation, energy metabolism, part of DNA	Rickets, poor growth, impaired reproduction, depraved appetite
Sodium (Na)	Acid-base balance, muscle contraction, nerve transmission, osmotic pressure, blood pH	Abnormal eating behavior (pica), urine licking, poor appetite, lower milk production
Chlorine (Cl)	Regulate osmotic pressure and acid-base balance, manufacture of hydrochloric acid	Loss of appetite, weakness, craving for salt, blood alkalosis
Potassium (K)	Osmotic pressure, acid-base balance, nerve transmission	Loss of hair glossiness, decreased feed intake
Magnesium (Mg)	Enzyme activator, bone and muscle, muscle contraction	Muscle hyper-irritability, salivation, convulsions, grass tetany
Sulfur (S)	Sulfur-containing amino acids, B-vitamins, cellulose digestion, acid-base balance	Reduce microbial growth, poor appetite

b. Micro minerals	Function	Deficiency signs
Cobalt (Co)	Synthesis of vitamin B12 by the rumen microbes	Poor appetite, anemia, rough hair coat
Copper (Cu)	Enzyme activation, blood synthesis, nervous system	Rough hair coat, change in hair color (grey or reddish), diarrhea, immune system impairment, mastitis
Iodine (I)	Synthesis of thyroxine (hormone)	Goiter, big neck in calves, reduced metabolic rate, poor reproductive performance
Iron (Fe)	Part of blood hemoglobin, enzyme systems, immune system function	Anemia
Manganese (Mn)	Growth, bone formation, enzyme activation	Impaired growth, poor reproduction, skeletal abnormalities
Selenium (Se)	Enzyme formation (glutathione peroxidase), protect cell membranes, immune function	Reproductive disorders, mastitis, immune system dysfunction, white muscle disease, retained placenta
Zinc (Zn)	Enzyme activation, repair of damaged tissue, immune system, teat keratin formation	Parakeratosis of the skin, elevation in somatic cell count, mastitis, hoof dysfunction, stiff joints

Source : Hoard's Dairyman Feeding guide/M. Hutjens

7.4.2 Mineral requirements of dairy cattle.

Although precise figures for mineral requirements of the different categories of dairy cattle in the tropical regions of the world are still not known there are enough research data available from which some general conclusions may be drawn as to dietary recommendations (NRC 2001). These requirements should be used only as rough guides since individual animal needs may differ from averages.

It is recognized that most of the requirements were not established in the case of zebu cattle or cattle living under tropical conditions. It is also recognized that with the introduction of exotic breeds of dairy cattle and cross-breeding production levels have increased, with a consequent increase in mineral requirements. In spite of these shortcomings, most researchers agree that the NRC mineral requirement standards is the best information currently available and that these requirements should be used until more precise data are formulated.

Table 5: Mineral requirements of dairy cattle.

Category	Cows						Heifers (age in months)			
	Dry (far off)	Dry (close up)	Fresh	Early	Mid	Late	6	12	18	24 (close up)
Macro elements required in % of DM										
Calcium (%)	0.44	0.48	0.79	0.60	0.61	0.62	0.47	0.41	0.44	0.40
Phosphorus(%)	0.22	0.26	0.42	0.38	0.35	0.32	0.25	0.23	0.18	0.23
Magnesium(%)	0.11	0.20	0.29	0.21	0.19	0.18	0.11	0.11	0.08	0.14
Chlorine (%)	0.13	0.20	0.20	0.29	0.26	0.24	0.11	0.12	0.10	0.16
Sodium (%)	1.10	0.14	0.34	0.22	0.23	0.22	0.08	0.07	0.12	0.10
Potassium (%)	0.51	0.62	1.24	1.07	1.04	1.07	0.47	0.48	0.46	0.55
Sulfur (%)	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20	0.20
Micro elements in mg per kg DM										
Iron	13-30									
Cobalt	0.11									
Copper	10-16									
Manganese	14-24									
Zinc	22-70									
Iodine	0.3-0.4									
Selenium	0.30									
Fluorine	Maximum concentration 30 mg/kg dry matter !									
Molybdenum	Maximum concentration 6 mg/kg dry matter !									
Vitamins in IU per day										
Vit. A (IU)	58000	60600	75000	75000	75000	75000	16000	24000	36000	60100
Vit. D (IU)	11700	12100	21000	21000	21000	21000	6000	9000	13500	10000
Vit. E (IU)	1168	1211	545	545	545	545	160	240	360	1202

Source: NRC 2001.

7.4.3 General remarks regarding mineral supplementation.

Economic return of mineral supplementation has been at least two to one in various studies done around the world. In Ethiopia commercial mineral mixtures are available which have been developed based on local conditions. For most commercial small holder dairy farms these will be good enough. Dairy cows should have permanent access to a box with a mineral block or mix.

7.4.4 Vitamins.

Dairy cattle require the same vitamins as non-ruminants. However, because of synthesis in the rumen and tissues, most vitamins are not needed in the diet of dairy cattle which have a functioning rumen. Rumen microbes synthesize enough B vitamins and vitamin K to meet the usual needs except for young calves. Vitamin C is synthesized in the tissues of cattle and, thus, is not needed in the diet. Dairy cattle of all ages must have a source of vitamins A and E in the diet. Likewise, vitamin D must be in the diet or synthesized in the skin under the influence of ultraviolet irradiation in sunlight.

7.5 Water requirements.

Water is essential for all categories of livestock. Whereas an animal deprived of food can survive for a considerable time, without water it will die in a few days. Preferably livestock should always be able to drink water when they want to. Unfortunately this situation is not always there and the farmer has to supply water to the animals. Under those conditions cows should be supplied with water at least 2 times per day.

Other factors which will influence the water intake are climatic conditions (hot and dry versus cold and wet) and the DM content of the supplied feedstuffs (hay versus fresh grass). Also the physiological status of the cow (lactating, pregnant or dry) & mineral content of the ration influences the intake.

In general the water requirement, including the water in the feedstuffs, of dairy cattle is as follows:

- Calves 5 – 15 litres per day
- Young stock (1 – 2 years of age) 15 – 35 litres per day
- Dry cows 30 – 60 litres per day
- Lactating cows
 - *up to 10 kg milk 30 – 60 litres per day
 - *up to 20 kg milk 70 – 100 litres per day
 - *up to 30 kg milk 90 – 150 litres per day

Water intake depends to a large extent on the ambient temperature and the body weight. Under hot conditions the intake will be higher.

7.6. Feeding guidelines for dairy crossbred cows.

- **After 7 months of lactation. Aim : bring cows in proper condition for drying off.**

Extra ration demands:

- Feed for body condition score 3.0 – 3.5
- Ideal dry period is 40 – 60 days (cows shorter, heifers at least 60 days)
- 2 weeks before drying off aim for a milk production of max 10 kg milk by reducing concentrate supplementation to 0 and/or by feeding low energy forage and reducing protein level in the ration (14% CP). BCS should not go down though! Enough water!
- **Dry cows (60 – 14 days before calving). Aim : train rumen muscles and maintain rumen papillae as much as possible.**

Extra ration demands:

- Voluminous and structure rich, 9 – 11 kg DM, feeding value of 575 gTDN/ kg DM
- Supply small amount of starch (e.g. 1 kg Barley)
- Supply enough protein in the ration (+ 12 % CP)
- K, Na and Ca poor to reduce incidence of milk fever
- Enough Mg

- **Dry cows (14 – 0 days before calving). Aim : smooth transfer from “dry” to “high milk production”.**

Extra ration demands:

- DM intake drops dramatically
- Energy and protein requirements go up very quickly (calf + colostrum production)
- Rumen flora must adjust to the feeds provided after calving : provide gradually the same basic ration as other high producing cows (650 gTDN/kg DM)
- Provide roughages ad lib (10% left over), every day cleaning of feeding trough
- Last week provide 2-3 kg concentrates in at least 2 portions (mixing concentrates and roughages by hand avoids fermentation problems), heifers 1-2 kg
- Total ration : CP level 14 %
- More starch
- Provide enough Mg, Se and Cu
- Water should not be too cold
- **Early lactation cows (0 – 20 days after calving). Aim : Careful start of the lactation, making sure that cows have a healthy start.**

Extra ration demands:

- Around calving no ration changes
- Top quality roughage (minimum of 10 – 15% of the ration dry matter should comprise of highly digestible forage like young grass and/or legume)
- Enough effective fibre to ensure optimum rumen fermentation
- Slowly increase concentrate supplementation (increase with 200 grams per day). Make sure animals eat plenty of roughage ! Cows that show healthy rumen function and good dry matter intake can have a faster increase in concentrate supplementation.
- CP level in ration 14%, this makes sure that cows slowly increase milk production (they are not “pushed” too much)
- Provide minerals/ vitamins according requirements

- **High production cows, 30 – 120/150 days in lactation. Aim : feed cows for maximum milk production and reproduction.**

Extra ration demands:

- Record milk production daily; for every 1.5 kg milk produced more provide 1 kg concentrate more
- Top quality roughage (minimum of 10 – 15% of the ration dry matter should comprise of highly digestible forage like young grass and/or legume)
- Never give more than 45% of total ration dry matter in the form of concentrates; generally 8 kg dry concentrates is maximum for crossbred cows weighing 500 kg.
- Provide a source of bypass protein (oil seed meal, fish meal, cereal bran, lignin containing legume) at a level of 10% of the dry matter intake
- Provide concentrate in portions of max. 2-3 kg
- CP level ration 16 %.

- **Low Production cows, > 150 days in lactation. Aim : produce cheap milk based on as much roughage as possible.**

Extra ration demands :

- Good quality roughage if available; DM intake from roughage can be high during this lactation stage.
- Limited concentrates
- Supplementation of concentrates above basic ration level : 1 kg concentrates/2 kg milk more.
- Later in lactation adjust concentrates in accordance to body condition
- CP level of ration around 15%
- End of lactation : body condition score of 3 – 3.5

8. Feed Improvement and feed storage.

There are different types of feed improvement methods to improve the nutrition content of feed such as urea straw treatment and multi-nutrient block. However, under smallholder condition, urea straw treatment and multi-nutrient block making are very common practices.

8.1 Urea-molasses straw treatment

Urea treatment is important for improving nutritive value of cereal straws and stovers. In addition to improving the nutrient content, urea - molasses treatment prevents loss of crop residue and consequently save the bulk which leads to improved utilization of feed resources.

Table 6: Requirements for preparing urea treatment of straw

Straw	Water (Liter)	Molasses (kg)	Urea (kg)
25	15	2.5	1
50	30	5	2
100	60	10	4

Procedure:

1. Make sure the bag is clean and has no holes
2. Chop the straw (3-5 cm)
3. Weigh urea and dissolve in water (stir solution until urea is completely dissolved)
4. Add molasses to urea and water solution and stir until completely mixed
5. Sprinkle (urea + molasses + water) solution on the straw and mix thoroughly. Good distribution is important.
6. Pack treated straw well after each addition in the bag by hand or foot
7. Tie open end of the bag tightly
8. Store in place where the bag is protected from possible damage

Note :

- Animal should feed other basal diet before feeding urea treated straw. Cows weighing 200-300 kg can feed 4-6 kg of urea treated straw in the morning and night
- Do not feed to mono gastric animals, i.e. horses, donkeys, or pigs and young ruminants less than six months of age (calves, kids and lambs).

8.2 Multi-nutrient block preparation

Multi-nutrient blocks (MNB) are a commonly used feed supplement for ruminants. MNB is made from different ingredients where each has its own contribution in the mixture. It is usually made up of molasses, urea, cement, wheat bran, protein rich by-products, salt and water which are mixed and processed to form a block. It provides fermentable

energy (usually from molasses), non-protein nitrogen (usually from urea), essential minerals and occasionally vitamins. Supplementation with Multi-Nutrient Blocks (MNB) can increase digestibility of fibrous feeds by up to 20%, increases nutrients the animal receives and can increase feed intake by 25 to 30%.

Table 7: Materials required for multi-nutrient block preparation

Ingredients	Alternative Ingredients Formulae						
	A	B	C	D	E	F	G
Wheat bran	25	25	23	25	35	22	35
Molasses	37	37	40	48	39	50	20
Urea	10	10	10	10	10	9	7.5
Salt	3	3	4	5	5	5	7.5
Cement		5	3	5	6	5	5
Limestone	10	5	10	5	5	9	5
Oil seed cake	15	15	10	2			
Clay							20
Total	100	100	100	100	100	100	100

Source: Holeta Research centre:

Procedure:

1. Weigh out correct amount of each ingredient
2. Pour molasses into a large bowl
3. Spread urea, salt and cement evenly across surface of molasses. Mix them thoroughly.
4. Add concentrate and ground oil seed cake to molasses mixture and mix well.
5. Pack the mixture into a mold. Apply enough pressure by hand to form solid mass and eliminate air pockets.
6. Remove the block from the mold and place in a well-ventilated area to cure (dry and harden). Curing can take a few days to a few weeks.
7. Once cured, the blocks are ready to be fed

Characteristics of a good Urea Molasses Multi-nutrient Block

A block is considered to be good when it fulfills the following characteristics:

- Ingredients are well distributed throughout the block
- It does not have lumps of urea
- It is hard enough not to be squashed between our fingers and should be resistant enough not to break when a person steps on it
- Our hands should feel the sticky molasses when we hold the block.

Precautions while supplementing Urea Molasses Multi-nutrient Block

It is essential to note the following while supplementing Urea Molasses Multi-nutrient Blocks:

- Feed to ruminants only (sheep, goats and cattle).
- Do not feed to mono gastric animal i.e., horses, donkeys, or pigs.
- Do not feed to young ruminants less than six months of age (calves, kids, lambs)
- Blocks should be used as a supplement and not as the basic ration
- A minimum of coarse forage in the rumen is essential
- Never give blocks to an emaciated animal with an empty stomach. There is the risk of urea poisoning due to excessive consumption
- The amount of blocks fed to sheep and goats should be limited to 100 grams/day while for cattle it should be limited to 700 grams/day;
- The blocks should never be supplied in ground form or dissolved in water as this can result in over consumption
- Supply sufficient amount of water *ad lib*



Figure 10: Examples of multi-nutrient blocks

8.3 Feed storage.

8.3.1 Causes of feed spoilage and losses

All types of ingredients, as well as completed compound feeds, require special care during storage to prevent deterioration in quality and loss. Good storage is essential because the value of the feed that you present to your animals depends on it. Feed spoils during storage. Whether it deteriorates quickly or slowly depends partly on its quality when you receive it but very largely on how it is stored on the farm.

Losses and deteriorative changes which occur during feed storage.

Environmental factors, such as moisture (feed moisture content and relative humidity), temperature, light, and oxygen influence deteriorative changes and losses in feedstuffs. These affect the feedstuff either directly or by influencing the rate of development of insects and fungi which consume the feed during storage.

The following are the major factors which affect the quality and weight of feedstuffs during storage:

Physical loss

Significant loss can occur as an accumulative effect of individually small, but regular theft. Less obvious are the losses caused by scavenging animals, particularly rats and mice. Feed stores are notorious breeding grounds for such animals. Temperature in stored feeds can sometimes increase sufficiently to cause fire. Fire can occur in stacked feeds. Feed stores are flammable, particularly if they are constantly full of fine atmospheric dust from grinding processes within the store or adjacent areas.

Water and heat damage

High levels of moisture content and relative humidity cause direct losses by making it difficult to use the material in its original form. More serious is the effect that high levels of product moisture and relative humidity have on insect infestation and the growth of fungi. High temperatures also affect the rate of loss and damage in feeds. High temperature in feeds may occur not only because of environment and the way in which they are stored but because of the heat generated by the growth of fungi and insects. Increases in temperature within large stacks of feed have been known to cause spontaneous combustion followed by fire losses. Increase in temperature may reduce availability of the amino acids in feeds.

Insect damage

Feeds are attractive places for insects including various species of moths, weevils and beetles, which consume the feed. Insects thrive better on ground materials.

Fungal damage

Higher temperatures and moisture levels favor increased fungal growth. Fungal growth itself encourages local rises in temperature and moisture content. Fungal growth causes weight loss, increases in temperature and moisture, staleness (off-flavor), discoloration and perhaps worst of all, some common species produce mycotoxins. Mycotoxins, the best known of which are called aflatoxins, are known to be toxic. Aflatoxins are toxic metabolites of *Aspergillus* fungi that can contaminate various food and feed products. When ingested, AFB1 is converted to aflatoxin M1 (AFM1) and is secreted in the milk of lactating mammals. In children, exposure to aflatoxins can cause stunting. In livestock, consumption of aflatoxin contaminated feed reduces productivity, which can result in serious economic losses.

Sorghum, maize and its by-products, groundnut, cottonseed, cassava, coconut and

sunflower are ingredients especially prone to contamination with mycotoxins. Many fungi are killed during the processing of ingredients but their spores are resistant and remain present to re-infect the material later if the environmental conditions become favorable for their development.

8.3.2 Proper feed handling and storage

Ingredients should be stored for as short a period as possible and compounded feeds used quickly, especially in tropical conditions. Vitamins should not be mixed with minerals before storage. Mixed feeds, which have undergone a heat treatment during production, such as steam pelleting, store better than other mixtures because many of the damaging factors will have been destroyed. Provide a building for storage which is secure and can be adequately locked. Ensure that its roof will protect the feed from rain and that surface water cannot enter the store. Provide it with ventilation points. Plan your ingredients purchases carefully so that you do not need to keep too large quantities in stock. If possible, raise sacks off the ground by stacking them on wooden pallets. Don't allow sacks to rest against the outer walls of the store – leave a space between the stacks and the wall.

9. Ration calculation.

9.1 Dry matter intake.

The dry matter intake depends on a number of factors:

a. The cow.

- **Level of milk yield.** A high-yielding cow takes in more kg DM than a low-yielding cow.
- **Weight.** A heavy cow has a higher DM intake than a light cow.
- **Age.** The DM intake of a heifer is around 2 kg less compared to an adult cow. As age increased dry matter (feed) intake decrease and milk yield decreased
- **Lactation stage.** On the last period Lactation stage dry matter intake decrease and milk yield decreased
- **Breed.** Dry matter intake is proportional to body weight of dairy cow; exotic breed DM intake is high than local breed.
- **Individual differences**

b. The feed.

- **Quality of the feed.** The better the quality of the roughage, the higher the intake will be.
- **Taste and smell.** Moulds, butyric acid, ammonia, etc. in roughage will reduce the intake. Also in pastures where manure has been applied will have a negative effect.
- **Digestibility.** Feedstuffs which digest slowly will remain longer in the rumen.
- **Quantity of concentrates fed.** Concentrates will reduce the roughage intake to some extent, but it will increase the total DM intake.

c. Feed supply.

- Grazing system: unlimited, limited or zero grazing.
- Rationed or “unlimited” feeding.
- Variation in the ration.
- Transition, e.g. from a summer to a winter ration, has to be implemented gradually.
- Frequency of feeding

d. Environment.

- Temperature. Particularly too high temperatures will reduce feed intake.
- Weather. In wet weather grass is less palatable, especially in the autumn season.
- The possibility for cows to rest and eat quietly, without disturbances. An overcrowded shed will lead to more fighting among cows which will affect the DM intake.
- Thumb rules for estimating the DM-intake from roughages are (*):
 - Poor quality roughages 1.5% of live weight
 - Average quality roughages 2.0% of live weight
 - Good quality roughages 2.5% of live weight

* Thumb rules are only valid if the animals are given unlimited roughages (roughage leftovers should be at least 5 -10%) ! When the roughage ration is supplemented with concentrates the total DM- intake will go up. The maximum dry matter intake from roughages and concentrates is around 3.5 % of her body weight.

The minimum amount of DM an average cow should have daily to meet maintenance and some production requirements is 2% of her live weight !



Figure 11: Three bags of napier (60 kg) is enough to feed 1 crossbreed cow for 1 day

9.2 Steps to be taken in calculating and balancing a ration.

A ration should be calculated step by step. The first step is:

- a. Estimate the average roughage quality available for the animals.
- b. With this estimated average roughage quality the average DM-intake can be calculated with the "thumb rules". If an animal does not get unlimited roughage then calculate with the actual quantities given. In this situation always check whether the quantities fed are meeting the minimum DM requirements of the animal. **In general, in case enough roughage is there, calculate with a dry matter intake of 2% of live weight out of roughage.**
- c. Calculate the kg TDN and g CP available in the roughage. If an animal is also getting a certain amount of concentrates, then add this feeding value as well (the roughages and concentrates which are "always" fed are also called the "basic ration").
- d. Calculate the maintenance and production (milk, growth or calf) requirement of the animal.
- e. Calculate whether the ration is in balance as far as energy is concerned. In general there will be an energy shortage. If so, then calculate how many kgs of concentrates are required. One can then use the **thumb rule of 1 kg concentrate for 2 kgs of milk.**
- f. Then check whether the protein in the total ration and the protein requirement are in balance. If not adjust the composition of the supplemented concentrate (or change, if possible, part of the roughages).

In case rations have to be calculated for high productive cows also check whether the maximum feed intake capacity has been reached/exceeded. Sometimes such cows can not be fed according to their requirement because the maximum feed intake has been reached. These cows usually produce extra milk out of their body reserves (a weight loss of 25 – 30 kgs during the first 2 - 3 months of lactation is quite common with the HF breed).

- a. Roughage is of poor quality.
- b. The cow will eat 1.5 % of her body weight of the straw, which is 6 kg DM.
- c. 6 kg DM straw provides the following amount of energy and protein :
 $6 \times 0.44 = 2.64 \text{ kg TDN}$
 $6 \times 36 = 216 \text{ g CP}$
- d. Maintenance requirement is:
3.1 kg TDN and 321 g CP
Production requirement for 4 kg milk, 4% fat is:
 $4 \times 0.32 = 1.28 \text{ kg TDN}$
 $4 \times 90 = 360 \text{ g CP}$
Total requirement of the cow is:
 $3.1 \text{ kg TDN (maint.)} + 1.28 \text{ kg TDN (product.)} = 4.38 \text{ kg TDN}$
 $321 \text{ g CP (maint.)} + 360 \text{ g CP (product.)} = 681 \text{ g CP}$

e. Total requirement minus what is available in the straw is:

4.38 kg TDN - 2.64 kg TDN = 1.74 kg TDN shortage

681 g CP - 216 g CP = 465 g CP shortage

Necessary concentrates (kg) to fulfill the requirements is:

1.74 kg TDN shortage./0.7 kg TDN (energy value 1 kg concentrates) = 2.5

To fulfill the energy requirement 2.5 kg concentrates are necessary.

The protein content per kg concentrate should be $465/2.5 = 186$.

The protein content of the concentrate is 180 which is roughly adequate.

9.3 Supplementation and substitution.

After calculating the TDN and g CP content of the basic ration it will become clear whether the ration needs supplementation. Sometimes farmers think that the supplementation is merely providing the animals with more roughage. However, "full is full". A cow (and other categories of cattle) can only digest a certain amount of roughage. If one provides more roughage of another kind the intake of the basic ration will be reduced with roughly the same quantity (depending on the quality). This effect is called "substitution". By substitution we understand the phenomenon that a feedstuff reduces the dry matter intake of another feedstuff because the rumen has already totally or partially been filled with feed.

Roughages more strongly prevent the intake of other products than concentrates. Concentrates digest more easily in the rumen, making space available for roughage. Substitution also depends on the quantity provided. In general, in case of concentrates, the first 2 – 3 kgs of concentrates will hardly have an effect on the intake of the roughage from the basic ration. More kgs will have an effect and this should be taken into account while calculating a ration. In the excel file for ration calculation this substitution effect is taken into account.



Figure 12: Concentrate supplementation during milking, maximum 2-3 kg.

9.4 Formulation of concentrate mixtures on the farm.

Example No. 1.

Suppose that a concentrate feed is required with a medium protein content of 180 g CP and at least 700 g TDN/ kg. The ingredients available are:

	Feeding value per kg product	
	kg TDN	g CP
Wheat bran	0.650	152
Sunflower heads	0.733	114
Cottonseed cake With hulls	0.755	242

First the difference between the desired CP content and the content of each ingredient is calculated:

Wheat bran	152 – 180	= - 28
Sunflower heads	114 – 180	= - 66
Cottonseed cake With hulls	242 – 180	= + 62

These differences must be combined in such a way that they add up to about zero. Then the overall mixture will have the desired CP content :

Wheat bran	- 28	x 2	- 56
Sunflower heads	- 66	x 1	- 66
Cottonseed cake With hulls+ 62	<u>x 2</u>		<u>+ 124</u>
	5		+2

The desired concentrate comprises of 2/5 wheat bran, 1/5 sunflower heads and 2/5 cottonseed cake with hulls. For checking whether the above answer is correct 5 kg mixture can be prepared:

	kg TDN	g CP
Wheat bran contributes	2 x 0.650 = 1.300	2 x 152 = 304
Sunflower heads contributes	1 x 0.733 = 0.733	1 x 114 = 114
Cottonseed contributes	2 x 0.755 = <u>1.510</u>	2 x 242 = <u>484</u>
Total :	3.543	902

Per kg mixture there is $3.543/5 = 709$ kg TDN and $902/5 = 181$ g CP. This meets the requirement.

If a 100 kg of the mixture is required then:

Wheat bran contributes	$2/5 \times 100 =$	40 kg product
Sunflower heads contribute	$1/5 \times 100 =$	20 kg product
Whole cotton seed contributes	$2/5 \times 100 =$	<u>40 kg product</u>
		100 kg mixture

The same calculation can be made with the feeding values on DM basis. In that case the quantities given should be converted from DM to "product", i.e. the water content of the feeds should be taken into account. However most dry ingredients of concentrate feeds have about the same DM content of 90%. The proportions of ingredients calculated on a DM basis will therefore remain the same as for the "product" basis.

Note that the above mixture has a high proportion of whole cottonseed, which has a high fat content. Cows should therefore not be fed more than 6 -7 kgs of this mixture. Higher intakes could upset the working of the rumen. The fibre content of this mixture is rather high, but the cost will probably be low.

Example No. 2.

The Pearson square method with 2 ingredients is also used to determine the proportion of 2 ingredients to achieve a specific nutrient content in the mixture

Example: Determine the right combination of lentil straw (7.7% CP) and tella atela (20.2% CP) to create a mixture containing 13% CP.

Steps in using the Pearson Square Method:

Step 1

- Draw a square with lines connecting the opposite corners.
- Write the desired percent of crude protein (13) in the center of the square.

Step 2

- Write the feeds to be used and their crude protein percent at the left hand corners of the square.

Step 3

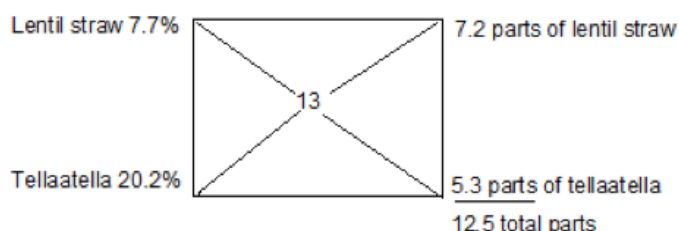
- Subtract the smaller number from the larger, along the diagonal lines. Write the differences at the opposite end of the diagonals.
- The difference between the percent protein in the lentil straw and the percent protein in the ration are the parts of tella atela needed.
- The difference between the percent protein in the tella atela and the percent protein in the ration are the parts of lentil straw needed.
- The sum of the numbers on the right equals the difference in the numbers on the left. This fact is used as a check to see if the square is set up correctly.

Divide the parts of each feed by the total parts to find the percent of each feed in the ration. Hence, the percentage of lentil straw and Tella atella to be mixed will be as follows.

Lentil straw = $(7.2/12.5) \times 100 = 57.6\%$

Tella atella = $(5.3/12.5) \times 100 = 42.4\%$.

The technique is applicable when the nutrient content of one of the two feeds is lower and that of the other is higher than the target nutrient level shown in the center of the square.



Example No. 3.

Because the cost of protein in a concentrate mixture can be quite high and the farmer might not have enough of the protein source (e.g. oilseeds) on the farm, it may be interesting to replace some of the vegetable protein with NPN in the form of urea. Urea has a very high CP value. Already a small amount of urea can therefore make quite a big difference to the CP content of a mixture. However, urea is poisonous to cattle when eaten in too great a quantity. And it can only be used by ruminants, so it should not be included in the ration for calves as their rumen is not yet developed.

As a general rule 1.5% urea in the dry matter of the concentrate/50 g per cow per day should be the maximum. Cows eating more than 3 kgs of concentrates should only have 1% urea added. Fertiliser grade urea should be finely ground using a mortar and pestle (or similar) and mixed thoroughly with the other ingredients of the mixture. If this is not done, a cow may consume large enough to kill it. There should always be enough easily-digested carbohydrate (e.g. maize) in a ration containing urea. Urea should first be mixed with a few kgs of maize bran for example before it is mixed thoroughly with the other ingredients.

Example No. 4.

Molasses is a useful supplement for dairy cattle. Its energy content is quite high (72% TDN) and it is usually quite cheap. With the inclusion of 2% urea it has a CP value of approximately 5%. Normally cows will only consume 2-3 kgs per day when given free choice. Higher intakes cause digestive problems, so molasses has only a limited role in rations for high productive cows. It is however valuable for low to medium producers which are meeting most of their protein requirements from other sources.

Other possibilities.

There are several other possible ingredients which can be used in concentrate rations for dairy cows. Poultry manure, if available, is useful for low producing cows but not for high producers due to its high NPN content. Poultry manure has a variable composition depending on the diet of the chickens, whether litter is used, and what kind of litter. Manure from battery layers has a CP value of around 18% while a common value for broiler manure is around 15%. The energy content of battery manure is only about

40%, but if maize bran is used as a litter, the energy content will be much higher.

It is also possible to use the pods of various indigenous fodder trees in concentrate rations. They have a CP value of around 15%.

10. Planning for dry season feeding.

An important feeding management objective is to have at all times enough quantity and preferably also quality roughages available to feed your animals all the year round. This requires careful planning which often is neglected. To make sure enough roughage is available in the dry season roughage has to be conserved during the growing season in the form of hay or silage besides dried crop residues. The following steps should be followed in this planning process:

- Determine the average herd composition during the dry season.
- Estimate the average roughage quality which will be available during the dry season.
- Estimate the average dry matter intake of each animal category depending on the average roughage quality.
- Based on the numbers calculate how much dry matter is required. Include minimum 5% wastage.
- Determine how much dry matter will be available from crop residues and communal grazing.
- Total amount from "d" – "e" = the amount of roughage to be conserved during the growing season or purchased.
- Estimate the dry matter yield per acre of the forage crop you want to use for hay or silage making. The amount of dry matter calculated under "f" divided by this dry matter yield determines the acreage to be grown.

11. Monitoring Dairy Cattle Nutrition Management.

For regular monitoring of dairy cattle nutrition management the following criteria and their benchmarks can be used. Benchmarks are based on a commercial small holder dairy farm with F1 crossbred cows producing around 3000 kg milk per lactation/10 kg milk per day.

Table 8: Criteria dairy cattle nutrition management

Criteria dairy cattle nutrition management	Score (1= bad, 5 = good)	Remarks
1. Herd clearly categorized into groups		Only in case of herds > 10 cows
2. Rations worked out for various ration groups		
3. Trained feeding supervisor		
4. Feed weighed in measurable units		
5. Selective feeding/ other feed/ feeding signals		
6. Rumen fill various feeding groups		
7. BCS various lactation stages		

Criteria dairy cattle nutrition management	Score (1= bad, 5 = good)	Remarks
8. Rumen function: . Cud chewing, more than 70% of animals during resting periods . 7 to 10 hrs/ day . >50 – 70 times chewing per regurgitated cud . Rumen movements (2 per minute) . Manure consistency . Checking manure for digestion		
9. Ration today is balanced ?		
10. Dry matter intake . Daily, intake drop max.5%		
11. Water and trough cleanliness		
12. Feeding frequency, push up and cleaning		
13. Sufficient feeding (feed leftover minimum 5 %/feeding trough never empty)		
14. Feed storage . Concentrates . Hay . Silage . Other		
15. . kg dry conc/cow/yr and day (production level of 3000 kg milk/yr: max. 1200 kg/yr and 4 kg/day) . kg dry conc/100 kg milk (max 40 kg/ 100 l milk) . Conc cost B/kg (max. Birr) . Roughage costs/cow/yr (depending on land area available per cow)		
16. Milk production: . Per cow per day according to expectation . Per cow per lactation according to expectation		
17. Metabolic problems incidence rate: . Milk fever, < 5% . Ketosis, < 5% . Rumen acidosis, 0% . Laminitis, < 10%		
18. Monitors used in commercial farms in Europe: . Feed efficiency (kg DM feed/kg milk) . Milk fat content . Milk protein content . Milk urea content		
19. . Enough space for cows to feed at same time . Enough space for drinking water . Floor feed manger/ trough 5 – 10 cm above standing place cows . Enough space for animals to escape from dominant cows . Comfortable resting place/cubicles . Animals stress free		
20. Planning done for dry season feeding		

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Annex 1: List of common feedstuffs in the (sub) tropics and their nutritive value for cattle

Feedstuffs	In 1 kg product			In 1 kg DM		Source
	DM g	TDN g	CP g	TDN g	CP g	
1 Cereals and other energy concentrates						
Maize grain-white	900	868	95	964	106	Tr. Feeds
Maize grain-yellow	880	868	109	964	121	Tr. Feeds
Maize grain	900	765	90	850	100	NRC
Sorghum grain	900	740	98	822	109	Tr. Feeds
Wheat grain	875	805	122	894	135	Tr. Feeds
Barley grain	900	765	117	850	130	NRC
Barley grain	900	765	93	850	103	Tr. Feeds
Tef grain	880	752	116	855	132	Feedipedia
Maize bran	900	695	52	770	58	E/C Africa
Wheat bran	900	650	152	722	169	Tr. Feeds
Wheat bran	900	630	157	700	174	NRC
Rice bran	900	680	130	756	144	NRC
Brewer's grain (sorghum)	320	230	77	720	240	E/C Africa
Brewer's grain, wet	225	166	63	736		Tr. Feeds
Brewer's grain, dry	900	603	250	278		NRC
Cassava meal (tapioca)	900	740	20	670	278	NRC
Potatoes	250	200	26	822	22	NRC
Blackstrap molasses	740	541	31	730	42	Tr. Feeds
Beet molasses	800	589	118	736	147	Tr. Feeds
Cane molasses	750	540	30	720	40	NRC
Citrus pulp	900	760	62	844	69	NRC
Sugar beet pulp	100	76	10	758	99	NRC
Maize gluten feed	900	654	239	727	266	Tr. Feeds
Bakery waste	910	820	110	901	121	NRC
Dried banana powder	830	614	45	740	54	NRC
Fat, animal/ vegetable blend	990	1752	0	1770	0	NRC
2 Legume seeds and oil seeds						
Cowpea seed	900	531	175	590	194	NRC
Cotton seed (whole)	900	880	203	978	226	NRC
Soya bean seed (whole)	900	822	372	913	413	NRC
Soya bean seed with hulls	910	844	240	928	264	Tr. Feeds
Sunflower seed with hulls	940	1118	116	1189	123	Tr. Feeds
Sunflower heads (whole)	900	733	114	814	127	Tr. Feeds
Velvet beans	900	848	233	942	259	Tr. Feeds
Velvet beans, in pods	890	720	132	810	148	E/C Africa
Whole groundnuts	930	997	202	1072	217	Tr. Feeds

Feedstuffs	In 1 kg product			In 1 kg DM		Source
	DM g	TDN g	CP g	TDN g	CP g	
3 Oil seed cakes and meals						
Cotton seed cake (with hulls)						
Groundnut oil cake, without hulls, Extr.	900	755	242	839	269	Tr. Feeds
Soya bean oil cake, with hulls	920	831	455	903	495	Tr. Feeds
Sunflower cake (without hulls)	900	760	396	844	440	Tr. feeds
Noug Cake	900	713	384	792	427	Tr. Feeds
4 Animal protein feeds	900	702	282	780	313	Feedipedia
Blood meal						
Fish meal						
Fish meal (herring)	910	603	800	670	880	Tr. Feeds
Meat and bone meal	940	725	606	770	645	E/C Africa
Feathers, poultry, hydrolyzed	900	707	697	785	774	NRC
Poultry manure	960	625	453	650	472	E/C Africa
5 Dry roughages/ crop residues (not pas- ture)	920	630	850	685	924	NRC
Sweet potato vines, hay	890	480	249	540	280	NRC
Cotton seed hulls						
Cowpea hay (with seeds)						
Cowpea straw (no seeds)	865	512	142	592	164	Tr. Feeds
Dolichos lablab hay (with seeds)	900	370	3	410	3	E/C Africa
Groundnut hay, mature	910	500	131	550	144	E/C Africa
Berseem hay, good quality	910	380	20	420	22	E/C Africa
	900	555	149	617	166	Tr. Feeds
Lucerne hay, late vegetative	920	470	118	510	128	E/C Africa
Lucerne hay, early bloom	900	500	135	550	150	Pakistan
Lucerne hay, midbloom						
Lucerne hay, full bloom	900	570	180	630	200	NRC
Lucerne hay, early bloom	900	540	162	600	180	NRC
	900	520	153	580	170	NRC
Maize cobs	900	500	135	550	150	NRC
Maize husk (cob sheath)	900	510	115	565	125	Tr. Feeds
Maize stover						
Sorghum stover	900	685	19	761	21	Tr. Feeds
	950	435	4	460	4	E/C Africa
Soyabean hay (mature)	850	425	50	500	59	NRC
Soyabean pod husks	900	450	36	500	40	Tr. Feeds
Soyabean straw						
	920	525	50	570	54	E/C Africa
Velvet bean hay, mature	870	350	10	400	11	E/C Africa
Wheat straw	870	330	15	380	17	E/C Africa
Rice straw						
Tef straw	900	480	104	530	115	E/C Africa
	900	400	32	440	36	NRC
Sugarcane bagasse	900	400	41	440	45	NRC
Sugarcane tops, fresh, post ripe	920	497	38	540	41	Feedipedia
Dry sugarcane leaves						
	910	255	19	280	21	NRC
	400	223	18	558	45	Tr. Feeds
	850	425	22	500	26	Tr. Feeds

Feedstuffs	In 1 kg product			In 1 kg DM		Source
	DM g	TDN g	CP	TDN g	CP g	
6 Dry roughages (pasture)						
Rhodes grass hay (early cut)	910	490	34	540	37	E/C Africa
Rhodes grass hay (first cutting)	870	425	32	490	37	Tr. Feeds
Rhodes grass hay (late cut)	910	430	0	470	0	E/C Africa
Starr grass hay	890	445	47	500	53	E/C Africa
Guinea grass (full seed)	910	410	15	450	16	E/C Africa
Napier grass hay, mature	900	365	68	405	75	Tr. Feeds
Natural pasture hay	930	455	4	490	5	E/C Africa
range grass hay	900	410	36	450	40	Pakistan
Rhodes grass/ Siratro hay	880	440	88	500	100	E/C Africa
Stylo hay (2 months regrowth)	760	440	112	580	147	E/C Africa
Berseem hay (good quality)	900	555	135	615	150	Tr. Feeds
Berseem hay, stemmy, few leaves	900	450	45	500	50	Pakistan
Lucerne hay, late vegetative	900	570	180	630	200	NRC
Lucerne hay, early bloom	900	540	162	600	180	NRC
Lucerne hay, mid bloom	900	520	153	580	170	NRC
Lucern hay, full bloom	900	500	135	550	150	NRC
Italian ryegrass hay, early vegetative	900	612	137	680	152	NRC
Italian ryegrass hay, early bloom	900	486	58	540	65	NRC
Perennial ryegrass hay, early bloom	900	522	54	580	60	NRC
Desho grass hay	900	435	36	483	40	Feedipedia
Vetch hay	900	576	180	640	200	Feedipedia
Pigeon Pea hay	900	576	131	600	145	Feedipedia
Desmodium Greenleaf hay	850	451	109	530	128	Feedipedia
7 Fresh pasture and roughages						
Sweet potato vines, fresh	150	100	20	665	135	E/C Africa
Sweet potato vines, fresh	85	60	15	705	175	Tr. Feeds
Maize, fresh, 8 weeks	160	100	16	625	100	Tr. Feeds
Green maize, milky stage	200	120	16	600	80	Pakistan
Sorghum, fresh, mature	230	115	10	500	45	Tr. Feeds
Sorghum, fresh, before flowering	200	115	16	575	80	E/C Africa
Sorghum-sudan grass (over 60 cm High to avoid prussic acid poisoning)	300	180	22	600	75	NRC
Napier grass, fresh, veg., 40 cm	200	128	20	640	98	Tr. Feeds
Napier grass, fresh, 80 cm	200	107	18	533	90	Tr. Feeds
Napier grass, fresh, early bloom, 240 cm	250	124	18	497	72	Tr. Feeds
Desho grass, young vegetative stage	263	147	17	558	65	Feedipedia
Rhodes grass, 1 month regrowth	200	128	18	640	90	E/C Africa
Rhodes grass, 2 months regrowth	230	135	11	590	50	E/C Africa
Rhodes grass, 3 months regrowth	260	115	5	440	20	E/C Africa
Leucaena (ipil ipil) leaves	250	168	72	670	290	NRC
Leucaena pods	240	208	63	867	261	Feedipedia
Maize without cob	350	175	23	500	65	Pakistan
Green oats (boot to milk stage)	200	120	34	602	169	NRC
Green wheat (boot to milk stage)	200	126	33	631	164	NRC

Feedstuffs	In 1 kg product			In 1 kg DM		Source
	DM g	TDN g	CP g	TDN g	CP g	
Naivasha starr grass, late veg., 25 cm	200	134	38	670	191	Tr. Feeds
Naivasha starr grass, early bl., 40 cm	280	162	31	578	110	Tr. Feeds
Naivasha starr grass, early bl., 60 cm	300	164	33	547	110	Tr. Feeds
Naivasha starr grass, standing hay	910	379	47	417	52	Tr. Feeds
Perennial rye grass, fresh, early veg.	150	105	27	700	180	Netherlands
Berseem, 1st cutting, early veg.	110	75	24	683	220	Tr. Feeds
Lucerne, early vegetative	110	75	27	686	245	Tr. Feeds
Lucerne, early bloom	200	131	41	655	203	Tr. Feeds
Lucerne, mid bloom	250	140	40	558	160	Tr. Feeds
Lucerne, fully flowering	300	144	36	480	120	Pakistan
soya bean, mid-bloom, fertilized	160	103	27	642	166	Tr. Feeds
Natural grass:						
Early growing season	300	190	25	630	82	E/C Africa
Mid-growing season	300	145	12	490	39	E/C Africa
Mid-dry season	550	135	0	250	0	E/C Africa
Tree lucern young leaves	200	160	50	800	250	Tropical For.
Vetch	200	140	46	700	230	Feedipedia
Pigeon Pea	300	195	57	650	190	Feedipedia
Desmodium Greenleaf	240	140	37	581	155	Feedipedia
8 Silages						
Rhodes grass silage	240	122	11	508	45	Tr. Feeds
Rhodes grass silage, early flowering	230	135	14	580	63	E/C Africa
Rhodes grass silage, full bloom	250	135	12	550	49	E/C Africa
Rhodes grass silage, mature	300	140	5	460	16	E/C Africa
maize silage, good	400	280	25	700	63	E/C Africa
maize silage, average	330	210	17	640	56	E/C Africa
maize silage, poor	270	150	9	550	35	E/C Africa
maize silage, milk stage	220	120	13	545	60	E/C Africa
grass silage, early bloom (Dutch quality)	450	300	70	667	156	Netherlands
sorghum silage	260	145	14	560	53	E/C Africa
sunflower silage	270	156	30	578	111	NRC
Napier (elephant) grass silage, 2 m high	240	115	5	480	21	E/C Africa
Dwarf elephant (Mott) grass silage	290	171	31	590	107	NRC

Resources

Remarks:

-Most of the data comes from research stations in East and Central Africa (indicated by E/C Africa), from the NRC and from the FAO publication Tropical Feeds and FAO Website "Feedipedia". Various other sources have been used as well.

-The list is not at all exhaustive. Our advice to the users of this manual and these feeding value tables is to also consult local research stations or other local sources of information and try to collect feeding value data for all products that are locally available.

-Feeding value data should not be taken too rigidly. Products known under one and the same name are seldom quite comparable and analysing techniques and standards may not be exactly the same in different parts of the world. This is to some extent reflected in our list. Our advice is to rely on the data which local research stations use.

-Grazing cows generally choose the most nutritious vegetation (i.e. selective grazing) when plant growth is relatively abundant. Feeding value data apply to the average product quality. Therefore cows may actually eat a product that is better than assumed. Also keep in mind that the TDN system is a "rough" system which has been designed for assessing fodder quality in temperate climates. It might overestimate the feeding values of tropical roughages. Nevertheless it is not recommended to feed dairy cows (and other livestock) more or less "blindly", i.e. without having an idea of the qualities of the product(s) that are included in feeding rations.

-The data refer to the "pure" product and not to mixtures including a large portion of weeds.

-Presumably the data refer to products grown on well fertilised soils.

Dairy Cattle Feeding and Nutrition management Training Guideline for Dairy Extension Workers



II. Dairy Cattle Feeding and Nutrition management Training Guideline

A. Module Book

Introduction

The Module Dairy Cattle Feeding and nutrition management is part of the Working Packages introduced By SNV Ethiopia as part of the EDGET Project. One of the EDGET project objectives is to make small holders dairy farms more aware about “commercial farming”. The Module Dairy Cattle Feeding and nutrition management will help the dairy extension workers to acquire the knowledge and skills to feed dairy cows for optimal milk production.

Feeding management is one of the most important aspects in a commercial dairy farm. It is not only determining milk production levels, but it also has a major impact on the health and fertility levels on the dairy farm. Proper feeding management also influences economic results of the dairy farm to a very large extend. Higher levels of milk production also ensure an efficient use of feed resources reducing the amount of greenhouse gas emissions per litre of milk.

Professional situation

As an extension worker, you have to advise smallholder farms on how to manage their crops and livestock in the most optimal way particularly the dairy enterprise. A smallholder dairy farmer in Ethiopia has to manage his/her dairy farm unit/ enterprise in the most optimal way given his/her specific circumstances. This also means that the farmer has to determine the objectives of his/her dairy enterprise, has to develop action plans for achieving these objectives, has to develop tools for monitoring and evaluation, in short applying the so-called “management cycle” also concerning dairy cattle feeding and nutrition management.

As an extension worker, you have the knowledge, skills and innovations to manage the feeding of all categories of dairy cows in the most optimal way and you have the motivation and “drive” to convince smallholder dairy farmers to improve on their knowledge and skills in order to improve the milk production level of their dairy enterprise. Feeding and nutrition management is the backbone of every dairy farm and is very important to make your dairy farm successful. Not only for the income on the short term but also to sustain the farm on the longer term. The Smallholder Farm Manager has to deal with feeding management issues like:

- Is there enough roughage in stock for the dry or winter period,
- How will I distribute the different feed resources to the different groups of animals,
- Which quality is needed for my breed of dairy cattle, young stock, dry cows, and milk producing animals and maybe also beef cattle,
- How to optimize the rations for a better animal health and (re)production,
- What will happen when rations are not well balanced, will it result in metabolic diseases, etc.

- What is the best feeding system/bunk management given my situation
- improve and conserve feed to increase feed intake and palatability

Not only calculations are important but also observing the animals, what signals are they giving to you as a farmer, how can we recognize and interpret them and what kind of information does it give you. And the most important: what are you going to do, so acting.

In this module, it is your task to develop your knowledge and skills in such a way that you are able to act properly yourself, and that you can guide your farmers, and assist them to optimize the total farm results. On dairy farms, worldwide farmers are looking for the ideal ration for their cows in their circumstances with their roughages and concentrate types. A higher milk production per cow not only requires better quality of roughages, but also a better cow management.

There is also a direct link between the supply of good quality and sufficient roughages, high milk production, greenhouse gas emissions, health and good fertility levels. All the knowledge and skills need to be present on the farm regarding nutritional issues. If only 1% is missing, it might influence your total (financial) result with more than 50%. In all of the above tasks you will consider the “gender” situation of the farm (who takes the decisions with regard to feeding of the livestock, who does the actual feeding on the farm, who decides on which concentrates, minerals and other feeding inputs to buy) in order to address the right target group for farmer training.

Required entry qualification

To take part in this module on Dairy Cattle Feeding and Nutrition Management you should comply with the following entry requirements:

- Competent in the English language
- Have completed other EDGET Program training modules successfully
- Competent in basic calculation skills
- Have basic insight/experience in managing small holder dairy farms

Specific objectives and related topics

- a. At the end of the course, participants can judge whether the situation on a farm is conducive for cows feeling comfortable with regard to feeding.

Related topics are:

- Cow behavior and impact on feeding management
 - Feed bunk management aspects including water
- b. At the end of the course, participants are able to assess the rumen function of dairy cows and their link with metabolic problems.

Related topics are:

- How the rumen functions
- Checking the rumen function using various “cow signals”
- c. At the end of the course participants are able to make an inventory of available feedstuffs on a dairy farm, assess their feeding value (dry matter content, energy level, protein level) using physical inspection and the use of a feeding value table and judge the quality of roughages using different criteria.

Related topics are:

- Composition of feedstuffs and their feeding value
- Criteria for judging the quality of roughages
- d. At the end of the course, participants are able to calculate dry matter intake of dairy cows, their feed requirements (energy, protein) and compose/ calculate rations to meet these requirements.

Related topics are:

- Calculation of dry matter intake and feed requirements
- Selection of feedstuffs for rations for different categories of dairy cows
- Ration calculation by hand and the use of an excel program
- Explanation of feed improvement and conservation
- General guidelines for feeding dairy cows in different stages of lactation
- Assessing whether a dairy farmer has enough roughages in stock to feed his animals during the dry season
- e. At the end of the Course, participants are able to assess dairy cattle nutrition management on dairy farms using a number of criteria and benchmark them.

Related topics are:

- “Monitors” and key performance indicators for proper dairy cattle nutrition management and their reference levels/ benchmarks

Assessment

During the course, one assessment will be conducted to measure the competence level of the participants to advise a smallholder farm on his/ her dairy cattle feeding and nutrition management performance. The assessment will be a group assignment. The group (maximum 4 persons) will have to visit an assigned smallholder dairy farm and implement the following tasks:

- a. Making a checklist for monitoring the feeding management.
- b. Visit of the assigned smallholder dairy farm, monitor the herd and the dairy unit and fill out this checklist.

- c. Develop at least two advices for improving the feeding management on the assigned farm.
- d. Assess the quantities of roughages available on the farm and calculate for how many days roughage supply is available.
- e. Based on the roughage supply assessment you prepare a feeding plan for this farm.
- f. Calculation by hand and the use of the excel sheet of amount of concentrate supplementation of to the ration of all cows present on the farm.
- g. Improve feed and conserve feed for dairy cattle and advice which animals should feed the improved feed
- h. Presentation of results

Activities

Below an overview of all activities related to this module are presented:

Day	Time	Subject
Day 1	1 st	-Introduction Module/ participants/ trainer/ brainstorm on the gender situation on small holder farms with regard to feeding management and introduction into dairy cattle feeding and nutrition management
	2 nd	-Feed resource availability -Feed bunk management aspects including water -Rumen function and assessment criteria
	3 rd and 4 th	Visit of small holder farm and implement following practical lessons: -Judging the quality of available roughages -Making an inventory of available feedstuffs and their quality/ feeding value -Assess "feed bunk" management on the farm -Assess the rumen function of cows on the farm -Homework (study relevant chapters of the manual)
Day 2	1 st	-Calculation of dry matter intake and feed requirements -Selection of feedstuffs for rations for different categories of dairy cows -Ration calculation by hand and the use of an excel program
	2 nd	-Ration calculation continued
	3 rd and 4 th	Visit of small holder dairy farm -Assessment of roughage supplies on the farm and for how many days supplies are sufficient to feed the animals -Feeding of the dairy cows present on the farm (weighing right quantities, chopping, cleaning manger, feeding, calculation of concentrate supplementation and feeding during milking -Conserve and improve dairy feed for supplementation -Homework
Day 3	1 st	-General guidelines for feeding dairy cows in different stages of lactation -Assessing whether a dairy farmer has enough roughages in stock to feed his animals during the dry season
	2 nd	-Monitors and key performance indicators for assessing dairy cattle nutrition management on a small holder dairy farm -Explanation of assignment/ assessment
	3 rd	-Visit of assigned small holder dairy farm and implement assignment

Day	Time	Subject
	4 th	-Assignment/ assessment continued
Day 4	1 st	Prepare for presentation
	2 nd	Presentation in presence of farmers
	3 rd	Brainstorm on farmer training in dairy cattle nutrition management taking into consideration the "gender situation" on small holder farms
	4 th	Course Evaluation and closing

Notes:

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B. Lesson matrix

Lesson Matrix	
Topic / Serial #	
Practical Lesson	Lesson 1 Dairy Cattle Feed and nutrition Management
Date	
Venue	
Duration	1 day/ Day 1 of the Course
Type of students	Dairy Extension Workers
Suggested number of students	16
Starting situation	Students have very little or no experience with Dairy Cattle Feeding and Nutrition Management
Learning Objectives:	The student is able to:
Skills	-Assess feeding environment and management aspects on farms -Assess quality of feedstuffs on farms, especially roughage quality -Calculate how many kg fresh/dry matter a cow will eat per day and quantity of roughage required to overcome the dry season -Assess rumen function and link with metabolic problems -Assess feed conservation
Knowledge	-Explain main animal feed resources available in the area -Explain how the rumen functions and the linkage with some metabolic diseases -Explain what impact good feeding management has on other management areas on dairy farms like health and fertility -Explain differences in feeding values of feedstuffs and how to assess the quality of roughages
Attitude	Convince farmers that dairy cattle feeding and nutrition management is a crucial part in dairy farm management influencing not only milk production but also the health and fertility levels on the farm and especially the income a farmer can earn

Time in minutes	Brief Content	Role trainer / didactical methods	Teaching aids	Role/activities Participants
15	-Opening of the course, getting acquainted with each other, explanation Module Book and Course Objectives	-Explaining/ guiding/ making trainees feel at ease	-Module Book	-Listening, introduction of themselves, asking questions, answering questions
30	-Sharing experiences with dairy cattle feeding and nutrition management -Gender situation on farms, who is doing the feeding, who is buying concentrates, etc. -Distribution Dairy Cattle Feeding and Nutrition Management Manual	-Asking for experiences to get an idea about student level and the gender situation in their working area -Introduction to dairy cattle feeding and nutrition management	-Dairy Cattle feeding and Nutrition Management Manual	-Sharing experiences -Check contents of Manual
75	-Feed resource assessment in relation to dairy requirement	-Sharing experiences/ lecturing/ PPT/ assignment	-Manual -PPT/ video -Questions	-Sharing experiences/ listening/ asking + answering questions
30	Break			
120	-Rumen function and assessment criteria, link with some metabolic problems	-Sharing experiences/ lecturing/ PPT/ assignment	-Manual -PPT/ video -Questions/ exercise 1	-Sharing experiences/ listening/ asking + answering questions
60	Lunch			
180	Visit of small holder farm and implement following practical lessons: -Judging the quality of available roughages -Making an inventory of available feedstuffs and their quality/ feeding value -Assess "feed bunk" management on the farm -Assess the rumen function of cows on the farm	-Travel to farm -Explaining assignments and worksheets: *Roughage quality assessment *Inventory feedstuffs *Assessment rumen function	-Small holder farm	-Visit small holder farm/ assess quality of roughages/ make inventory of feedstuffs/ assess rumen function/ ask clarity questions
30	-Summary/ reflection/ homework	-Summary theory/ testing knowledge -Common made mistakes -Study chapter 1/2/3/4/5 -Finish assignment rumen function	-PPT -Manual -Assignments -Homework study Manual	Answering and asking questions

Lesson Matrix	
Topic / Serial #	
Practical Lesson	Day 2 Dairy Cattle Feed and Feeding Management
Date	
Venue	
Duration	1 day/day 2 of the Course
Type of students	Dairy Extension Workers
Suggested number of students	16
Starting situation	Students have very little or no experience with Dairy Cattle Feed and feeding Management
Outcomes	The student is able to:
Skills	-Calculate how many kg fresh/ dry matter a cow will eat per day and quantity of roughage required to overcome the dry season -Calculate energy and protein requirement of dairy cows -Calculate rations for dairy cows/ cow groups manually and with the use of an excel file -Feed a cow/ group of cows in line with requirements
Knowledge	-Explain energy and protein requirements of dairy cows -Explain mineral/vitamin and water requirements of dairy cows -Explain how much a cow can eat in dry matter and fresh matter -Explain how to calculate a ration
Attitude	Convince farmers that dairy cattle feed and feeding management is a crucial part in dairy farm management influencing not only milk production but also the health and fertility levels on the farm and especially the income a farmer can earn

Time in minutes	Brief Content	Role trainer / didactical methods	Teaching aids	Role/activities Participants
120	-Discussion Homework and chapter 1 – 5 -Calculation of dry matter intake and feed requirements -Selection of feedstuffs for rations for different categories of dairy cows -Ration calculation by hand and the use of an excel program	-Discussion results assignment -Answering questions chapter 1 – 5 -Explaining ration calculation -Give assignment ration calculation by hand -Students present -Explanation excel file -Students exercise	-Manual -Answer sheet assignment -PPT ration calculation -Assignment ration calculation by hand -White board for students to present answers -Excel ration calculation file	-Answering/ asking -Listening -Making assignments -Exercising excel file
30	Break			
120	Ration calculation continued	Ration calculation continued	Ration calculation continued	Ration calculation continued
60	Lunch			
180	Visit of small holder dairy farm -Assessment of roughage supplies on the farm and for how many days supplies are sufficient to feed the animals -Feeding of the dairy cows present on the farm (weighing right quantities, chopping, cleaning manger, feeding, calculation of concentrate supplementation and feeding during milking)	-Guidance and coaching	Excursion farms	-Collection of information, measuring supplies, weighing feed, chopping, feeding cows
30	-Summary/ reflection/ homework	-Summary theory/ testing knowledge -Common made mistakes -Study chapter 6/7 -Finish assignment ration calculation	-PPT -Manual -Assignments -Homework study Manual	Answering and asking questions

Lesson Matrix	
Topic / Serial #	
Practical Lesson	Day 3 Dairy Cattle Feed and nutrition Management
Date	
Venue	
Duration	1 day/ Day 3 of the Course
Type of students	Dairy Extension Workers
Suggested number of students	16
Starting situation	Students have very little or no experience with Dairy Cattle feed and feeding Management
Outcomes	The student is able to:
Skills	-Assess feeding environment and "bunk" management aspects on farms -Assess quality of feedstuffs on farms, especially roughage quality -Assess rumen function and link with metabolic problems -Monitor and evaluate quality of feeding management on farms and advise on improvements -Assess water supply/ mineral supply -Assess feed improvement -Assess feed conservation -Write report on findings and recommendations -Present findings and recommendations to farmers
Knowledge	-Explain preparation of urea mineral multi-nutrient block -Explain urea straw treatment -Explain hay and silage making -Give general guidelines for feeding dairy cows in different stages of lactation
Attitude	Convince farmers that dairy cattle feeding and nutrition management is a crucial part in dairy farm management influencing not only milk production but also the health and fertility levels on the farm and especially the income a farmer can earn

Time in minutes	Brief Content	Role trainer / didactical methods	Teaching aids	Role/activities Participants
?	-feed improvement and conservation methods -Preparation of urea molasses multi nutrient block straw treatment, hay and silage making	-Sharing experiences/ lecturing/ PPT/ assignment	-Manual -PPT/ video -Questions/ exercise 1	-Sharing experiences/ listening/ asking + answering questions
?	Visit of small holder farm and implement following practical lessons: -Judging the quality of available hay and silage -Assess feed improvement system and making inventory of available improved feed and their quality/ feeding value -Assess "feed bunk" management on the farm -Assess feed conservation methods	-Travel to farm -Explaining assignments and worksheets: *urea treated straw quality assessment *quality assessment molasses multi nutritional block *Assessment hay and silage making *assess EM technology	-Small holder farm	-Visit small holder farm/ assess quality of MMNB urea straw treated and prepared hay and silage
120	-Discussion homework -General guidelines for feeding dairy cows in different stages of lactation -Assessing whether a dairy farmer has enough roughages in stock to feed his animals during the dry season + assignment	-Discussing/ asking -Explaining and discussion -Explaining and discussion + explaining assignment	-Manual -PPT -Assignment	-Asking/ discussing/ listening/ making assignment and presentation
60	Break			
120	-Monitors and key performance indicators for assessing dairy cattle feeding and nutrition management on a small holder dairy farm, making assignment -Explanation of assignment/ assessment	Explaining/ discussion/ give assignment (included in the PPT) Explaining assignment/ assessment/ making of groups	-Manual -PPT -Assignment	-Listening/ asking/ making assignment/ discussing -Working on assignment
60	Lunch			
180	-Visit of assigned small holder dairy farm and implement assignment	Guiding/ coaching	-Assignment	-Students collecting and processing information
30	-Summary/ reflection/ advises for assignment	-Summary theory/ testing knowledge -Common made mistakes -Study chapter 8/9/10 -Finish assignment	-PPT -Manual -Assignment -Homework study Manual	Answering and asking questions

Lesson Matrix	
Topic / Serial #	
Practical Lesson	Day 4 Dairy Cattle Feed and nutrition Management
Date	
Venue	
Duration	1 day/ Day 4 of the Course
Type of students	Dairy Extension Workers
Suggested number of students	16
Starting situation	Students have very little or no experience with Dairy Cattle feed and feeding Management
<u>Outcomes</u>	The student is able to:
Skills	-Write report on findings and recommendations -Present findings and recommendations to farmers
Knowledge	-Give general guidelines for feeding dairy cows in different stages of lactation -Explain how feeding management can be monitored and evaluated
Attitude	Convince farmers that dairy cattle feed and feeding management is a crucial part in dairy farm management influencing not only milk production but also the health and fertility levels on the farm and especially the income a farmer can earn

Time in minutes	Brief Content	Role trainer / didactical methods	Teaching aids	Role/activities Participants
120	Students prepare for presentation	-Guidance and coaching	-target farms	-Preparing presentation
60	Break and arrival of farmers			
120	Presentation assignment results	-Assessing performance	-Projector/ white board/ flip overs	-Group presentations
60	Lunch (students and farmers)			
60	-Assessment result per group	-Reflection on assignment results and assessment score	-Discussion per group	-Reflection and discussing assessment result
60	-Brainstorm on farmer training -Special focus on gender, whether the male or female members are involved in feeding dairy cattle and take decisions with regard to feeding management of the farm, etc.	-Lead discussion and gives ideas how to train farmers in the topic; -Who are the target group for the training, only men or only women or both -What lesson material to use and at what level to train	-Experiences during the Course -Experiences with the small holder farms in the working area	-Active participation in the brain storm session
30	-Evaluation of course	-Explaining evaluation form	-Evaluation form	-Filling evaluation form
30	Closing	-Final Remarks -Awarding certificates	-Certificates	-Listening and receiving their certificate

C. Checklist for monitoring feeding management

Feed

		Good	Sufficient	Poor
Hay	Quality	0	0	0
	Palatability	0	0	0
	Heating up/fungi	0	0	0

Remarks:

		Good	Sufficient	Poor
Other products	Quality	0	0	0
	Palatability	0	0	0
	Heating up/ fungi	0	0	0

Remarks:

Cows

		Too thin	Good	Too fat
Body condition	High productive	0	0	0
	Middle productive	0	0	0
	Low productive	0	0	0
	Dry	0	0	0
	Fresh, < 60 days	0	0	0

Remarks: _____

		Good	Sufficient	Poor
General	Hair coat	0	0	0
	Rumen fill	0	0	0
	Rumen function	0	0	0
	Hooves	0	0	0
	Other:	0	0	0

Remarks: _____

		Good	Sufficient	Poor
Manure	Consistency	0	0	0
	Colour	0	0	0
	Digestion	0	0	0

Remarks: _____

Ration

		Good	Sufficient	Poor
Feeding hygiene	Cleanliness trough	0	0	0
Drinking water	Hygiene/ quality	0	0	0
Composition on	Ration calculation	0	0	0
Paper	Minerals	0	0	0

Remarks:

D. Judgement results last milk production registration

	Good	Sufficient	Poor
Standard cow production (SCP)	0	0	0
SCP results over the year	0	0	0
SCP level various production groups	0	0	0
Milk fat %	0	0	0
Milk fat % level of the various			
Production groups	0	0	0
Protein %	0	0	0
Protein % level of the various			
Production groups	0	0	0
Urea level milk	0	0	0

Remarks:

E. ANALYSIS OF POSSIBLE METABOLIC DISORDERS

Calculate the various percentages and compare with desired levels with the help of the last milk production registration.

Month	Number of cows 0 – 60 days in lactation	% of cows with lower than expected milk yield	Fat minus Protein > 1,25 (%)	Fat minus protein < 1 (%)	Protein < 3 (%)
January					
February					
March					
April					
May					
June					
July					
August					
September					
October					
November					
December					
Levels which indicate a problem might be there		> 10 % Cows start lactation poorly	> 45 % Negative energy balance	> 38 % Rumen acidosis	> 18 % Negative energy balance/ too low feed intake



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